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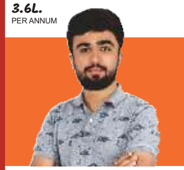
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(SCIENCE STREAM)
(CLASS - XII)

Part - A : Time : 1 Hour / Marks : 50
Part - B : Time : 2 Hours / Marks : 50

(Part - A)

Time : 1 Hour] *[Maximum Marks : 50*

Instructions :

1) There are 50 objective type (M.C.Q.) questions in Part - A and all questions are compulsory.
2) The questions are serially numbered from 1 to 50 and each carries 1 mark.
3) Read each question carefully, select proper alternative and answer in the OMR sheet.
4) The OMR sheet is given for answering the questions. The answer of each question is represented by (A) O, (B) O, (C) O and (D) O. Darken the circle ● of the correct answer with ball-pen.
5) Rough work is to be done in the space provided for this purpose in the Test Booklet only.
6) Set No. of Question Paper printed on the upper- most right side of the Question Paper is to be written in the column provided in the OMR sheet.
7) Use of simple calculator and log table are allowed, if required.
8) Notations used in this question paper have proper meaning.

1) The number of arbitrary constants in the general solution of a differential equation of fourth order are ________.

(A) 2

(B) 0

(C) 3

(D) 4

Rough Work

Rough Work

2) For the differential equation $y \log y\, dx - x\, dy = 0$, the general solution is ________.

(A) $y = e^{cx}$

(B) $x = e^{-cy}$

(C) $y = e^{-cx}$

(D) $x = e^{cy}$

3) The homogeneous differential equation

$\left(1 + e^{x/y}\right)dx + e^{x/y}\left(1 - \frac{x}{y}\right)dy = 0$ can be solved by taking the substitution ________.

(A) $v = yx$

(B) $y = vx$

(C) $x = vy$

(D) $x = v$

4) Find angle θ between the vectors $\vec{a} = \hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$.

(A) $\pi - \cos^{-1}\left(\frac{1}{3}\right)$

(B) $\cos^{-1}\left(\frac{1}{3}\right)$

(C) $\pi - \cos^{-1}\left(\frac{2}{3}\right)$

(D) $\cos^{-1}\left(\frac{2}{3}\right)$

Rough Work

5) Find the area of the parallelogram whose adjacent sides are determined by the vectors $\vec{a} = \hat{i} - \hat{j} + 3\hat{k}$ and $\vec{b} = 2\hat{i} - 7\hat{j} + \hat{k}$.

(A) 15

(B) $15\sqrt{2}$

(C) $\frac{15}{\sqrt{2}}$

(D) 30

6) The value of $\hat{i}\cdot(\hat{j}\times\hat{k})+\hat{j}\cdot(\hat{i}\times\hat{k})+\hat{k}\cdot(\hat{i}\times\hat{j})+\hat{j}\cdot(\hat{j}\times\hat{k}) =$ ______________.

(A) -1

(B) 0

(C) 1

(D) 3

7) Find the direction cosines of the vector $\hat{i} - 2\hat{j} + 3\hat{k}$.

(A) $\frac{-1}{\sqrt{14}}, \frac{-2}{\sqrt{14}}, \frac{-3}{\sqrt{14}}$

(B) 1, -2, 3

(C) $\frac{1}{\sqrt{14}}, \frac{-2}{\sqrt{14}}, \frac{3}{\sqrt{14}}$

(D) $\frac{1}{14}, \frac{2}{14}, \frac{3}{14}$

8) Find a vector of magnitude 5 units and parallel to the resultant of the vectors $\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{b} = \hat{i} - 2\hat{j} + \hat{k}$.

(A) $\frac{3\sqrt{10}}{2}\hat{i} - \frac{10\sqrt{2}}{2}\hat{j}$

(B) $\frac{5}{\sqrt{51}}\hat{i} - \frac{5}{\sqrt{51}}\hat{j} - \frac{35}{\sqrt{51}}\hat{k}$

(C) $\frac{3\sqrt{10}}{2}\hat{i} + \frac{10\sqrt{2}}{2}\hat{j} + \frac{\sqrt{2}}{2}\hat{k}$

(D) $\frac{3\sqrt{10}}{2}\hat{i} + \frac{\sqrt{10}}{2}\hat{j}$

Rough Work

9) $(\vec{a}+\vec{b})\cdot(\vec{a}+\vec{b})=|\vec{a}|^2+|\vec{b}|^2$ if and only if __________, $(\vec{a}\neq\vec{0}, \vec{b}\neq\vec{0})$.

(A) $\vec{a}, \vec{b}$ are perpendicular

(B) $\vec{a}, \vec{b}$ are in same direction

(C) $\vec{a}, \vec{b}$ are in opposite direction

(D) $\vec{a}, \vec{b}$ are neither parallel nor perpendicular

10) If the lines $\frac{1-x}{3}=\frac{7y-14}{2p}=\frac{z-3}{2}$ and $\frac{7-7x}{3p}=\frac{y-5}{1}=\frac{6-z}{5}$ are at right angle then the value of $p=$ __________.

(A) $\frac{70}{11}$

(B) 70

(C) $-\frac{70}{11}$

(D) -70

11) Find the Cartesian equation of the line through the point $(5, 2, -4)$ and which is parallel to the vector $3\hat{i}+2\hat{j}-8\hat{k}$.

(A) $\frac{x+5}{3}=\frac{y+2}{2}=\frac{z-4}{-8}$

(B) $\frac{x-5}{3}=\frac{y-2}{2}=\frac{z+4}{-8}$

(C) $\frac{x-5}{3}=\frac{y-2}{2}=\frac{z-4}{-8}$

(D) $\frac{x-5}{-3}=\frac{y-2}{-2}=\frac{z+4}{-8}$

Rough Work

12) Find the angle between the pair of lines $\frac{x}{2}=\frac{y}{2}=\frac{z}{1}$ and $\frac{x-5}{4}=\frac{y-2}{1}=\frac{z-3}{8}$.

(A) $\pi-\cos^{-1}\left(\frac{2}{3}\right)$ (B) $\cos^{-1}\left(\frac{2}{3}\right)$

(C) $-\cos^{-1}\left(\frac{2}{3}\right)$ (D) $\sin^{-1}\left(\frac{2}{3}\right)$

13) For linear programming problem the objective function $Z = 3x + 9y$, if the corner points of the feasible region are (0, 10), (5, 5), (15, 15) and (0, 20), then the maximum value of Z is __________.

(A) 90 (B) 60

(C) 0 (D) 180

14) The solution of the linear programming problem, minimize $Z = 3x + 4y$, subject to the constraints $x + y \leq 4$, $x \geq 0$, $y \geq 0$ is __________

(A) 16 (B) 12

(C) 28 (D) 0

C

15) If $2\,P(A)=P(B)=\frac{5}{13}$ and $P(A/B)=\frac{2}{5}$ then $P(A\cup B)=$________.

(A) $\frac{10}{13}$ (B) $\frac{11}{13}$

C

(C) $\frac{11}{26}$ (D) $\frac{10}{26}$

16) If A and B are any two events such that

P(A) + P(B) – P(A and B) = P(A), then ________.

(A) P(A/B) = 1 (B) P(B/A) = 1

(C) P(B/A) = 0 (D) P(A/B) = 0

Rough Work

01

17) The relation R in the set R of real numbers, defined as $R = \{(a,b) : a < b\}$ is ________.

(A) transitive but neither reflexive nor symmetric

(B) symmetric but neither reflexive nor transitive

(C) reflexive and symmetric but not transitive

(D) reflexive and transitive but not symmetric

18) The function $f : N \to N$, defined by $f(x) = \begin{cases} x+1, \text{if } x \text{ is odd} \\ x-1, \text{if } x \text{ is even} \end{cases}$ is ________.

(A) many-one and onto

(B) one-one and onto

(C) one-one but not onto

(D) neither one-one nor onto

19) A function $f : R \to R$, $f(x) = 4x + 3$, $f^{-1}(x) =$ ________.

(A) $\frac{x-4}{3}$ (B) $\frac{x+4}{3}$

(C) $\frac{x+3}{4}$ (D) $\frac{x-3}{4}$

Rough Work

20) $\tan^{-1}\left(-\sqrt{3}\right)-\sec^{-1}(-2)=$ ________.

(A) π

(B) $-\frac{2\pi}{3}$

(C) $-\pi$

(D) $\frac{2\pi}{3}$

21) $\sin\left(\tan^{-1}x\right), |x|<1=$ ________.

(A) $\frac{1}{\sqrt{1-x^2}}$

(B) $\frac{x}{\sqrt{1-x^2}}$

(C) $\frac{1}{\sqrt{1+x^2}}$

(D) $\frac{x}{\sqrt{1+x^2}}$

22) Write the simplest form of $\cot^{-1}\left(\frac{1}{\sqrt{x^2-1}}\right), x>1$.

(A) $-\sec^{-1}x$

(B) $\sec^{-1}x$

(C) $\operatorname{cosec}^{-1}x$

(D) $-\operatorname{cosec}^{-1}x$

Rough Work

23) $\cos^{-1}\left(\cos\frac{13\pi}{6}\right)+\tan^{-1}\left(\tan\frac{7\pi}{6}\right) =$ ________.

A

(A) $\frac{\pi}{3}$ (B) π

(C) $\frac{\pi}{6}$ (D) 0

24) If A is a matrix of order $m \times n$ and B is a matrix such that AB' and $B'A$ are both defined, then order of matrix B is ________.

(A) $n \times n$ (B) $m \times m$

(C) $n \times m$ (D) $m \times n$

25) For any two matrices A and B of order 3×3, we have ________.

(A) $AB \neq BA$

(B) $AB = BA$

(C) $AB = O$

(D) $AB = I$

26) If $A = \begin{bmatrix} \alpha & \beta \\ \gamma & -\alpha \end{bmatrix}$ is such that $A^2 = I$, then ________.

(A) $1-\alpha^2+\beta\gamma=0$

(B) $1+\alpha^2+\beta\gamma=0$

(C) $1-\alpha^2-\beta\gamma=0$

(D) $1+\alpha^2-\beta\gamma=0$

Rough Work

27) If A and B are skew symmetric matrices of the same order, then $(AB)' =$ __________.

(A) $A'B'$

(B) BA

(C) $-A'B'$

(D) $-BA$

28) If A is an invertible matrix of order 2, then determinant of A^{-1} is __________

(A) $\dfrac{1}{\det(A)}$ (B) det(A)

(C) 1 (D) 0

29) If area of triangle is 35 sq. units with vertices (2, –6), (5, 4) and (k, 4), then k is __________.

(A) –2 (B) 12

(C) –12, –2 (D) 12, –2

30) For a matrix $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$, $(A^{-1})^2 =$ __________.

(A) $\begin{bmatrix} -4 & 0 & 0 \\ 0 & -9 & 0 \\ 0 & 0 & -16 \end{bmatrix}$ (B) $\begin{bmatrix} 4 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 16 \end{bmatrix}$

(C) $\begin{bmatrix} \frac{1}{4} & 0 & 0 \\ 0 & \frac{1}{9} & 0 \\ 0 & 0 & \frac{1}{16} \end{bmatrix}$ (D) $\begin{bmatrix} -\frac{1}{4} & 0 & 0 \\ 0 & -\frac{1}{9} & 0 \\ 0 & 0 & -\frac{1}{16} \end{bmatrix}$

Rough Work

31) If A and B are matrices of order 3×3 and $|A| = 5$, $|B| = 3$, then $|3AB| =$ ________.

(A) 15 (B) 45

(C) 135 (D) 405

32) If $x = \sin y$, then $\frac{d^2y}{dx^2} =$ ________, $(0 < x < 1)$.

(A) $\frac{-x}{(1-x^2)^{3/2}}$ (B) $\frac{1}{(1-x^2)^{3/2}}$

(C) $\frac{x}{(1-x^2)^{3/2}}$ (D) $\frac{-1}{(1-x^2)^{3/2}}$

33) For $xy = e^{x-y}$, $\frac{dy}{dx} =$ ________

(A) $\frac{y(x-1)}{x(y+1)}$ (B) $\frac{x(y+1)}{y(x-1)}$

(C) $\frac{y(y+1)}{x(x-1)}$ (D) $\frac{y(x+1)}{x(y-1)}$

34) If $x = at^2$, $y = 2at$, then $\frac{dy}{dx} =$ ________.

(A) $\frac{x}{2y}$ (B) $\frac{x}{y}$

(C) $\frac{y}{2x}$ (D) $\frac{y}{x}$

35) The radius of a circle is increasing at the rate of 0.7 cm/s. What is the rate of increase of its circumference?

Rough Work

(A) 14π cm/s
(B) 1.4π cm/s
(C) 0.14π cm/s
(D) -1.4π cm/s

36) Function $y = 6 - 9x - x^2$ is strictly increasing in the interval ________.

(A) $\left(-\infty, \frac{9}{2}\right)$
(B) $\left(0, -\frac{9}{2}\right)$
(C) $(-\infty, 0)$
(D) $\left(-\infty, -\frac{9}{2}\right)$

37) What is the maximum value of the function $\sin x + \cos x$?

(A) $\sqrt{2}$
(B) 1
(C) 2
(D) 0

38) The point is on the curve $x^2 = 2y$ which is nearest to the point (0, 5) is ________.

(A) $(2\sqrt{2}, 0)$
(B) $(2\sqrt{2}, 4)$
(C) (0, 0)
(D) (2, 2)

Rough Work

39) If $\frac{d}{dx}(f(x)) = 4x^3 - \frac{3}{x^4}$ such that $f(2) = 0$, then $f(x)$ is ________.

(A) $x^3 + \frac{1}{x^4} + \frac{129}{8}$ (B) $x^4 + \frac{1}{x^3} - \frac{129}{8}$

(C) $x^4 + \frac{1}{x^3} + \frac{129}{8}$ (D) $x^3 + \frac{1}{x^4} - \frac{129}{8}$

40) $\int \frac{\sin(\tan^{-1} x)}{1+x^2} dx =$ ____________ $+ C$.

(A) $-\sin(\tan^{-1} x)$ (B) $\sin(\tan^{-1} x)$

(C) $-\cos(\tan^{-1} x)$ (D) $\cos(\tan^{-1} x)$

41) $\int \frac{x}{\sqrt{x+4}} dx, x > -4 =$ ________ $+ C$.

(A) $\frac{2}{3}\sqrt{x+4}(x-8)$ (B) $-\frac{2}{3}\sqrt{x+4}(x-8)$

(C) $\frac{1}{3}\sqrt{x+4}(x-8)$ (D) $-\frac{1}{3}\sqrt{x+4}(x-8)$

A

42) $\int \frac{e^x(1+x)}{\sin^2(x \cdot e^x)} dx =$ ________ $+ C$.

(A) $\tan(x \cdot e^x)$ (B) $\cot(x \cdot e^x)$

(C) $-\tan(x \cdot e^x)$ (D) $-\cot(x \cdot e^x)$

Rough Work

43) $\int e^{x} \cdot \sec x(1+\tan x)\,dx =$ ______ $+ C$.

(A) $e^{x} \cdot \sec x$

(B) $e^{x} \cdot \cos x$

(C) $e^{x} \cdot \sin x$

(D) $e^{x} \cdot \tan x$

44) $\int_{\pi/6}^{\pi/3} \frac{dx}{1+\sqrt{\tan x}} =$ ______ $+ C$.

(A) $\frac{\pi}{3}$

(B) $\frac{\pi}{6}$

(C) $\frac{\pi}{12}$

(D) 0

45) $\int \frac{1-\cos x}{1+\cos x}\,dx =$ ______ $+ C$.

D

(A) $2\tan\frac{x}{2}+x$

(B) $-\tan\frac{x}{2}-x$

(C) $-2\tan\frac{x}{2}-x$

(D) $2\tan\frac{x}{2}-x$

46) $\int \tan^{8} x \cdot \sec^{4} x\,dx =$ ______ $+ C$.

(A) $\frac{\tan^{11} x}{11}-\frac{\tan^{9} x}{9}$

(B) $\frac{\tan^{11} x}{11}+\frac{\tan^{9} x}{9}$

(C) $\frac{\tan^{9} x}{9}+\frac{\tan^{7} x}{7}$

(D) $\frac{\tan^{9} x}{9}-\frac{\tan^{7} x}{7}$

47) Find the area bounded by the curve $y = \cos x$ between $x = 0$ and $x = \frac{3\pi}{2}$.

C

(A) 2

(B) 1

(C) 3

(D) 4

48) Find the area of the region bounded by the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$.

(A) 36π

(B) 6π

(C) 13π

(D) 24π

49) The area bounded by the curve $y = x|x|$, X - axis and the ordinates $x = -1$ and $x = 1$ is given by _______.

(A) $\frac{1}{3}$

(B) 0

(C) $\frac{2}{3}$

(D) $\frac{4}{3}$

50) The degree of the differential equation

$$\left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{dy}{dx}\right)^2 + \sin\left(\frac{dy}{dx}\right) + 1 = 0$$ is _________.

(A) 3

(B) 1

(C) 2

(D) not defined

Rough Work

(SCIENCE STREAM)
(CLASS - XII)

(Part - B)

Time : 2 Hours] *[Maximum Marks : 50*

Instructions :

1) Write in a clear legible handwriting.
2) There are three sections in Part - B of the question paper and total 1 to 27 questions are there.
3) All questions are compulsory. Internal options are given.
4) The numbers at right side represent the marks of the question.
5) Start new section on new page.
6) Maintain sequence.
7) Use of simple calculator and log table is allowed, if required.
8) Use the graph paper to solve the problem of L.P.

SECTION - A

■ Answer any 8 questions from the questions given below from 1 to 12. (Each question carries 2 marks.) **[16]**

1) Prove that : $3\cos^{-1}x = \cos^{-1}\left(4x^3 - 3x\right); x \in \left[\frac{1}{2}, 1\right]$. **[2]**

2) Prove that : $\tan^{-1}\sqrt{x} = \frac{1}{2}\cos^{-1}\left[\frac{1-x}{1+x}\right]; x \in [0,1]$. **[2]**

3) If the function $f(x) = \begin{cases} \frac{k\cos x}{\pi - 2x}, & \text{if } x \neq \frac{\pi}{2} \\ 3, & \text{if } x = \frac{\pi}{2} \end{cases}$ is continuous at $x = \frac{\pi}{2}$, then find the value of k. **[2]**

4) Find $\int \frac{dx}{e^x - 1}$. [2]

5) Sketch the graph of $y = |x+3|$ and evaluate $\int_{-6}^{0} |x+3| \, dx$. [2]

6) Find the area of the region bounded by the line $y = 3x + 2$, the X - axis and the ordinates $x = -1$ and $x = 1$. [2]

7) Find the solution of the differential equation [2]

$$x\frac{dy}{dx} + 2y = x^2 \log x$$

8) If $\vec{a}, \vec{b}, \vec{c}$ are unit vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, find the value of $\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a}$. [2]

9) Find the vector equation of the line passing through the point (1, 2, –4) and perpendicular to the two lines : [2]

$$\frac{x-8}{3} = \frac{y+19}{-16} = \frac{z-10}{7} \text{ and } \frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5}$$

10) Show that the line through the points (4, 7, 8), (2, 3, 4) is parallel to the line through the points (–1, –2, 1), (1, 2, 5). [2]

11) Given that the two numbers appearing on throwing two dice are different. Find the probability of the event 'the sum of numbers on the dice is 4'. [2]

12) Probability of solving specific problem independently by A and B are $\frac{1}{2}$ and $\frac{1}{3}$ respectively. If both try to solve the problem independently, find the probability that exactly one of them solves the problem. [2]

SECTION - B

Answer any 6 questions from the questions given below from 13 to 21. (Each question carries 3 marks.) **[18]**

13) Let $A = R - \{3\}$ and $B = R - \{1\}$. Consider the function $f : A \to B$ defined by $f(x) = \left(\frac{x-2}{x-3}\right)$. Is f one-one and onto? Justify your answer. [3]

14) Express the matrix $\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ as the sum of a symmetric and a skew symmetric matrices. [3]

15) If $A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$, then verify that $A \cdot \operatorname{adj} A = |A| \cdot I$. Also find A^{-1}. [3]

16) If $x = a\left(\cos t + \log \tan \frac{t}{2}\right), y = a \sin t$ then find $\frac{d^2y}{dx^2}$. [3]

17) Prove that $y = \frac{4\sin\theta}{(2+\cos\theta)} - \theta$ is an increasing function of θ in $\left[0, \frac{\pi}{2}\right]$. [3]

18) If with reference to the right handed system of mutually perpendicular unit vectors $\hat{i}, \hat{j}$ and $\hat{k}$, $\vec{\alpha} = 3\hat{i} - \hat{j}, \vec{\beta} = 2\hat{i} + \hat{j} - 3\hat{k}$, then express $\vec{\beta}$ in the form $\vec{\beta} = \vec{\beta}_1 + \vec{\beta}_2$, where $\vec{\beta}_1$ is parallel to $\vec{\alpha}$ and $\vec{\beta}_2$ is perpendicular to $\vec{\alpha}$. [3]

19) Find the shortest distance between the lines l_1 and l_2 whose vector equations are

$$\vec{r} = \hat{i} + \hat{j} + \lambda\left(2\hat{i} - \hat{j} + \hat{k}\right)$$

$$\vec{r} = 2\hat{i} + \hat{j} - \hat{k} + \mu\left(3\hat{i} - 5\hat{j} + 2\hat{k}\right).$$ [3]

20) Solve the following Linear Programming problem graphically : [3]

Minimize $Z = 200x + 500y$

Subject to constraints

$$x + 2y \geq 10$$
$$3x + 4y \leq 24$$
$$x \geq 0, y \geq 0$$

21) A bag contains 4 red and 4 black balls, another bag contains 2 red and 6 black balls. One of the two bags is selected at random and a ball is drawn from the bag which is found to be red. Find the probability that the ball is drawn from the first bag. [3]

SECTION - C

■ **Answer any 4 questions from the questions given below from 22 to 27. (Each question carries 4 marks.)** **[16]**

22) If $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$, prove that $A^3 - 6A^2 + 7A + 2I = O$ and deduce A^{-1}. **[4]**

23) Solve the following system of equations by matrix method **[4]**

$$x + y + z = 6$$
$$y + 3z = 11$$
$$x + z = 2y$$

24) If $y = e^{a\cos^{-1}x}, -1 \le x \le 1$, show that $\left(1-x^2\right)\frac{d^2y}{dx^2} - x\frac{dy}{dx} - a^2y = 0$. **[4]**

25) Show that of all the rectangles inscribed in a given fixed circle, the square has the maximum area. **[4]**

26) Evaluate : $\int_0^{\pi/4} \log(1 + \tan x)\, dx$. **[4]**

27) In a culture, the bacteria count is 1,00,000. The number is increased by 10% in 2 hours. In how many hours will the count reach 2,00,000, if the rate of growth of bacteria is proportional to the number present? **[4]**

SAMPLE QUESTION PAPER

PHYSICS

Subject Code – 042

CLASS – XII

Maximum Marks: 70 **Time Allowed: 3 hours**

General Instructions

(1) There are 33 questions in all. All questions are compulsory.

(2) This question paper has five sections: Section A, Section B, Section C, Section D and Section E.

(3) All the sections are compulsory.

(4) **Section A** contains **sixteen questions, twelve MCQ and four Assertion Reasoning based of 1 mark each**, **Section B** contains **five questions of two marks each**, **Section C** contains seven questions of three marks each, **Section D** contains **two case study-based questions of four marks each** and **Section E** contains **three long answer questions of five marks each**.

(5) There is no overall choice. However, an internal choice has been provided in one question in Section B, one question in Section C, one question in each CBQ in Section D and all three questions in Section E. You have to attempt only one of the choices in such questions.

(6) Use of calculators is not allowed.

(7) You may use the following values of physical constants where ever necessary

i. $c = 3 \times 10^8$ m/s

ii. $m_e = 9.1 \times 10^{-31}$ kg

iii. $m_p = 1.7 \times 10^{-27}$ kg

iv. $e = 1.6 \times 10^{-19}$ C

v. $\mu_0 = 4\pi \times 10^{-7}$ T m $\boldsymbol{A^{-1}}$

vi. $h = 6.63 \times 10^{-34}$ J s

vii. $\varepsilon_0 = 8.854 \times 10^{-12}$ $\boldsymbol{C^2N^{-1}m^{-2}}$

viii. Avogadro's number = 6.023 X $\mathbf{10^{23}}$ per gram mole

[SECTION – A] **(16x1=16 marks)**

Q1. A uniform electric field pointing in positive X-direction exists in a region. Let A be the origin, B be the point on the X-axis at x = +1 cm and C be the point on the Y-axis at y = +1 cm. Then the potential at points A, B and C satisfy.

(A) $V_A < V_B$ (B) $V_A > V_B$. (C) $V_A < V_C$ (D) $V_A > V_C$

Q2. A conducting wire connects two charged conducting spheres such that they attain equilibrium with respect to each other. The distance of separation between the two spheres is very large as compared to either of their radii.

The ratio of the magnitudes of the electric fields at the surfaces of the two spheres is

(A) $\frac{r_1}{r_2}$ (B) $\frac{r_2}{r_1}$ (C) $\frac{r_2^2}{r_1^2}$ (D) $\frac{r_1^2}{r_2^2}$

Q3. A long straight wire of circular cross section of radius 'a' carries a steady current I. The current is uniformly distributed across its cross section. The ratio of magnitudes of the magnetic field at a point a/2 above the surface of wire to that of a point a/2 below its surface is

(A) 4:1 (B) 1:1 (C) 4: 3 (D) 3 :4

Q4. The diffraction effect can be observed in

(A) sound waves only (B) light waves only

(C) ultrasonic waves only (D) sound waves as well as light waves

Q5. A capacitor consists of two parallel plates, with an area of cross-section of 0.001 m^2, separated by a distance of 0.0001 m. If the voltage across the plates varies at the rate of 10^8 V/s, then the value of displacement current through the capacitor is

(A) $8.85 \times 10^{-3} A$ (B) $8.85 \times 10^{-4} A$ (C) $7.85 \times 10^{-3} A$ (D) $9.85 \times 10^{-3} A$

Q6. In a series LCR circuit, the voltage across the resistance, capacitance and inductance is 10 V each. If the capacitance is short circuited the voltage across the inductance will be

(A) 10 V (B) $10\sqrt{2}$ V (C) $10/\sqrt{2}$ V (D) 20 V

Q7. Correct match of column I with column II is

C-I (waves)	**C-II (Production)**
(1) Infra-red	P . Rapid vibration of electrons in aerials
(2) Radio	Q . Electrons in atoms emit light when they move from higher to lower energy level.
(3) Light	R . Klystron valve
(4) Microwave	S . Vibration of atoms and molecules

(A) 1-P, 2-R, 3-S, 4-Q (B) 1-S, 2-P, 3-O, 4-R

(C) 1-Q, 2-P, 3-S, 4-R (D) 1-S. 2-R, 3-P, 4-Q

Q8. The distance of closest approach of an alpha particle is d when it moves with a speed V towards a nucleus.

Another alpha particle is projected with higher energy such that the new distance of the closest approach is $d/2$. What is the speed of projection of the alpha particle in this case?

(A) V /2 (B) $\sqrt{2}$ V (C) 2 V (D) 4 V

Q9. A point object is placed at the centre of a glass sphere of radius 6 cm and refractive index 1.5. The distance of virtual image from the surface of the sphere is

(A) 2 cm (B) 4 cm (C) 6 cm (D) 12 cm

Q10. Colours observed on a CD (Compact Disk) is due to

(A) Reflection (B) Diffraction (C) Dispersion (D) Absorption

Q11. The number of electrons made available for conduction by dopant atoms depends strongly upon

(A) doping level (B) increase in ambient temperature

(C) energy gap (D) options (a) and (b) both

Q12. If copper wire is stretched to make its radius decrease by 0.1%, then the percentage change in its resistance is approximately

(A) –0.4% (B) +0.8% (C) +0.4% (D) +0.2%

For Questions 13 to 16, two statements are given –one labelled Assertion (A) and other labelled Reason (R). Select the correct answer to these questions from the options as given below.

A. If both Assertion and Reason are true and Reason is the correct explanation of Assertion.
B. If both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
C. If Assertion is true but Reason is false.
D. If both Assertion and Reason are false.

Q13. Assertion (A): On increasing the current sensitivity of a galvanometer by increasing the number of turns may not necessarily increase its voltage sensitivity.

Reason(R) : The resistance of the coil of the galvanometer increases on increasing the number of turns.

Q14. Assertion (A): In a hydrogen atom there is only one electron but its emission spectrum shows many lines.

Reason (R): In a given sample of hydrogen there are many atoms each containing one electron; hence many electrons in different atoms may be in different orbits so many transitions from higher to lower orbits are possible.

Q15. Assertion (A): Nuclei having mass number about 60 are least stable**..**

Reason (R): When two or more light nuclei are combined into a heavier nucleus then the binding energy per nucleon will decrease.

Q16. Assertion (A): de Broglie's wavelength of a freely falling body keeps decreasing with time.

Reason (R): The momentum of the freely falling body increases with time.

[SECTION – B] (05x2=10 marks)

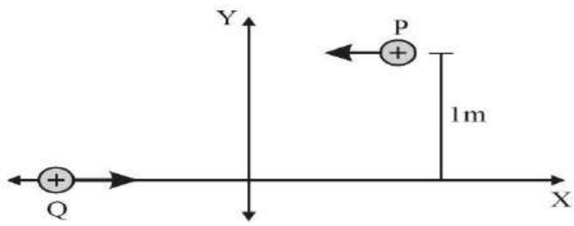
Q17. A platinum surface having work function 5.63 eV is illuminated by a monochromatic source of 1.6×10^{15} Hz. What will be the minimum wavelength associated with the ejected electron.

Q18. (I) In Young's double-slit experiment using monochromatic light of wavelength λ, the intensities of two sources is I. What is the intensity of light at a point where path difference between wave front is λ/4?

OR

(II) A beam of light consisting of two wavelengths, 4000 Å and 6000 Å, is used to obtain interference fringes in a Young's double-slit experiment. What is the least distance from the central maximum where the dark fringe is obtained?

Q19. P and Q are two identical charged particles each of mass 4×10^{-26} kg and charge 4.8×10^{-19} C, each moving with the same speed of 2.4×10^{5} m/s as shown in the figure. The two particles are equidistant (0.5 m) from the vertical Y -axis. At some instant, a magnetic field B is switched on so that the two particles undergo head-on collision.



Find –

(I) the direction of the magnetic field and

(II) the magnitude of the magnetic field applied in the region.

(for VI candidates)

A proton is moving with speed of 2×10^{5} m s^{-1} enters a uniform magnetic field B = 1.5 T. At the entry velocity vector makes an angle of 30° to the direction of the magnetic field. Calculate

(a) the pitch of helical path described by the charge

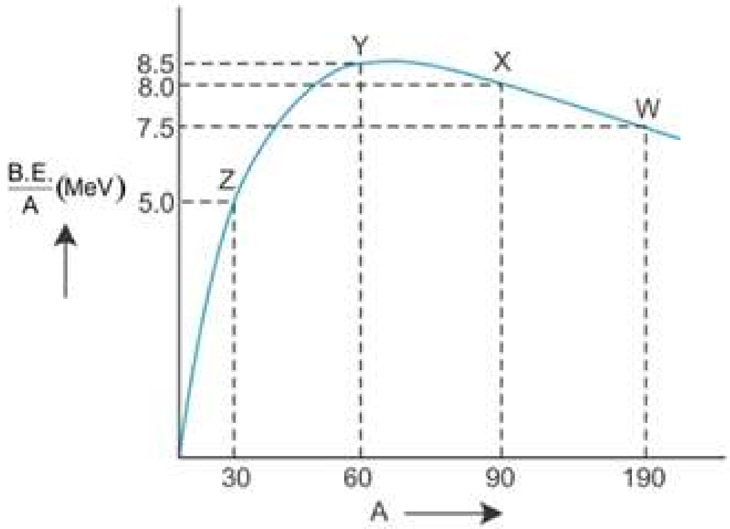
(b) Kinetic energy after completing half of the circle.

Q.20. Binding energy per nucleon vs mass number curve for nuclei is shown in the figure. W, X, Y and Z are four nuclei indicated on the curve. Identify which of the following nuclei is most likely to undergo

(i) Nuclear Fission

(ii) Nuclear Fusion.

Justify your answer.



(for V.I. Candidates)

Binding energy per nucleon and mass number of the following nuclei are given in the below table

Nuclei	Binding energy per nucleon (MeV)	Mass number
W	7.5	190
X	8.0	90
Y	8.5	60
Z	5.0	30

Identify which of the following nuclei is most likely to undergo

(i) Nuclear Fission

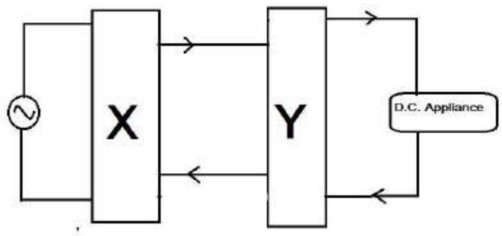
(ii) Nuclear Fusion.

Justify your answer.

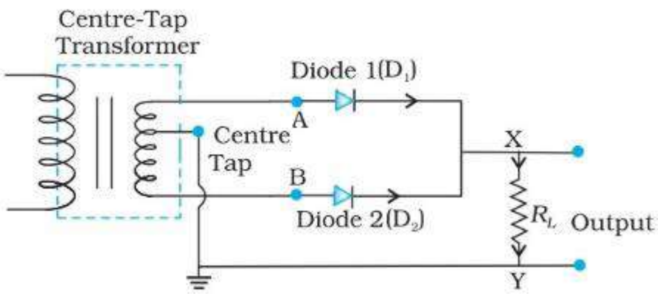
Q21. What should be the radius 'r' of nearest possible orbits of satellite of mass 'm' revolving around the planet of mass 'M' as per Bohr Postulates in terms of m, M, G, h where G is Gravitational constant and h is plank's constant.

[SECTION – C] **(07x3=21 marks)**

Q22. (I) Identify the circuit elements X and Y as shown in the given block diagram and draw the output waveforms of X and Y.



(II) If the centre tapping is shifted towards Diode D1 as shown in the diagram, draw the output waveform of the given circuit.



(for V.I. candidates)

Which device is used to convert AC into DC. State it's underlying principle and explain its working. If the frequency of input AC to this device is 60 Hz, then what will be frequency of the output of this device.

Q23. Find the expression for the capacitance of a parallel plate capacitor of plate area A and plate separation d when (I) a dielectric slab of thickness t and (II) a metallic slab of thickness t, where (t < d) are introduced one by one between the plates of the capacitor. In which case would the capacitance be more and why?

Q24. (I) Draw a ray diagram for the formation of image by a Cassegrain telescope.

(II)Why these types of telescopes are preferred over refracting type telescopes. (Write 2 points)

(for V.I. Candidates)

A Cassegrain telescope is built with an arrangement of two mirrors placing them 20 mm apart. If the radius of curvature of the large mirror is 200mm and the small mirror is 150mm, where will the final image of an object at infinity be?

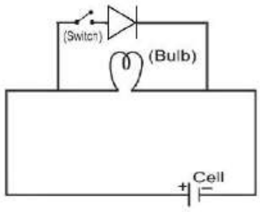
Q25. (I) Draw the energy band diagram for P-type semiconductor at (i) T=0K and (ii) room temperature.

(II)In the given diagram considering an ideal diode, in which condition will the bulb glow

(a) when the switch is open

(b) when the switch is closed

Justify your answer.



(for V.I. Candidates)

Explain briefly how

(i) barrier potential is formed in p-n junction diode.

(ii) Width of depletion region of the diode is affected when it is (a) forward biased, (b) reverse biased.

Q26. A boy is holding a smooth, hollow and non-conducting pipe vertically with charged spherical ball of mass 10 g carrying a charge of +10 mC inside it which is free to move along the axis of the pipe. The boy is moving the pipe from East to West direction in the presence of magnetic field of 2T. With what minimum velocity, should the boy move the pipe such that the ball does not move along the axis. Also determine the direction of the magnetic field.

Q27. A light ray entering a right-angled prism undergoes refraction at the face AC as shown in Fig. 1.

(I) What is the refractive index of the material of the prism in Fig. 1?

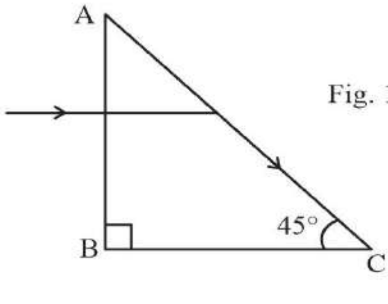


Fig. 1

(II) (a) If the side AC of the above prism is now surrounded by a liquid of refractive index $\frac{2}{\sqrt{3}}$, as shown in Fig. 2, determine if the light ray continues to graze along the interface AC or undergoes total internal reflection or undergoes refraction into the liquid.

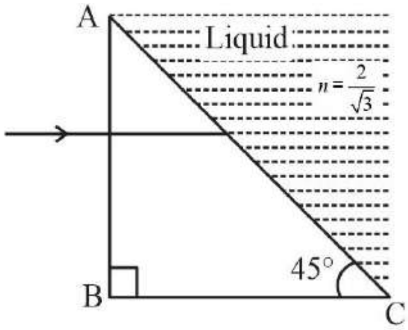


Fig. 2

(b) Draw the ray diagram to represent the path followed by the incident ray with the corresponding angle values.

(Given, $sin^{-1}(\frac{\sqrt{2}}{\sqrt{3}}) = 54.6^{\circ}$)

(for V.I. candidates)

A ray of light is incident on an equilateral prism at an angle 3/4 th of the angle of the prism. If the ray passes symmetrically through the prism, find the (a) angle of minimum deviation, and (b) refractive index of the material of the prism.

Q28. (I) State Gauss's theorem in electrostatics. Using this theorem, derive an expression for the electric field due to an infinitely long straight wire of linear charge density λ.

OR

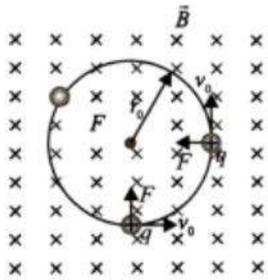
(II) (a) Define electric flux and write its SI unit.

(b) Use Gauss's law to obtain the expression for the electric field due to a uniformly charged infinite plane sheet of charge.

[SECTION D] **(02x4=08 marks)**

Q29. **Case Study Based Question**: **<u>Motion of Charge in Magnetic Field</u>**

An electron with speed $v_o << c$ moves in a circle of radius r_o in a uniform magnetic field. This electron is able to traverse a circular path as the magnetic force acting on the electron is perpendicular to both v_o and B ,as shown in the figure. This force continuously deflects the particle sideways without changing its speed and the particle will move along a circle perpendicular to the field. The time required for one revolution of the electron is T_o



(i) If the speed of the electron is now doubled to $2v_o$. The radius of the circle will change to

(A) $4r_o$ (B) $2\ r_o$ (C) r_o (D) $r_o/2$

(ii) If $v = 2v_o$, then the time required for one revolution of the electron (T_o) will change to

(A) $4\ T_o$ (B) $2\ T_o$ (C) T_o (D) $T_o/2$

(iii) A charged particles is projected in a magnetic field . The acceleration of the particle is found to be . Find the value of x.

(A) $4\ ms^{-2}$ (B) $-4\ ms^{-2}$ (C) $-2\ ms^{-2}$ (D) $2\ ms^{-2}$

(iv) If the given electron has a velocity not perpendicular to B, then trajectory of the electron is

(A) straight line (B) circular (C) helical (D) zig-zag

OR

If this electron of charge (e) is moving parallel to uniform magnetic field with constant velocity v, the force acting on the electron is

(A) Bev (B) Be/v (C) B/ev (D) Zero

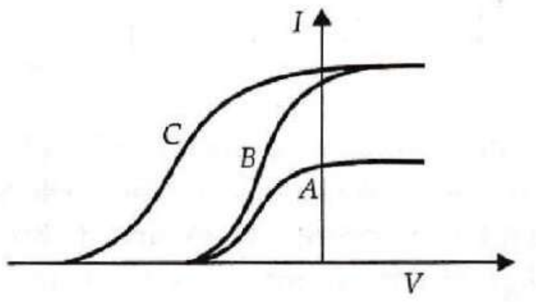
Q30. Case Study Based Question: Photoelectric effect

It is the phenomenon of emission of electrons from a metallic surface when light of a suitable frequency is incident on it. The emitted electrons are called photoelectrons.

Nearly all metals exhibit this effect with ultraviolet light but alkali metals like lithium, sodium, potassium, cesium etc. show this effect even with visible light. It is an instantaneous process i.e. photoelectrons are emitted as soon as the light is incident on the metal surface. The number of photoelectrons emitted per second is directly proportional to the intensity of the incident radiation.

The maximum kinetic energy of the photoelectrons emitted from a given metal surface is independent of the intensity of the incident light and depends only on the frequency of the incident light. For a given metal surface there is a certain minimum value of the frequency of the incident light below which emission of photoelectrons does not occur.

(I) In a photoelectric experiment plate current is plotted against anode potential.



(A) A and B will have same intensities while B and C will have different frequencies

(B) B and C will have different intensities while A and B will have different frequencies

(C) A and B will have different intensities while B and C will have equal frequencies

(D) B and C will have equal intensities while A and B will have same frequencies.

(II) Photoelectrons are emitted when a zinc plate is

(A) Heated (B) hammered

(C) Irradiated by ultraviolet light (D) subjected to a high pressure

(III) The threshold frequency for photoelectric effect on sodium corresponds to a wavelength of 500 nm. Its work function is about

(A) $4\text{x}10^{-19}$ J (B) 1 J (C) $2\text{x}10^{-19}$ J (D) $3\text{x}10^{-19}$ J

(IV) The maximum kinetic energy of photoelectrons emitted from a surface when photons of energy 6 eV fall on it is 4 eV. The stopping potential is

(A) 2 V (B) 4 V (C) 6 V (D) 10 V

OR

The minimum energy required to remove an electron from a substance is called its

(A) work function (B) kinetic energy (C) stopping potential (D) potential energy

[SECTION E] (03X5=15)

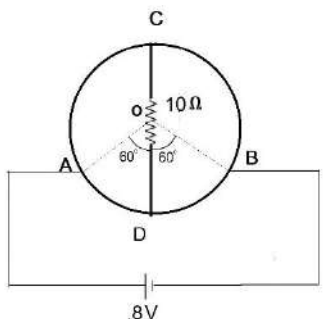
Q31. (I) a) Write two limitations of ohm's law. Plot their I-V characteristics.

b) A heating element connected across a battery of 100 V having an internal resistance of 1 Ω draws an initial current of 10 A at room temperature 20.0 °C which settles after a few seconds to a steady value. What is the power consumed by battery itself after the steady temperature of 320.0 °C is attained? Temperature coefficient of resistance averaged over the temperature range involved is 3.70×10^{-4} °C^{-1}.

OR

(II) a) Using Kirchhoff's laws obtain the equation of the balanced state in Wheatstone bridge.

b) A wire of uniform cross-section and resistance of 12 ohm is bent in the shape of circle as shown in the figure. A resistance of 10 ohms is connected to diametrically opposite ends C and D. A battery of emf 8V is connected between A and B. Determine the current flowing through arm AD.



(for V.I. Candidates)

(II) a) Using Kirchhoff's laws obtain the equation of the balanced state in Wheatstone bridge.

b) What do you understand by 'sensitivity of Wheatstone bridge' ? How the sensitivity of wheatstone bridge can be increased?

Q32. (I) Explain briefly, with the help of a labelled diagram, the basic principle of the working of an a.c. generator. In an a.c. generator, coil of N turns and area A is rotated at an angular velocity ω in a uniform magnetic field B. Derive an expression for the instantaneous value of the emf induced in coil. What is the source of energy generation in this device?

OR

(II) a) With the help of a diagram, explain the principle of a device which changes a low ac voltage into a high voltage . Deduce the expression for the ratio of secondary voltage to the primary voltage in terms of the ratio of the number of turns of primary and secondary winding. For an ideal transformer, obtain the ratio of primary and secondary currents in terms of the ratio of the voltages in the secondary and primary coils.

b) Write any two sources of the energy losses which occur in actual transformers.

c) A step-up transformer converts a low input voltage into a high output voltage. Does it violate law of conservation of energy? Explain.

Q33. (I) a) A giant refracting telescope at an observatory has an objective lens of focal length 15 m. If an eyepiece of focal length 1.0 cm is used, what is angular magnification of the telescope in normal adjustment?

b) If this telescope is used to view the moon, what is the diameter of the image of the moon formed by the objective lens? The diameter of the moon is 3.48×10^6 m, and the radius of lunar orbit is 3.8×10^8 m.

OR

(II) A compound microscope consists of an objective lens of focal length 2.0 cm and an eyepiece of focal length 6.25 cm separated by a distance of 15 cm. How far from the objective should an object be placed in order to obtain the final image at

a) the least distance of distinct vision (25 cm) and
b) infinity? What is the magnifying power of the microscope in each case?

PHYSICS QUESTION PAPER
CLASS-XII

Time : 3.00 Hours | ***Maximum Marks : 100***

Instructions :

1. There are **4** sections and total **60** questions in this question paper.
2. Symbols used in this question paper have their usual meanings.
3. Log table or simple electronic calculator can be used.
4. Write new section from a new page.

SECTION - A

Question Nos. ***1*** *to* ***16*** *are multiple choice questions. Each carries* ***ONE*** *mark.*
Choose correct answer (A, B, C, D) from the given alternatives and write it. **16**

1. The dimensions of Permitivity $[\varepsilon_0]$ are ....
Take Q as dimension of charge

(A) $M^{-1}L^{3}T^{-2}Q^{-2}$ (B) $M^{-1}L^{-3}T^{2}Q^{2}$

(C) $M^{-1}L^{2}T^{-3}Q^{-1}$ (D) $M^{1}L^{-2}T^{-2}Q^{-2}$

2. One variable capacitor is connected to a 100 V battery. If the capacitance is increased from 2 μF to 10 μF, then the change in energy in the above system will be ....

(A) 2×10^{-2} J (B) 2.5×10^{-2} J

(C) 4×10^{-2} J (D) 6.5×10^{-2} J

3. Maximum power in a 0.5 Ω resistance connected with two batteries of 2V *emf* and 1Ω internal resistance in parallel is

(A) 2 W (B) 1.28 W

(C) $\frac{8}{9}$ W (D) 3.2 W

4. A wire of $2\,m$ is bent in the form of a circular loop. The magnetic moment of the loop when 1 A current flows through it is .... Am^2.

(A) 2π (B) $\frac{1}{\pi}$

(C) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$

5. Magnetization intensity for Vacuum is

(A) Zero (B) Negative

(C) Positive (D) ∞

6. The relative permeability of a diamagnetic substance is

(A) negative (B) very large

(C) less than one (D) small but greater than 1.

7. An inductor stores energy in its ....

(A) electric field (B) conducting wire

(C) magnetic field (D) electric and magnetic field.

8. Which of the following option for L, C and R does not give us the dimension of frequency?

(A) $\frac{C}{L}$ (B) $\frac{1}{\sqrt{LC}}$

(C) $\frac{R}{L}$ (D) $\frac{1}{RC}$

9. The maximum value of $\vec{E}$ in an electromagnetic wave is equal to $18\ Vm^{-1}$, then the maximum value of $\vec{B}$ will be equal to

(A) 4×10^{-6} T (B) 6×10^{-8} T

(C) 9×10^{-9} T (D) 11×10^{-11} T

10. Focal length of the lens of the eye is changed by

(A) Cornea (B) Ciliary muscles

(C) Retina (D) Crystallic lens

11. Which of the following phenomenon is not possible for Sound?

(A) Polarization (B) Reflection
(C) Interference (D) Diffraction

12. A proton and an α-particle are passed through same potential difference. If their initial velocity is zero, the ratio of their de-Broglie's wave-length after getting accelerated is

(A) 1 : 1 (B) 1 : 2
(C) 2 : 1 (D) $2\sqrt{2}:1$

13. The frequency of characteristic X-ray determines .... property of the target.

(A) Atomic weight (B) Atomic number
(C) Melting point (D) Conductivity

14. Half life of a radio active element is 5 min. In 20 min. the ....% of the substance will remain undecayed.

(A) 6.25 (B) 25
(C) 75 (D) 93.75

15. Complete the following reaction :

$_{92}U^{235} + _0n^1 \rightarrow .... + _{38}Kr^{90} + ...$

(A) $_{54}Xe^{143}$, $3\ _0n^1$ (B) $_{54}Xe^{145}$
(C) $_{57}Xe^{142}$ (D) $_{54}Xe^{142}$, $_0n^1$

16. The value of the depletion capacitance .... on increasing the reverse bias.

(A) increases (B) decreases
(C) becomes zero (D) does not change

SECTION - B

*Questions **17** to **32** are very short questions each carrying **ONE** mark.* **16**

17. Write Coulomb's Inverse Square law.

18. Define Static Electric Potential.

OR

Write dimensional formula of Capacitance.

19. How many electrons will pass in one second through a conducting wire carrying current equal to 0.64 A ? $e = 1.6 \times 10^{-19}$ C.

OR

What is Ohmic loss ?

20. What is called Toroid?

OR

What is called Gyro-magnetic ratio?

21. What is called Permanent magnets?

22. What is called Inductor? Draw its Circuit symbol?

23. Write principle of a Transformer.

24. How much current does leg or lead the voltage in A.C. circuit with only inductor?

25. What is Mie-Scattering?

26. Write Huygen's principle.

27. What is Polarised light?

28. Slope of a graph of $V_0e \rightarrow f$ gives which physical quantity?

OR

Write De-Broglie hypothesis.

29. Name the series which falls in ultra violet region of Hydrogen spectra.

30. Fill in the blank :

1 n Ci = ... Bq

31. What is called Doping?

OR

Write boolean equation of 'NOT' gate.

32. In case of a transistor, write formula showing relation between I_E, I_C, and I_B.

SECTION - C

*Question Nos. **33** to **48** are short answer type questions.*
*Each question carries **TWO** marks.*

33. Using Gaussian law, derive formula for electric field due to an infinitely long straight charged wire (line charge).

$$\vec{E} = \frac{\lambda}{2\pi\varepsilon_0} \cdot \frac{1}{r}\, \hat{r}.$$

OR

Obtain formula for capacitance of a parallel plate capacitor.

34. Write Kirchhoff's First Law and obtain equation using necessary figure.

35. Explain charging process in Lead storage cell (accumulator), using necessary circuit.

36. Obtain formula for force (Lorentz force) acting an a charge moving in a magnetic and electric field.

37. Write down points of comparison between an electric dipole and a magnetic dipole.

38. Write two definitions of Mutual Inductance and on which factors it depends?

39. Define real power for A.C. circuit and hence obtain formula for power.

$P = V_{rms}\ I_{rms} \cos\delta$ for A.C. series circuit.

40. Derive Gaussian formula for concave mirror using appropriate figure.

41. For a Prism, derive formula $i + e = A + \delta$.

42. Write down condition for constructive and destructive interference in terms of Path difference and Phase difference.

43. Write four characteristics of Photon.

OR

Write four uses of a Photocell.

44. Write limitations of Bohr model.

OR

Write four uses of LASER Light.

45. Define Radio activity and obtain Exponential law for radio-active decay.

46. Name the parameters of a Transistor and define them.

47. Draw logic circuit of AND gate and describe two cases of input and output of AND gate.

48. Explain Analog and Digital communication.

OR

Explain Simplex and Duplex communication.

SECTION - D

Question Nos. ***49*** *to* ***60*** *are short answer type questions.*
Each question carries ***THREE*** *marks.*

49. Q amount of charge is uniformly distributed over some body. How should the body be divided into two parts, so that the force acting between the two parts is maximum for a given separation between them?

50. A drop of water (spherically shaped) has 3×10^{-10} C amount of charge residing on it. 500 V electric potential exists on its surface. Calculate the radius of this drop. Two such drops (having identical charge and radius) combine to form a new drop. Calculate the electric potential on the surface of the new drop.

$K = 9 \times 10^{9}$ SI.

51. 29.1×10^{-2} A current is obtained when a 5Ω resistor is connected with a battery of unknown internal resistance *r* and unknown *emf*.

14.7×10^{-2} A current is obtained, if the above battery is connected to 10Ω resistor. Calculate the *emf* and internal resistance of the battery.

52. A battery having an *emf* ε and an internal resistance r is connected with a resistance R. Prove that the power in the external resistance is maximum when $R = r$.

53. An electron in an atom is revolving round the nucleus in a circular orbit with a speed of $10^7\ ms^{-1}$. If the radius of the orbit is $10^{-10}\ m$, find the resulting magnetic field at the center. $e = 1.6 \times 10^{-19}\ C$; $\mu_0 = 4\pi \times 10^{-7}\ T\ m\ A^{-1}$

OR

A toroidal core with 3000 turns has inner and outer radii of 0.11 m and 0.12 m respectively. When a current of 700 mA is passed, then the magnetic field produced in the core is 2.5 T. Find the relative permeability of the core ($\mu_0 = 4\pi \times 10^{-7}\ T\ m\ A^{-1}$).

54. Two long solenoids are of equal length l and the smaller solenoid having cross-sectional area a is placed within the larger solenoid in such a way that their axes coincide. Find the mutual inductance of the system.

OR

For an A.C. circuit comprising of L-C-R in series, L = 10 H, W=100 rad s^{-1}, R = 100 Ω and Power Factor is equal to 0.5. Calculate the capacitance of the circuit.

55. A 1000 W bulb is kept at the centre of a spherical surface and is at a distance of 10 m from the surface.

Find E_0, B_0 (the maximum electric and magnetic field strengths) and I (the intensity of the waves). Take the working efficiency of the bulb as 2.5% and consider it as a point source. $\varepsilon_0 = 8.85 \times 10^{-12}$ SI and $C = 3 \times 10^8\ ms^{-1}$.

56. When a linear object is placed infront of a convex mirror, image of the $\frac{1}{4}$ th size of object is formed. Calculate the object distance and the image distance. This linear object is kept perpendicular to the axis.

OR

The ratio of intensities of light emerging from two sources is α. For the interference pattern produced by them, prove that

$$\frac{I_{max} + I_{min}}{I_{max} - I_{min}} = \frac{1+\alpha}{2\sqrt{\alpha}}$$

where I_{max} = Intensity of bright fringe, and

I_{min} = Intensity of dark (faint) fringe.

57. Wave length of Light incident on a photo-sensitive surface is reduced from 4000 A to 360 nm. Find the change in stopping potential.

$h = 6.625 \times 10^{-34}$ Js.

58. Calculate the quantum number for which the radius of the orbit of electron in Be^{3+} would be equal to that for the ground state of electron in Hydrogen atom. Also compare the energy of the two states.

OR

Mass of a ${}_{17}Cl^{35}$ nucleus is 34.9800 u. If mass of a proton is 1.00783 u and neutron is 1.00866 u, find the binding energy of ${}_{17}Cl^{35}$ nucleus. Take 1 u = 931 MeV.

59. The base current changes by 200 μA when a 200 mV signal is applied at the input of a CE amplifier. If the output voltage is equal to 2 volt, what is the voltage gain?

60. Height of TV tower is 10^2 m. If the average population density is 1000 / km^2, how many people can observe the programmes of this station?

(Radius of the Earth = 6400 km)

CHEMISTRY QUESTION PAPER
CLASS-XII

Time : 3.00 Hours] ***[Maximum Marks : 100***

Instructions :

(1) This question paper contains **60** questions. **All** are **compulsory**.

(2) Write each new section with a new page and write answers of the questions in their order.

(3) Write your answers according to instructions pointwise and to the point. Draw figure and give reactions wherever required.

(4) Section-A contains **1** to **16** multiple choice questions, each of **1** mark.

(5) Section-B contains **17** to **32** very short answered questions, each of **1** mark.

(6) Section-C contains **33** to **48** short answered questions, each of **2** marks.

(7) Section-D contains **49** to **60** long answered questions, each of **3** marks.

(8) Use log-table provided by Board, or a simple calculator provided for your calculations.

(9) Use Pencil for figure drawing and Blue-pen for your writing in the Answer Book.

(10) Constants :

R = 1.987 Cal./ mole °K.

= 8.314 Joule / mole °K.

SECTION-A **16**

1. 2 litre of a solution contains 5 mole solute and 45 mole solvent. Mole fraction of solute is ....

(A) 5.0 (B) 10.0

(C) 0.5 (D) 0.10

2. The equation to determine the change in free energy alongwith the change in pressure-volume change at constant temperature is ....

(A) $\Delta G = nRT \log \frac{P_2}{P_1}$

(B) $\Delta G = nRT \log \frac{V_2}{V_1}$

(C) $\Delta G = nRT\, l_n \frac{V_1}{V_2}$

(D) $\Delta G = nRT \log \frac{P_2}{P_1}$

3. Unit of rate constant of 3rd order reaction is ....

(A) $\frac{(\text{Litre})^2}{(\text{Mole})^2\ \text{sec}}$

(B) sec.^{-1}

(C) $\left(\frac{\text{Mole}}{\text{Litre}}\right)^{-1} \cdot \text{sec}^{-1}$

(D) $\left(\frac{\text{Litre}}{\text{Mole}}\right)^{2} .\text{sec}$

4. Catalyst which convert alcohol directly into gasolene.

(A) ZSM-5

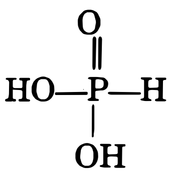
(B) Zinc stearate

(C) Zinc Blende

(D) PHBV

5. Which structure indicate Phosphinic acid?

(A)



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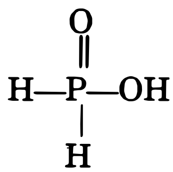

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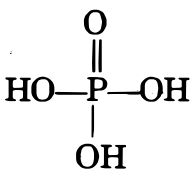
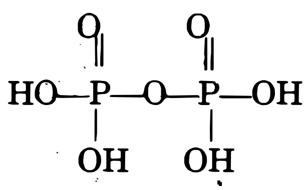


(D)



6. Transition ion of which compound has maximum magnetic moment?

(A) $MnSO_4$

(B) $Cr_2(SO_4)_3$

(C) $FeSO_4$

(D) $CuSO_4$

7. Which is the structural formula of Sodium tris oxalato ferrate (III) ?

(A) $Na[Fe(Ox)_3]$

(B) $Na_2[Fe(Ox)_3]$

8. Which character will exhibit α particles?
(A) Al foil like thin paper can stop it.
(B) Only more thick Al strip can stop it.
(C) 15-20 cm thick Al strip can stop it.
(D) 15-20 cm thick tin metal strip can stop it.

9. *l*-Epinephrine is how much more effective to raise blood pressure than its *d* isomer?
(A) 500 times (B) 20 times
(C) 50 times (D) 10 times

10. The use of a substance obtained by hydrolysis of ethylene oxide in presence of H_2SO_4 at 80°C is .....
(A) As a rubber solvent (B) as a filler
(C) for Nylon fibres (D) for terrylene fibres.

11. Which of the following substance undergoes Aldol condensation?
(A) C_6H_5CHO (B) $CH_3 \cdot CHO$
(C) H.CHO (D) All of above.

12. Which of the following has highest boiling point?
(A) Ethanol (B) Ethanal
(C) Glycerol (D) Ethyl amine

13. What type of the polymer Novolac is?
(A) Branched (B) Linear
(C) Cross-linked (D) Thermoplastic

14. Joining of which Carbon of Glucose unit form Starch?
(A) C-1 and C-2 (B) C-1 and C-3
(C) C-1 and C-4 (D) C-1 and C-5

15. Which of the following is LAS?

(A) $CH_3 \cdot (CH_2)_x - C_6H_4 - SO_3^- Na^+$

(B) $CH_3 - (CH_2)_{11} - O\ SO_3^- Na^+$

(C) $CH_3 - (CH_2)_{15} - N^+ (CH_3)_3 \cdot Cl^-$

(D) $CH_3 \cdot (CH_2)_{10} - CH_2 - O - SO_3^+ - Na^-$

16. The substance having no specific molecular formula but useful in manufacturing of Machinery is ......

(A) Fe_2S_3 (B) $FeSO_4$

(C) $Fe_2(SO_4)_3$ (D) Fe_3C

SECTION-B

16

17. Who proved the de-Broglie's principle experimentally? How?

18. Draw the structure of Diamond and write its type of hybridization.

19. Define : Colligative properties.

OR

Define : Osmosis.

20. What will be the change in entropy, when 18 grams of Water is converted into its vapour at 100°C temperature?
Heat of Vapourisation of Water is 9720 cal/mole.

21. Define : Cell Potential

22. Write uses of SnO_2 (two).

23. Write any one chemical reaction to prepare Chlorine in laboratory.

24. State the type of classification of ligands exist in complex ion $[Pt(en)_2Cl_2]^{2+}$

25. Calculate ratio of Neutron and Proton of the element obtained by emission of α– particle from ${}^{232}_{90}Th$.

26. What are Isomorphous?

27. Write Van't-Hoff rule in terms of Stereo-isomers.

28. Draw electronic structure of Methanol.

29. Write : Decarboxylation reaction of Acetic acid.

OR

Write : Reaction to prepare Acetic anhydride from Acetic acid.

30. Diazotisation reaction is carried out at a low temperature. Why?

31. How Dextran is produced? State its use.

32. What are Nucleoside and Nucleotide?

SECTION-C

32

33. Explain the evolution of Spin Quantum Number.

34. Explain Raoult's law for the solution which possess solute gas in a liquid solvent.

35. Write short note on : Substitutional Solid Solution.

36. Write difference between Electro-chemical cell and Electrolytic cell - (state 4 points).

37. Explain the equation of average rate of reaction by graph showing the change in concentration of reactants and products with time.

38. The first order reaction takes 20 minutes to complete 15% of its concentration, calculate what time will be required to complete its 75% concentration?

OR

The rate constant of a reaction is 2.0×10^{-3}min.$^{-1}$ at 27 °C. If the temperature is increased by 20 °C, its value becomes three times. Calculate energy of activation.

39. On which principle the Langmuir adsorption isotherm depends? Write its hypothesis.

40. Uses of adsorption (any four).

41. Give chemical reactions for preparation of K_2Cr_2O7.

OR

How photographic plate is prepared? Explain the preparation.

42. Explain structure of complex ion in $K_2[Ni(CN)_4]$ on the basis of hybridisation.

OR

State magnetic moment of metal ion present in $[Fe(H_2O)_6]^{3+}$ and $[Fe(CN)_6]^{3-}$. Give reason for their different values.

43. Complete reactions :

(1) $^{27}_{13}Al + ^{4}_{2}He \rightarrow$ .... $+ ^{1}_{0}n$

(2) $^{239}_{94}Pu$ $(\alpha, \beta) \rightarrow$ ....

44. What is called Nuclides? State the element Z=90 atomic weight with 230 and 228 in the form of nuclides.

45. Write short note on Di-saccharides.

46. Give structural formula of α-amino acid obtained by hydrolysis of Protein. Write names of any two amino-acids occur in nature.

47. Explain : Mordant Dyes.

OR

How Ceramics are obtained? Write names of ceramics used in cutting and grinding tools.

48. Write short note on :

Synthetic Sweetners.

SECTION-D — 36

49. Describe energy band model. Explain the various electrical conductivity observed in substances on the basis of this theory.

50. Write short note on :

(1) Ferromagnetic substances.

(2) Anti-ferromagnetic substances.

51. At 25°C K_p for the given reaction is 1.792×10^{12}. Calculate its entropy change.

ΔS . R=1.987 Cal./K.

$2NO_{(g)} + O_{2(g)} \rightarrow 2NO_{2(g)}$ $\Delta H = -7.77$ K.Cal.

52. At 25°C, the potential of the following given cell is 0.71 V. Calculate the ionic product of Water (K_w).

$Pt / H_2(1\ atm.) / KOH\ (0.1M) // HCl\ (0.1\ M) / H_2\ (1\ atm.) / Pt$

OR

At 25°C, the potential of the following cell is 1.041 V; calculate the pH of HCl solution. $E^0\ Ag^+ / Ag = 0.8\ V.$

$Pt / H_2\ (1\ atm.) / HCl\ (x\ M) // Ag^+\ (0.01\ M) / Ag_{(s)}.$

53. Name the oxy-acids of Phosphorus, giving their molecular and structural formula (any Six).

OR

Describe Contact process for manufacturing of H_2SO_4, stating chemical reactions. Also give electronic structural formula of H_2SO_4.

54. Discuss the magnetic properties of Transition ions or compounds. The experimental values of magnetic moment of some compounds differ than their theoretical values. Why?

OR

What is Actinide series ?

State properties and uses of the elements of this series.

55. Define Chelates. Give structures of Optical isomers of the following.

(1) $[Cr(Ox)_3]^{3-}$

(2) $[CrCl_2(NH_3)_2\ en]^+$

OR

Write application of Complex compounds.

56. Explain importance of Stereo-chemistry.

57. *(i)* Aliphatic compounds containing –OH group are neutral but Aromatic compounds containing –OH group are acidic. Why?

(ii) Explain Reimer-Tiemann reaction.

58. Explain by giving chemical reaction for the intermediate obtained by reaction of Methyl magnesium iodide with Ethanal and Propanone which give alcohols on their hydrolysis.

OR

Explain Condensation reaction of Aldehyde and Ketone compounds by reactions only.

59. *(i)* Write Conversion :

Ethyl acetate from Acetamide.

(ii) Complete the reaction :

$$\text{Propane} \xrightarrow[\text{400° C}]{\text{Fuming } HNO_3}$$

60. Write preparation of Vulcanised rubber. State its properties and uses.

CHEMISTRY THEORY (043)

Max. Marks:70 **Time: 3 hours**

GENERAL INSTRUCTIONS:

Read the following instructions carefully.

(a) There are **33** questions in this question paper with internal choice.
(b) SECTION A consists of 16 multiple-choice questions carrying 1 mark each.
(c) SECTION B consists of 5 short answer questions carrying 2 marks each.
(d) SECTION C consists of 7 short answer questions carrying 3 marks each.
(e) SECTION D consists of 2 case-based questions carrying 4 marks each.
(f) SECTION E consists of 3 long answer questions carrying 5 marks each.
(g) All questions are compulsory.
(h) Use of log tables and calculators is not allowed.

SECTION A

The following questions are multiple-choice questions with one correct answer. Each question carries 1 mark. There is no internal choice in this section.

1 Ammonolysis of ethyl chloride followed by reaction of the amine so formed with 1 mole of methyl chloride gives an amine that 1

a. reacts with Hinsberg reagent to form a product soluble in an alkali.
b. on reaction with Nitrous acid, produced nitrogen gas.
c. reacts with Benzenesulphonyl chloride to form a product that is insoluble in alkali.
d. does not react with Hinsberg reagent.

2 Which one of the following has the highest dipole moment? 1

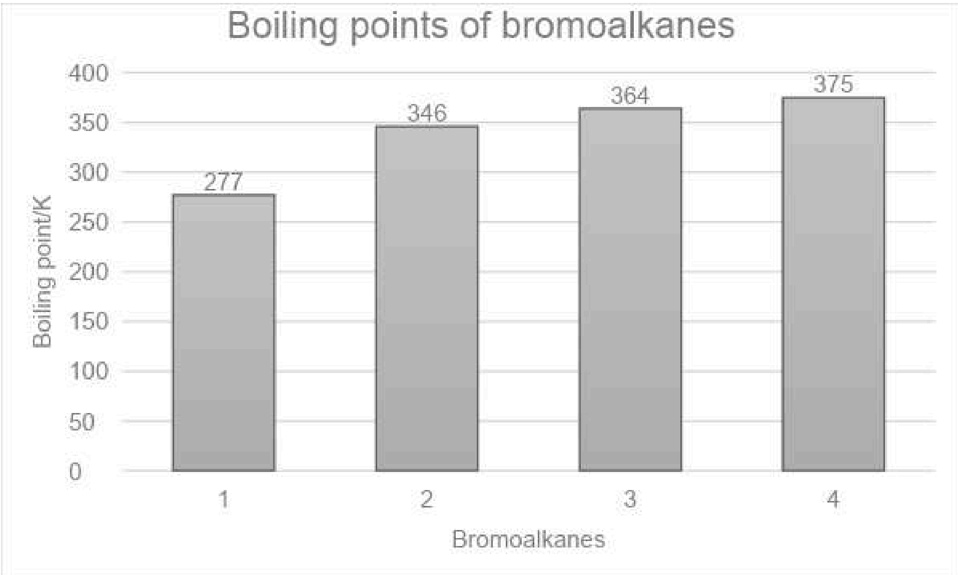
a. CH_3F
b. CH_3Cl
c. CH_3I
d. CH_3Br

3 Match the properties given in column I with the metals in column II 1

Column I	Column II
(i) Actinoid having configuration [Rn] $5f^7 6d^1 7s^2$	(A) Ce
(ii) Lanthanoid which has $4f^{14}$ electronic configuration in +3 oxidation state.	(B) Lu
(iii) Lanthanoid which show +4 Oxidation state	(C) Cm

a. (i)-(C), (ii)-(B), (iii)-(A)
b. (i)-(C), (ii)-(A), (iii)-(B)
c. (i)-(A), (ii)-(B), (iii)-(C)
d. (i)-(B), (ii)-(A), (iii)-(C)

4 Study the graph showing the boiling points of bromoalkanes and identify the compounds. 1



a. 1 = Bromomethane, 2= 2-Bromobutane, 3= 1-Bromobutane,
   4= 2-Bromo-2-methylpropane

b. 1 =1-Bromobutane, 2= 2-Bromo-2-methylpropane, 3= 2-Bromobutane,
   4= Bromomethane

c. 1 = Bromomethane, 2=1-Bromobutane, 3= 2-Bromo-2-methylpropane,
   4= 2-Bromobutane,

d. 1 =Bromomethane, 2= 2-Bromo-2-methylpropane, 3=2- Bromobutane,
   4= 1-Bromobutane

(for visually challenged learners)
Which of the following haloalkanes has the highest boiling point?
a. 2-Bromo-2-methylpropane
b. 2-Bromobutane
c. Bromomethane
d. 1-Bromobutane

5 The initial concentration of R in the reaction R□P is 4.62 x 10^{-2} mol/L. What is the half life for the reaction if k = 2.31x 10^{-2} $molL^{-1}s^{-1}$ 1

a. 30 s
b. 3 s
c. 1 s
d. 10 s

6 When $C_6H_5COOCOCH_3$ is treated with H_2O , the product obtained is : 1

a. Benzoic acid and ethanol
b. Benzoic acid and ethanoic acid
c. Acetic Acid and phenol
d. Benzoic anhydride and methanol

7 1

Formulation of Cobalt(III) Chloride-Ammonia Complexes

Colour	Formula	Solution conductivity corresponds to
Yellow	$[Co(NH_3)_6]^{3+}3Cl^-$	Y
Purple	$[CoCl(NH_3)_5]^{2+}2Cl^-$	1:2 electrolyte
Green	X	1:1 electrolyte

'X' and 'Y' in the above table are:

a. X=$[Co(NH_3)_6]^{2+}3Cl^-$, Y= 1:3
b. X= $[Co(NH_3)_4Cl_2]^+Cl,^-$ Y= 1:3
c. X=$[Co(NH_3)_4Cl_2]^+Cl^-$, Y= 1:1
d. X=$[Co(NH_3)_4Cl_2]^{3+}3Cl^-$, Y= 1:1

8 Which of the following contains only β-D- glucose as its monosaccharide unit: 1

a. Sucrose
b. Cellulose
c. Starch
d. Maltose

9 Which one of the following sets correctly represents the increase in the paramagnetic property of the ions? 1

a. $Ti^{3+} < Fe^{2+} < Cr^{3+} < Mn^{2+}$
b. $Ti^{3+} < Mn^{2+} < Fe^{2+} < Cr^{3+}$
c. $Mn^{2+} < Fe^{2+} < Cr^{3+} < Ti^{3+}$
d. $Ti^{3+} < Cr^{3+} < Fe^{2+} < Mn^{2+}$

10 A first-order reaction is found to have a rate constant, $k = 5.5 \times 10^{-1}4\ s^{-1}$. The time taken for completion of the reaction is: 1

a. 1.26×10^{13} s
b. 2.52×10^{13} s
c. 0.63×10^{13} s
d. It never goes to completion

11 A student was preparing aniline in the lab. She took a compound “X” and reduced it in the presence of Ni as a catalyst. What could be the compound “X” 1

a. Nitrobenzene
b. 1-Nitrohexane
c. Benzonitrile
d. 1-Hexanenitrile

12 Which of the following compound gives an oxime with hydroxylamine: 1

a. CH_3COCH_3
b. CH_3COOH
c. $(CH_3CO)_2O$
d. CH_3COCl

13 **Assertion (A):** $[Mn(CN)_6]^{3-}$ has a magnetic moment of two unpaired electrons while $[MnCl_6]^{3-}$ has a paramagnetic moment of four unpaired electrons. 1
Reason (R): $[Mn(CN)_6]^{3-}$ is inner orbital complexes involving d^2sp^3 hybridisation,on the other hand, $[MnCl_6]^{3-}$ is outer orbital complexes involving sp^3d^2 hybridisation.

Select the most appropriate answer from the options given below:
a. Both A and R are true and R is the correct explanation of A
b. Both A and R are true but R is not the correct explanation of A.
c. A is true but R is false.
d. A is false but R is true.

14 **Assertion (A)**: For strong electrolytes, there is a slow increase in molar conductivity with dilution and can be represented by the equation 1

$$\Lambda_m^\circ = \Lambda_m - A\ c^{1/2}$$

Reason (R): The value of the constant ‘*A*’ for NaCl, $CaCl_2$, and $MgSO_4$ in a given solvent and at a given temperature is different.

Select the most appropriate answer from the options given below:
a. Both A and R are true and R is the correct explanation of A
b. Both A and R are true but R is not the correct explanation of A.
c. A is true but R is false.
d. A is false but R is true.

15 **Assertion (A)** Glucose does not form the hydrogensulphite addition product with $NaHSO_3$. 1
Reason (R): Glucose exists in a six-membered cyclic structure called pyranose structure.

Select the most appropriate answer from the options given below:
a. Both A and R are true and R is the correct explanation of A
b. Both A and R are true but R is not the correct explanation of A.
c. A is true but R is false.
d. A is false but R is true.

16 **Assertion (A):** The half- life for a zero order reaction is independent of the initial concentration of the reactant. 1
Reason (R): For a zero order reaction, Rate = k

Select the most appropriate answer from the options given below:
a. Both A and R are true and R is the correct explanation of A
b. Both A and R are true but R is not the correct explanation of A.
c. A is true but R is false.
d. A is false but R is true.

SECTION B

This section contains 5 questions with internal choice in one question. The following questions are very short answer type and carry 2 marks each.

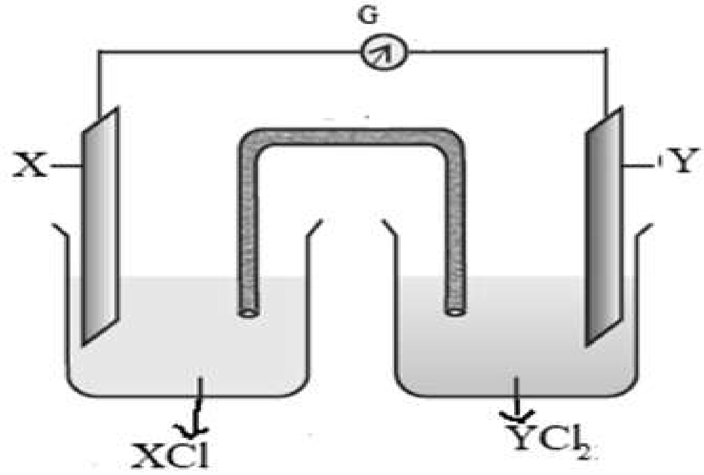
17 a. Nitrogen gas is soluble in water. At temperature 293 K, the value of K_H is 76.48 kbar . How would the solubility of nitrogen vary (increase, decrease or remain the same) at a temperature above 293 K , if the value of K_H rises to 88.8 kbar. 1
b. Chloroform (b.p. 61.2°C) and acetone (b.p. 56°C) are mixed to form an azeotrope. The mole fraction of acetone in this mixture is 0.339. Predict whether the boiling point of the azeotrope formed will be (i) 60°C (ii)64.5 °C or (iii)54 °C. Defend your answer with reason. 1

OR

a. A soda bottle will go flat (loose its fizz) faster in Srinagar than in Delhi. Is this statement correct? Why or why not? 1
b. How does sugar help in increasing the shelf life of the product? 1

18 a. Write the IUPAC name of the following complex: $K[Cr(H_2O)_2(C_2O_4)_2]H_2O$ 1
b. Name the metal present in the complex compound of
(i) Haemoglobin (ii) Vitamin B-12 ½+½

19 Observe the following cell and answer the questions that follow:



a. Represent the cell shown in the figure. 1
b. Name the carriers of the current in the salt bridge ½
c. Write the reaction taking place at the anode. ½

(for visually challenged learners)
For the cell represented as:

$Mg(s)/Mg^{2+}(aq)//Ag^{+}(aq)/Ag(s)$ 1
a. Identify the cathode and the anode 1
b. Write the overall reaction

20 Complete the following reactions by writing the major and minor product in each case (any 2)

a. $CH_3CH_2Br + KCN \rightarrow$ 1

b. $CH_3CH_2CH = CH_2$ + HBr □ 1

c. $(CH_3)_2CHCHClCH_3$ + alc KOH $\rightarrow$ 1

21 The presence of Carbonyl group in glucose is confirmed by its reaction with hydroxylamine. Identify the type of carbonyl group present and its position. Give a chemical reaction in support of your answer. 1 1

SECTION C

This section contains 7 questions with internal choice in one question. The following questions are short answer type and carry 3 marks each.

22 a. Write down the reaction occurring on two inert electrodes when electrolysis of copper chloride is done. What will happen if a concentrated solution of copper sulphate is replaced with copper chloride? 2

b. Write an expression for the molar conductivity of aluminium sulphate at infinite dilution according to Kohlrausch law. 1

23 Account for the following:

a. The lowest oxide of transition metal is basic, and the highest is acidic. 1
b. Chromium is a hard metal while mercury is a liquid metal 1
c. The ionisation energy of elements of the 3d series does not vary much with increasing atomic number. 1

24

a. Give the chemical reaction involved when p-nitrotoluene undergoes Etard reaction. 1
b. Why does Benzoic acid exist as a dimer in an aprotic solvent? 1
c. Benzene on reaction with methylchloride in the presence of anhydrous $AlCl_3$ forms toluene. What is the expected outcome if benzene is replaced by benzoic acid? Give a reason for your answer. 1

OR

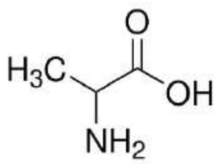
An organic compound 'X', does not undergo aldol condensation. However 'X' with compound 'Y' in the presence of a strong base react to give the compound 1,3-diphenylprop-2-en-1-one.

a. Identify 'X' and 'Y' 1
b. Write the chemical reaction involved. 1
c.Give one chemical test to distinguish between X and Y. 1

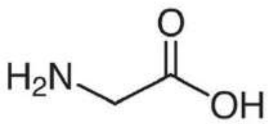
25

a. Give the structure of all the possible dipeptides formed when the following two amino acids form a peptide bond. 2

Alanine



Glycine



b. Keratin, insulin, and myosin are a few examples of proteins present in the human body. Identify which type of protein is keratin and insulin and differentiate between them based on their physical properties. 1

26 Neeta was experimenting in the lab to study the chemical reactivity of alcohols. She carried out a dehydration reaction of propanol at 140°C to 180°C. Different products were obtained at these two temperatures.

a. Identify the major product formed at 140°C and the substitution mechanism followed in this case. 1+½ 1+½

b. Identify the major product formed at 180°C and the substitution mechanism followed in this case.

27 Various isomeric haloalkanes with the general formula C_4H_9Cl undergo hydrolysis reaction. Among them, compound "A" is the most reactive through S_N1 mechanism. Identify "A" citing the reason for your choice. Write the mechanism for the reaction. 3

28 The equilibrium constant of cell reaction :

$Sn^{4+}(.aq) + Al(s) \rightarrow Al^{3+} + Sn^{2+}(aq)$ is 4.617×10^{184}, at 25 °C

a. Calculate the standard emf of the cell. 2

(Given: $\log 4.617 \times 10^{184} = 184.6644$)

b. What will be the E^o of the half cell Al^{3+}/Al, if E^o of half cell Sn^{4+}/Sn^{2+} is 0.15 V. 1

SECTION D

The following questions are case-based questions. Each question has an internal choice and carries 4 (2+1+1) marks each. Read the passage carefully and answer the questions that follow.

29 Dependence of the rate of reaction on the concentration of reactants, temperature, and other factors is the most general method for weeding out unsuitable reaction mechanisms. The term mechanism means all the individual collisional or elementary processes involving molecules (atoms, radicals, and ions included) that take place simultaneously or consecutively to produce the observed overall reaction. For example, when hydrogen gas reacts with bromine, the rate of the reaction was found to be proportional to the concentration of H_2 and to the square root of the concentration of Br_2. Furthermore, the rate was inhibited by increasing the concentration of HBr as the reaction proceeded. These observations are not consistent with a mechanism involving bimolecular collisions of a single molecule of each kind. The currently accepted mechanism is considerably more complicated, involving the dissociation of bromine molecules into atoms followed by reactions between atoms and molecules:

It is clear from this example that the mechanism cannot be predicted from the overall stoichiometry.

(source: Moore, J. W., & Pearson, R. G. (1981). *Kinetics and mechanism*. John Wiley & Sons.)

a. Predict the expression for the rate of reaction and order for the following:

$H_2 + Br_2$ □ 2 HBr 1

What are the units of rate constant for the above reaction? 1

b. How will the rate of reaction be affected if the concentration of Br_2 is tripled? 1

OR

What change in the concentration of H_2 will triple the rate of reaction?

c. Suppose a reaction between A and B, was experimentally found to be first order with respect to both A and B. So the rate equation is: 1

Rate = k[A][B]

Which of these two mechanisms is consistent with this experimental finding? Why?

Mechanism 1

A → C + D (slow)

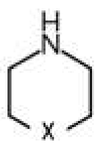
B + C → E (fast)

Mechanism 2

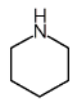
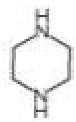
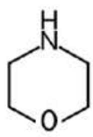
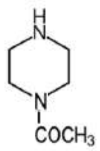
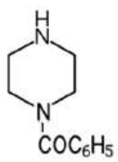
A + B → C + D (slow)

C → E (fast)

30 Amines are basic in nature. The pK_b value is a measure of the basic strength of an amine. Lower the value of pK_b, more basic is the amine. The effect of substituent on the basic strength of amines in aqueous solution was determined using titrations. The substituent "X" replaced "$-CH_2$" group in piperidine (compound 1) and propylamine $CH_3CH_2CH_2NH_2$, (compound 2).

Compound 1:  Compound 2: $HXCH_2CH_2NH_2$

The experimental data is tabulated below:

Substituent "X"	**Electro-negativity of X**	**substituted piperidine compound**	**pK$_a$**	**Substituted propylamine compound**	**pK$_a$**
CH_2	2.55		11.13	$CH_3CH_2CH_2NH_2$	10.67
NH	3.12		9.81	$NH_2CH_2CH_2NH_2$	10.08
O	3.44		8.36	$HOCH_2CH_2NH_2$	9.45
CH_3CON	3.6		7.94	$CH_3CONHCH_2CH_2NH_2$	9.28
C_6H_5CON	3.7		7.78	$C_6H_5CONHCH_2CH_2NH_2$	—

(source: Hall Jr, H. K. (1956). Field and inductive effects on the base strengths of amines. *Journal of the American Chemical Society*, *78*(11), 2570-2572.)

Study the above data and answer the following questions:

a. Plot a graph between the electronegativity of the substituent vs pK$_b$ value of the corresponding substituted propyl amine (given that pK$_a$ + pK$_b$ =14). Is there any relation between the electronegativity of the substituent and its basic strength? 2

b. The electronegativity of the substituent "C_6H_5CON" is 3.7, what is the expected pKa value of compound $C_6H_5CONHCH_2CH_2NH_2$? 1

(i) 9.9   (ii) 9.5   (iii) 9.3   (iv) 9.1

c. The pKa value of the substituted piperidine formed with substituent “X” is found to be 8.28. What is the expected electronegativity of “X” 1

(i)3.5 (ii)3.4 (iii)3.8 (iv) 3.1

OR

What is the most suitable pKa value of the substituted propylamine formed with substituent “X” with electronegativity 3.0

(i)10.67 (ii)10.08 (iii)10.15 (iv)11.10

(for visually challenged learners)

a.How does the electronegativity of the substituent affect the pK_b value and the basic strength of the substituted propyl amine (given that $pK_a + pK_b = 14$).? Give a reason to support your answer. 2

b. The electronegativity of the substituent “C_6H_5CON” is 3.7, what is the expected pKa value of compound $C_6H_5CONHCH_2CH_2NH_2$? 1

(i) 9.9   (ii) 9.5   (iii) 9.3   (iv) 9.1

c. The pKa value of the substituted piperidine (compound 1) formed with substituent “X” is found to be 8.28. What is the expected electronegativity of “X” 1

(i)3.5 (ii)3.4 (iii)3.8 (iv) 3.1

OR

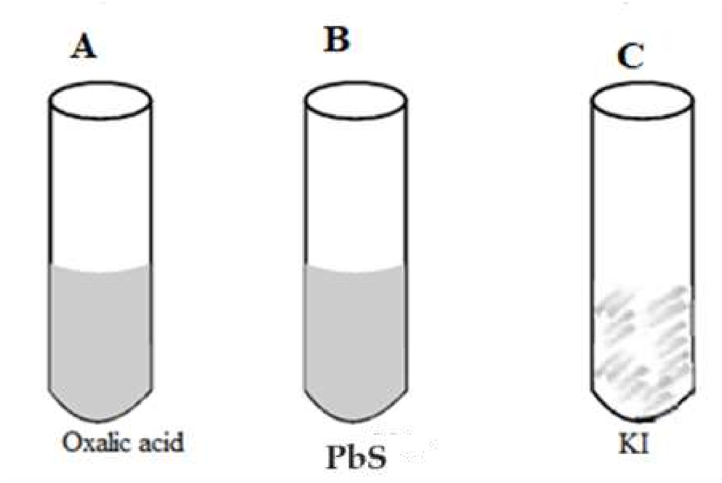
What is the most suitable pKa value of the substituted propylamine formed with substituent “X” with electronegativity 3.0

(i)10.67 (ii)10.08 (iii)10.15 (iv)11.10

SECTION E

The following questions are long answer types and carry 5 marks each. All questions have an internal choice.

31 a. A purple colour compound A, which is a strong oxidising agent and used for bleaching of wool, cotton, silk and other textile fibres was added to each of the three test tubes along with H_2SO_4. It was followed by strong heating.



In which of the above test tubes; A,B or C:

(i) Violet vapours will be formed 1

(ii) The bubbles of gas evolved will extinguish a burning matchstick. Write an equation for each of the above observations. 1

b. A metal ion M^{n+} of the first transition series having d^5 configuration combines with three didentate ligands. Assuming $\Delta_0 < P$:

(i) Draw the crystal field energy level diagram for the 3d orbital of this complex. 1

(ii) What is the hybridisation of M^{n+} in this complex and why? 1

(iii) Name the type of isomerism exhibited by this complex. 1

OR

a. Using, Valence Bond Theory identify A, B, C, D, E and F in the following table

S.No	Complex	central metal ion	configuration of metal ion	Hybridization of Metal ion	Geometry of the Complex	Number Of Unpaired Electron	Magnetic Behaviour
i	$[CoF_4]^{2-}$	**A**	$3d^7$	sp^3	tetrahedral	**B**	Paramagnetic
ii	$[Cr(H_2O)_2C_2O_4)_2]$	Cr^{3+}	$3d^3$	**C**	octahedral	3	**D**
iii	$[Ni(CO)_4]$	Ni	$3d^84s^2$	**E**	**F**	0	Diamagnetic

3

b. Write the ionic equations for the reaction of acidified $K_2Cr_2O_7$ with (i)H_2S and (ii)$FeSO_4$ 2

32 a. Give reasons for the following:

(i)The reaction of ethanol with acetyl chloride is carried out in the presence of pyridine . 1

(ii) Cresols are less acidic than phenol. 1

b. Williamson's process is used for the preparation of ethers from alkyl halide. Identify the alkyl bromide and sodium alkoxide used for the preparation of 2- Ethoxy-3-methylpentane 1

c. Convert:

(i) Toluene to 3-nitrobenzoic acid. 1

(ii) Benzene to m-nitroacetophenone. 1

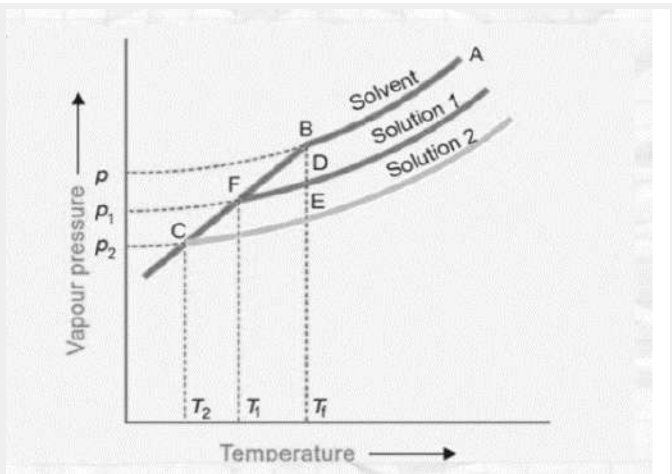
OR

a. Out of formic acid and acetic acid, which one will give the HVZ reaction? Give a suitable reason in support of your answer and write the chemical reaction involved. 2

b. Alcohols are acidic but they are weaker acids than water. Arrange various isomers of butanol in the increasing order of their acidic nature. Give a reason for the same. 1

c. An organic compound A which is a Grignard reagent is used to obtain 2-methylbutan-2-ol on reaction with a carbonyl compound 'B' . Identify A' and 'B'. Write the equation for the reaction between A and B. 2

33 a. An experiment was carried out in the laboratory, to study depression in freezing point. 1M aqueous solution of $Al(NO_3)_3$ and 1 M aqueous solution of glucose were taken. From the given figure identify solution 1 and solution 2. Give a plausible reason for your answer. 2



b. The osmotic pressure of a solution of cane sugar was found to be 2.46 atm at 300 K. If the solution was diluted five times, calculate the osmotic pressure at the same temperature. 3
How can the osmotic pressure of the given cane sugar solution be decreased without changing its volume? Give a reason for your answer.

OR

a. While giving intravenous injections to the patients, the doctors take utmost care of the concentration of the solution used. Why is it necessary to check the concentration of the solution? 2

b. A solution of phenol was obtained by dissolving $2X\ 10^{-2}$ kg of phenol in 1 kg of benzene. Experimentally it was found to be 73 % associated. Calculate the depression in the freezing point recorded. 3

(for visually challenged learners)

a. Which of the two solutions : 1M aqueous solution of $Al(NO_3)_3$ or 1M aqueous solution of glucose will show a greater depression in freezing point? Give a plausible reason for your answer. 2

b. The osmotic pressure of a solution of cane sugar was found to be 2.46 atm at 300 K. If the solution was diluted five times, calculate the osmotic pressure at the same temperature. 3

How can the osmotic pressure of the given cane sugar solution be decreased without changing its volume? Give a reason for your answer.

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a. While giving intravenous injections to the patients, the doctors take utmost care of the concentration of the solution used. Why is it necessary to check the concentration of the solution? 2

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Std : 12th (Science)

PAPER SET - 1

Subject : Chemistry

Marks : 100

PART - A (MCQ)

- **There are 50 objective type (M.C.Q.) questions in Part-A and all questions are compulsory.**

(1) Which of the following aqueous solutions should have the highest boiling point ?

(A) 1.0 M NaOH (B) 1.0 M Na_2SO_4 (C) 1.0 M NH_4NO_3 (D) 1.0 M KNO_3

(2) The unit of ebulioscopic constant is ....

(A) K kg mol^{-1} or K (m molality)$^{-1}$ (B) mol kg^{-1} or K^{-1} (molality)

(C) kg mol^{-1} K^{-1} or K^{-1} (molality)$^{-1}$ (D) K mol kg^{-1} K (molality)

(3) The values of Van't Hoff factors for KCl, NaCl and K_2SO_4, respectively, are ....

(A) 2, 2 and 2 (B) 2, 2 and 3 (C) 1, 1 and 2 (D) 1, 1 and

(4) We have three aqueous solutions of NaCl labelled as 'A', 'B' and 'C' with concentrations 0.1 M, 0.01 M and 0.001 M, respectively. The value of Van't Hoff factor for these solutions will be in the

(A) $i_A < i_B < i_C$ (B) $i_A > i_B > i_C$ (C) $i_A = i_B = i_C$ (D) $i_A < i_B > i_C$

(5) Van't Hoff factor i is given by the expression ....

(A) $i = \frac{\text{Normal molar mass}}{\text{Abnormal molar mass}}$ (B) $i = \frac{\text{Abnormal molar mass}}{\text{Normal molar mass}}$

(C) $i = \frac{\text{Observed colligative property}}{\text{Calculated colligative property}}$ (D) $i = \frac{\text{Calculated colligative property}}{\text{Observed colligative property}}$

(6) Which of the following binary mixtures will have same composition in liquid and vapour phase ?

(A) Benzene - Toluene (B) Water-Nitric acid

(C) Water-Ethanol (D) n-Hexane - n-Heptane

(7) Which cell will measure standard electrode potential of copper electrode ?

(A) $Pt_{(s)} | H_{2(g,\,0.1\,bar)} | H^{+}_{(aq,\,1M)} || Cu^{2+}_{(aq,\,1M)} | Cu$

(B) $Pt_{(s)} | H_{2(g,\,1\,bar)} | H^{+}_{(aq,\,1M)} || Cu^{2+}_{(aq,\,2M)} | Cu$

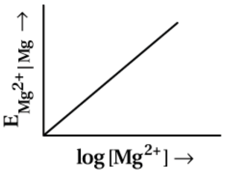
(C) $Pt_{(s)} | H_{2(g,\,1\,bar)} | H^{+}_{(aq,\,1M)} || Cu^{2+}_{(aq,\,1M)} | Cu$

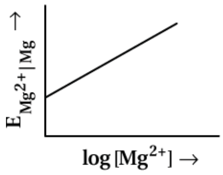
(D) $Pt_{(s)} | H_{2(g,\,1\,bar)} | H^{+}_{(aq,\,0.1M)} || Cu^{2+}_{(aq,\,1M)} | Cu$

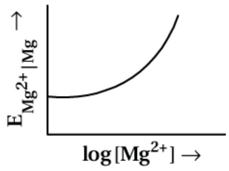
(8) Electrode potential for Mg electrode varies according to the equation :

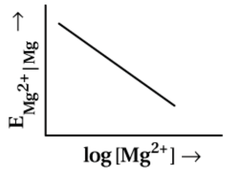
$$E_{Mg^{2+}|Mg} = E^{\ominus}_{Mg^{2+}|Mg} - \frac{0.59}{2}\log\frac{1}{[Mg^{2+}]}$$

The graph of $E_{Mg^{2+}|Mg} \rightarrow \log[Mg^{2+}]$ is

(A) 

(B) 

(C) 

(D) 

(9) An electrochemical cell can behave like an elctrolytic cell when_________.

(A) $E_{cell} = 0$ (B) $E_{cell} > E_{ext}$ (C) $E_{ext} > E_{cell}$ (D) $E_{cell} > E_{ext}$

(10) Which compound gives color ?

(A) Cu^{+} (B) Ht^{+3}

(C) Zn^{+2} (D) Cr^{+3}

(11) The quantity of charge required to obtain one mole of aluminium from Al_2O_3 is ....

(A) 1F (B) 6F (C) 3F (D) 2F

(12) $\Lambda^{o}_{m}(H_2O)$ is equal to ....

(A) $\Lambda^{o}_{m(HCl)} + \Lambda^{o}_{m(NaOH)} - \Lambda^{o}_{m(NaCl)}$ (B) $\Lambda^{o}_{m(HNO_3)} + \Lambda^{o}_{m(NaNO_3)} - \Lambda^{o}_{m(NaOH)}$

(C) $\Lambda^{o}_{m(HNO_3)} + \Lambda^{o}_{m(NaOH)} - \Lambda^{o}_{m(NaNO_3)}$ (D) $\Lambda^{o}_{m(NH_4OH)} + \Lambda^{o}_{m(HCl)} - \Lambda^{o}_{m(NH_4Cl)}$

(13) Assertion : Copper sulphate can be stored in zinc vessel.

Reason : Zinc is less reactive than copper.

(A) Both assertion and reason are true and the reason is the correct explanation of assertion.

(B) Both assertion and reason are true and the reason is not the correct explanation of assertion.

(C) Assertion is true but the reason is false.

(D) Both assertion and reason are false.

(14) The role of a catalyst is to change ....

(A) Gibbs energy of reaction (B) enthalpy of reaction

(C) activation energy of reaction (D) equilibrium constant

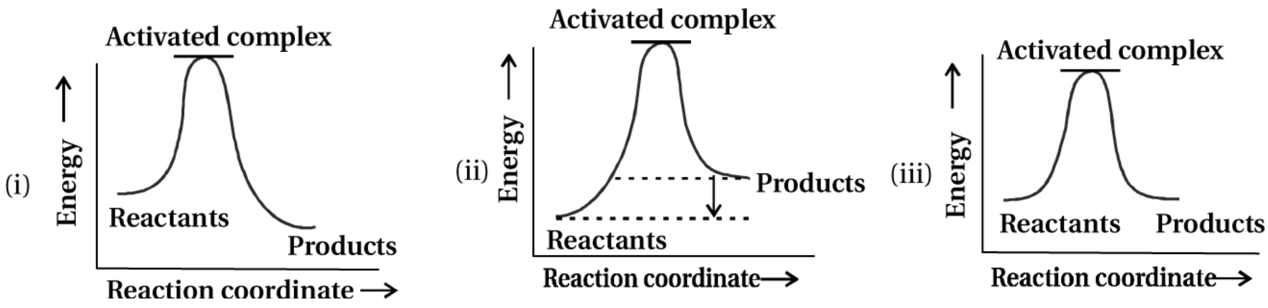
(15) Which of the following expressions is correct for the rate of reaction given below ?

$$5Br^{-}_{(aq)} + BrO^{-}_{3(aq)} + 6H^{+}_{(aq)} \rightarrow 3Br_{2(aq)} + 3H_2O_{(l)}$$

(A) $\frac{\Delta[Br^-]}{\Delta t} = 5\frac{\Delta[H^+]}{\Delta t}$ (B) $\frac{\Delta[Br^-]}{\Delta t} = \frac{6}{5}\frac{\Delta[H^+]}{\Delta t}$

(C) $\frac{\Delta[Br^-]}{\Delta t} = \frac{5}{6}\frac{\Delta[H^+]}{\Delta t}$ (D) $\frac{\Delta[Br^-]}{\Delta t} = 6\frac{\Delta[H^+]}{\Delta t}$

(16) Which of the following graphs represents exothermic reaction ?



(A) only (i) (B) only (ii) (C) only (iii) (D) (i) and (ii)

(17) A first order reaction is 50% completed in 1.26×10^{14} s. How much time would it take for 100% completion ?

(A) 1.26×10^{15} s (B) 2.52×10^{14} s (C) 2.52×10^{28} s (D) infinite

(18) Mark the incorrect statements.

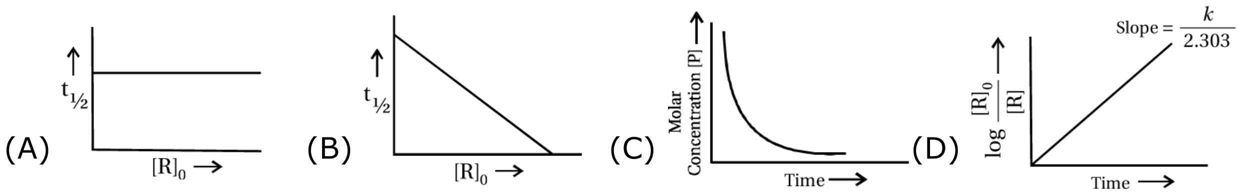
(A) Catalyst provides an alternative pathway to reaction mechanism.

(B) Catalyst raises the activation energy.

(C) Catalyst lowers the activation energy.

(D) Catalyst alters enthalpy change of the reaction.

(19) Which of the following graphs is correct for a first order reaction ?



(20) The magnetic nature of elements depends on the presence of unpaired electrons. Identify the configuration of transition element, which shows highest magnetic moment.

(A) $3d^7$ (B) $3d^5$ (C) $3d^8$ (D) $3d^2$

(21) Which of the following oxidation state is common for all lanthanoids ?

(A) +2 (B) +3 (C) +4 (D) +5

(22) $KMnO_4$ acts as an oxidising agent in acidic medium. The number of moles of $KMnO_4$ that will be needed to react with one mole of sulphide ions in acidic solution is ....

(A) $\frac{2}{5}$ (B) $\frac{3}{5}$ (C) $\frac{4}{5}$ (D) $\frac{1}{5}$

(23) $KMnO_4$ acts as an oxidising agent in alkaline medium. When alkaline $KMnO_4$ is treated with KI, iodide ion is oxidised to ....

(A) I_2 (B) IO^- (C) IO^-_3 (D) IO^-_4

(24) Which of the following actinoids show oxidation states upto (+7) ?

(A) Am (B) Pu (C) U (D) Np

(25) Which of the following will not act as oxidising agents ?

(A) CrO_3 (B) MoO_3 (C) WO_3 (D) $CrO^{2}{}_4^{-}$

(26) The colour of the coordination compounds depends on the crystal field splitting. What will be the correct order of absorption of wave-length of light in the visible region, for the complexes, $[Co(NH_3)_6]^{3+}$, $[Co(CN)_6]^{3-}$, $[Co(H_2O)_6]^{3+}$

(A) $[Co(CN)_6]^{3-} > [Co(NH_3)_6]^{3+} > [Co(H_2O)_6]^{3+}$

(B) $[Co(NH_3)_6]^{3+} > [Co(H_2O)_6]^{3+} > [Co(CN)_6]^{3-}$

(C) $[Co(H_2O)_6]^{3+} > [Co(NH_3)_6]^{3+} > [Co(CN)_6]^{3-}$

(D) $[Co(CN)_6]^{3-} > [Co(NH_3)_6]^{3+} > [Co(H_2O)_6]^{3+}$

(27) When 0.1 mol $CoCl_3(NH_3)_5$ is treated with excess of $AgNO_3$, 0.2 mol of AgCl are obtained. The conductivity of solution will correspond to.....

(A) 1:3 electrolyte (B) 1:2 electrolyte (C) 1:1 electrolyte (D) 3:1 electrolyte

(28) When 1 mole of $CrCl_3 \cdot 6H_2O$ is treated with excess of $AgNO_3$, 3 mol of AgCl are obtained. The

(A) $[CrCl_3(H_2O)_3] \cdot 3H_2O$ (B) $[CrCl_2(H_2O)_4] \cdot 2H_2O$

(C) $[CrCl\,(H_2O)_5]\,Cl_2 \cdot H_2O$ (D) $[Cr(H_2O)_6]\,Cl_3$

(29) The CFSE for octahedral $[CoCl_6]^{4-}$ is 18,000 cm^{-1}. The CFSE for tetrahedral $[CoCl_4]^{2-}$ will be ....

(A) 18,000 cm^{-1} (B) 16,000 cm^{-1} (C) 8,000 cm^{-1} (D) 20,000 cm^{-1}

(30) Which is the following options are correct for $[Fe(CN)_6]^{3-}$ complex ?

(A) d^2sp^3 hybridisation (B) sp^3d^2 hybridisation

(C) Paramagnetic (D) Diamagnetic

(31) Which of the following alcohols will yield the corresponding alkyl chloride on reaction with concentrated HCl at room temperature ?

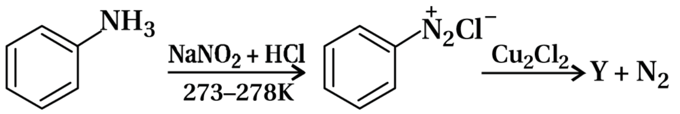
(A) $CH_3CH_2 - CH_2 - OH$

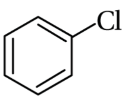
(B) $CH_3CH_2 - \underset{\underset{CH_3}{|}}{CH} - OH$

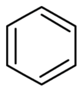
(C) $CH_3CH_2 - \underset{\underset{CH_3}{|}}{CH} - CH_2OH$

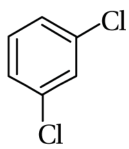
(D) $CH_3CH_2 - \overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}} - OH$

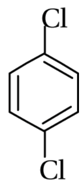
(32) Identify the compound Y in the following reaction.



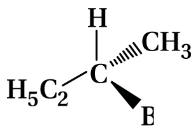
(A) 

(B) 

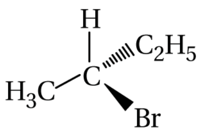
(C) 

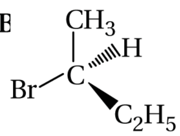
(D) 

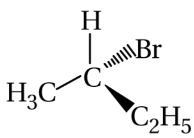
(33) Which of the following structures is enantiomeric with the molecule (A) given below :

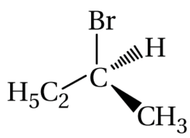


(A)

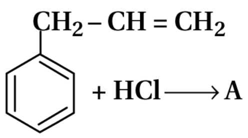
(A) 

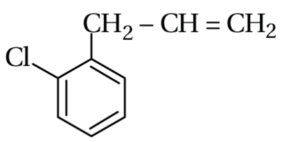
(B) 

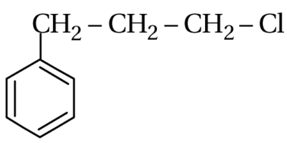
(C) 

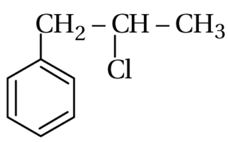
(D) 

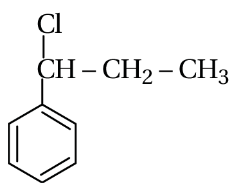
(34) What is 'A' in the following reaction ?



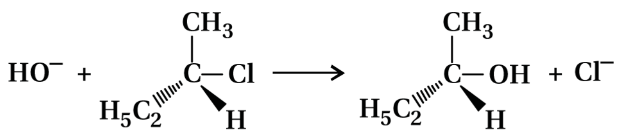
(A) 

(B) 

(C) 

(D) 

(35) Which of the following statements are correct about the kinetics of this reaction ?



(i) (ii) (iii) (iv)

(A) The rate of reaction depends on the concentration of only (ii)

(B) The rate of reaction depends on concentration of both (i) and (ii)

(C) Molecularity of reaction is one

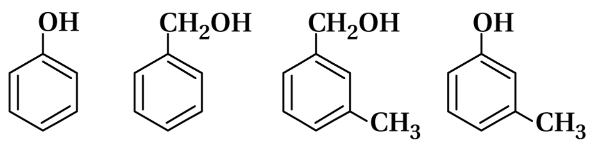
(D) Molecularity of reaction is two

(36) Mono-chlorination of toluene in a sunlight followed by hydrolysis with aq. NaOH yields.......

(A) o - Cresol (B) m - Cresol

(C) 2, 4-Dihydroxytoluene (D) Benzyl alcohol

(37) Which of the following compounds is aromatic alcohol ?

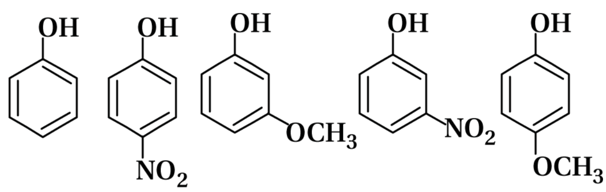


(A) (B) (C) (D)

(A) (A), (B), (C) an (D) (B) (A) and (B)

(C) (B) and (C) (A)

(38) Mark the correct order of decreasing acidic strength of the following compounds :



(I) (II) (III) (IV) (V)

(A) V > IV > II > I > III (B) II > IV > I > III > V

(C) IV > V > III > II > I (D) V > IV > III > II > I

(39) Arrange the following compounds in increasing order of boiling point :

Propan-1-ol, Butan-1-ol, Butan-2-ol, Pentan-1-ol

(A) Propan-1-ol, Butan-2-ol, Butan-1-ol, Pentan-1-ol

(B) Propan-1-ol, Butan-1-ol, Butan-2-ol, Pentan-1-ol

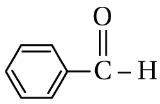
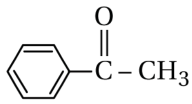
(C) Pentan-1-ol, Butan-2-ol, Butan-1-ol, Propan-1-ol

(D) Pentan-1-ol, Butan-1-ol, Butan-2-ol, Propan-1-ol

(40) Phenol can be distinguished from ethanol by the reaction with ....

(A) Br_2/water (B) Na (C) Neutral $FeCl_3$ (D) All of the above

(41) Which of the following compounds is most reactive towards nucleophilic addition reactions ?

(A) $CH_3-\overset{O}{\overset{||}{C}}-H$ (B) $CH_3-\overset{O}{\overset{||}{C}}-CH_3$ (C)  (D) 

(42) The correct order of increasing acidic strength is....

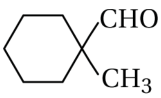
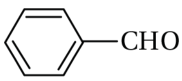
(A) Phenol < Ethanol < Chloroacetic acid < Acetic acid

(B) Ethanol < Phenol < Chloroacetic acid < Acetic acid

(C) Ethanol < Phenol < Acetic acid < Chloroacetic acid

(D) Chloroacetic acid < Acetic acid < Phenol < Ethanol

(43) Cannizzaro's reaction is not given by....

(A)  (B)  (C) HCHO (D) CH_3CHO

(44) $$CH_3-C\equiv CH \xrightarrow[1\%\ HgSO_4]{40\%\ H_2SO_4} A \xrightarrow{\text{Isomerisation}} CH_3-\overset{\displaystyle O}{\overset{||}{C}}-CH_3$$

Structure of 'A' and type of isomerism in the above reaction are respectively.

(A) Prop-1-en-2-ol, metamerism

(B) Prop-1-en-1-ol, tautomerism

(C) Prop-2-en-2-ol, geometrical isomerism

(D) Prop-2-en-2-ol, tautomerism

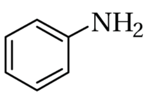
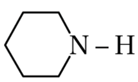
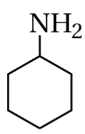
(45) Compounds A and C in the following reaction are ....

$$CH_3CHO \xrightarrow[\text{(ii) } H_2O]{\text{(i) } CH_3MgBr} (A) \xrightarrow[\Delta]{H_2SO_4} (B) \xrightarrow[\text{oxidation}]{\text{Hydroboration}} (C)$$

(A) identical (B) positional isomers

(C) functional isomers (D) optical isomers

(46) Which of the following is the weakest Bronsted base ?

(A)  (B)  (C)  (D) CH_3NH_2

(47) Amongst the given set of reactants, the most appropriate for preparing 2° amine is ....

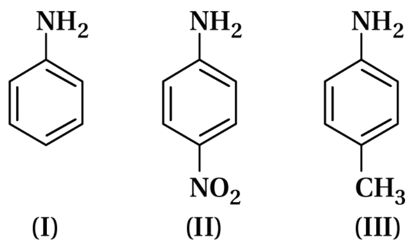
(A) $2° R - Br + NH_3$

(B) $2° R - Br + NaCN$ followed by H_2/Pt

(C) $1° R - NH_2 + RCHO$ followed by H_2/Pt

(D) $1°R - Br$ (2 mol) + potassium phthalimide followed by $H_3O^+/heat$

(48) The correct increasing order of basic strength for the following compounds is ....

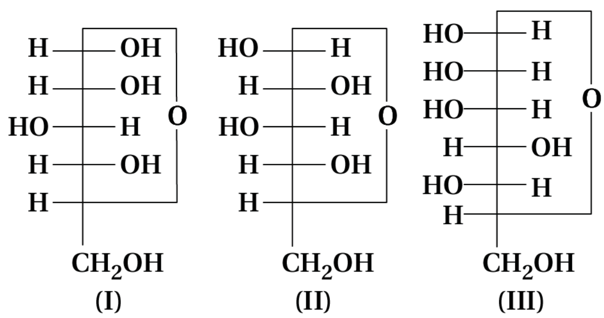


(A) II < III < I (B) III < I < II (C) III < II < I (D) II < I < III

(49) Nucleic acids are the polymers of ....

(A) Nucleosides (B) Nucleotides (C) Bases (D) Sugars

(50) Three cyclic structures of monosaccharides are given below which of these are anomers.



(A) I and II (B) II and III

(C) I and III (D) III is anomer of I and II

PART – B

SECTION - A

- **Answer questions. Each question 2 marks. (Any 8)** **(16)**

(51) At 27° C temperature, 36 gm glucose is in 1 litre aqueous solution has π = 4. 98 bar. Find out concentration if π = 1.52 bar at same temperature.

(52) Henry's law constant for the molality of methane in benzene at 298 K is 4.27×10^5. Calculate the solubility of methane in benzene at 298 K under 760 mm Hg.

(53) Give Reason : $TiCl_3$ is paramagnetic while $TiCl_4$ is diamagnetic.

(54) Explain: "Lanthanoid contraction."

(55) Draw the isomers of $[Co(NH_3)_3(NO_2)_3]$ complex.

(56) Give conversion: Ethene into butane.

(57) Explain : Lucas test.

(58) Explain silver mirror test and Fehling test.

(59) Write only chemical reaction for three different methods to form alkane from acetone.

(60) Write carbyl amine test.

(61) Explain Hoffmann reaction by two examples.

(62) Give only reaction to prepare glucose from sugar and starch.

SECTION - B

- **Answer questions. Each question 3 marks. (Any 6)** **(18)**

(63) Discuss biological and industrial importance of osmosis.

(64) In electrolysis of $CuCl_2$, 2.0 ampere current is passed through solution at 1 bar pressure and 300 K temperature for 1 hour, so calculate how much product is obtained at anode and cathode ? [Cu = 63.5, Cl = 35.5, R = 0.08314]

(65) Derive the formula of first order reaction for, (i) Rate constant K, (ii) Half life period $t\frac{1}{2}$ (graph is not required)

(66) Explain geometrical isomerism in four coordinated compounds.

(67) Give one example of each for Swarts reaction, Wurtz reaction and Finkelstein reaction of alkyl halide.

(68) Give two reactions that show the acidic nature of phenol. Compare acidity of phenol with that of ethanol.

(69) Write Wolff-Kishner and Clemmensen reduction of aldehyde and ketone.

(70) Give conversion in three steps : Nitrobenzene into chlorobenzene.

(71) What are the expected products of hydrolysis of lactose ?

SECTION - C

- **Answer questions. Each question 4 marks. (Any 4)** **(16)**

(72) Write reactions occurring at anode and cathode in dry cell and lead storage cell.

(73) During nuclear explosion, one of the products is ^{90}Sr with half-life of 28.1 years. If 1 mg ^{90}Sr was absorbed in the bones of a newly born baby instead of calcium, how much of it will remain after 10 years and 60 years if it is not lost metabolically.

(74) Write reaction to prepare $KMnO_4$ and its two properties.

(75) Explain on the basis of valence bond theory that $[Fe(H_2O)_6]^{3+}$ is paramagnetic where as $[Fe(CN)_6]^{4-}$is diamagnetic.

(76) Explain in detail about mechanism of SN_2 reaction.

(77) Write the reactions to prepared phenol from Aniline and Cumine.

Best of Luck

Std : 12^{th} (Science)

Subject : Chemistry

PAPER SET - 2

Marks : 100

PART - A (MCQ)

- **There are 50 objective type (M.C.Q.) questions in Part-A and all questions are compulsory.**

(1) Which of the following aqueous solutions should have the highest boiling point ?

(A) 1.0 M NaOH (B) 1.0 M Na_2SO_4 (C) 1.0 M NH_4NO_3 (D) 1.0 M KNO_3

(2) In comparison to a 0.01 M solution of glucose, the depression in freezing point of a 0.01 M $MgCl_2$ solution is ....

(A) the same (B) about twice

(C) about three times (D) about six times

(3) At a given temperature, osmotic pressure of a concentrated solution of a substance ....

(A) is higher than that at a dilute solution.

(B) is lower than that of a dilute solution.

(C) is same as that of a dilute solution.

(D) can not be compared with osmotic pressure of dilute solution.

(4) The values of Van't Hoff factors for KCl, NaCl and K_2SO_4, respectively, are ....

(A) 2, 2 and 2 (B) 2, 2 and 3 (C) 1, 1, and 2 (D) 1, 1 and 1

(5) We have three aqueous solutions of NaCl labelled as 'A', 'B' and 'C' with concentrations 0.1 M, 0.01 M and 0.001 M, respectively. The value of Van't Hoff factor for these solutions will be in the order ....

(A) $i_A < i_B < i_C$ (B) $i_A > i_B > i_C$ (C) $i_A = i_B = i_C$ (D) $i_A < i_B > i_C$

(6) K_H value for $Ar_{(g)}$, $CO_{2(g)}$, $HCHO_{(g)}$ and $CH_{4(g)}$ are 40.39, 1.83×10^{-5} and 0.413 respectively. Arrange these gases in the order of their increasing solubility.

(A) $HCHO < CH_4 < CO_2 < Ar$ (B) $HCHO < CO_2 < CH_4 < Ar$

(C) $Ar < CO_2 < CH_4 < HCHO$ (D) $Ar < CH_4 < CO_2 < HCHO$

(7) Which cell will measure standard electrode potential of copper electrode ?

(A) $Pt_{(s)} \mid H_{2(g,\ 0.1\ bar)} \mid H^{+}_{(aq,\ 1M)} \mid\mid Cu^{2+}_{(aq,\ 1M)} \mid Cu$

(B) $Pt_{(s)} \mid H_{2(g,\ 1\ bar)} \mid H^{+}_{(aq,\ 1M)} \mid\mid Cu^{2+}_{(aq,\ 2M)} \mid Cu$

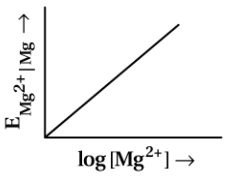
(C) $Pt_{(s)} \mid H_{2(g,\ 1\ bar)} \mid H^{+}_{(aq,\ 1M)} \mid\mid Cu^{2+}_{(aq,\ 1M)} \mid Cu$

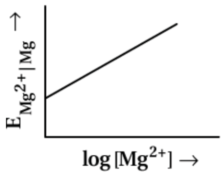
(D) $Pt_{(s)} \mid H_{2(g,\ 1\ bar)} \mid H^{+}_{(aq,\ 0.1M)} \mid\mid Cu^{2+}_{(aq,\ 1M)} \mid Cu$

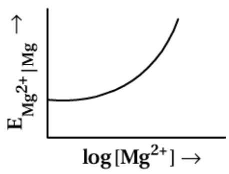
(8) Electrode potential for Mg electrode varies according to the equation :

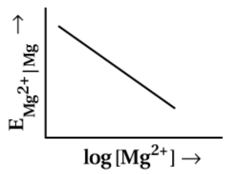
$$E_{Mg^{2+}|Mg} = E^{\ominus}_{Mg^{2+}|Mg} - \frac{0.59}{2} \log \frac{1}{[Mg^{2+}]}$$

The graph of $E_{Mg^{2+}|Mg} \to \log [Mg^{2+}]$ is

(A) 

(B) 

(C) 

(D) 

(9) The difference between the electrode potentials of two electrodes when no current is drawn through the cell is called_______.

(A) Cell potential (B) Cell emf

(C) Potential difference (D) Cell voltage

(10) Using the data given below find out the strongest reducing agent.

$E^{\ominus}_{Cr_2O_7^{2-}|Cr^{3+}} = 1.33V;\ E^{\ominus}_{Cl_2|Cl^-} = 1.36V$

$E^{\ominus}_{MnO_4^-|Mn^{2+}} = 1.51V;\ E^{\ominus}_{Cr^{3+}|Cr} = -0.74V$

(A) Cl^- (B) Cr (C) Cr^{3+} (D) Mn^{2+}

(11) Most oxidising species.

(A) Cr^{3+} (B) MnO_4^- (C) $Cr_2O_4^{2-}$ (D) Mn^{2+}

(12) $\Lambda^o_m(NH_4OH)$ is equal to ....

(A) $\Lambda^o_{m(NH_4OH)} + \Lambda^o_{m(NH_4Cl)} - \Lambda^o_{(HCl)}$

(B) $\Lambda^o_{m(NH_4Cl)} + \Lambda^o_{m(NaOH)} - \Lambda^o_{(NaCl)}$

(C) $\Lambda^o_{m(NH_4Cl)} + \Lambda^o_{m(NaCl)} - \Lambda^o_{m(NaCH)}$

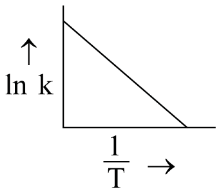
(D) $\Lambda^o_{m}{}^{(NaOH)} + \Lambda^o_{(NaCl)} - \Lambda^o_{m(NH_4Cl)}$

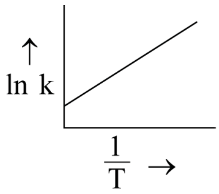
(13) Consider a first order gas phase decomposition reaction given below : $A_{(g)} \to B_{(g)} + C_{(g)}$ The initial pressure of the system before decomposition of A was p_i. After lapse of time 't', total pressure of the system increased by x units and became 'p_i'. The rate constant k for the reaction is given as________.

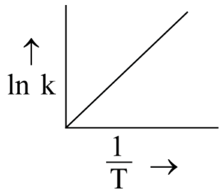
(A) $k = \frac{2.303}{t} \log \frac{p_i}{p_i - x}$ (B) $k = \frac{2.303}{t} \log \frac{p_i}{2p_i - p_t}$

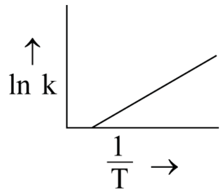
(C) $k = \frac{2.303}{t} \log \frac{p_i}{2p_i + p_t}$ (D) $k = \frac{2.303}{t} \log \frac{p_i}{p_i + x}$

(14) 245) According to Arrhenius equation rate constant k is equal to A $e^{-Ea/RT}$. Which of the following options represents the graph of ln k vs $\frac{1}{T}$?

(A) 

(B) 

(C) 

(D) 

(15) Which of the following expressions is correct for the rate of reaction given below ?

$5Br^-_{(aq)} + BrO^-_{3(aq)} + 6H^+_{(aq)} \rightarrow 3Br_{2(aq)} + 3H_2O_{(l)}$

(A) $\frac{\Delta[Br^-]}{\Delta t} = 5\frac{\Delta[H^+]}{\Delta t}$ (B) $\frac{\Delta[Br^-]}{\Delta t} = \frac{6}{5}\frac{\Delta[H^+]}{\Delta t}$

(C) $\frac{\Delta[Br^-]}{\Delta t} = \frac{5}{6}\frac{\Delta[H^+]}{\Delta t}$ (D) $\frac{\Delta[Br^-]}{\Delta t} = 6\frac{\Delta[H^+]}{\Delta t}$

(16) A first order reaction is 50% completed in 1.26×10^{14} s. How much time would it take for 100% completion ?

(A) 1.26×10^{15} s (B) 2.52×10^{14} s (C) 2.52×10^{28} s (D) infinite

(17) Which of the following oxidation state is common for all lanthanoids ?

(A) +2 (B) +3 (C) +4 (D) +5

(18) Which of the following reactions are disproportionation reactions ?

(a) $Cu^+ \rightarrow Cu^{2+} + Cu$

(b) $MnO^{2-}_4 + 4H^+ \rightarrow 2MnO^-_4 + MnO_2 + 2H_2O$

(c) $2KMnO_4 \rightarrow K_2MnO_4 + MnO_2 + O_2$

(d) $2MnO^-_4 + 3Mn^{2+} + 2H_2O \rightarrow 5MnO_2 + 4H^+$

(A) a, b (B) a, b, c (C) b, c, d (D) a, d

(19) $KMnO_4$ acts as on oxidising agent in acidic medium. The number of moles of $KMnO_4$ that will be needed to react with one mole sulphide icons in acidic solution is ________

(A) $\frac{2}{5}$ (B) $\frac{3}{5}$ (C) $\frac{4}{5}$ (D) $\frac{1}{5}$

(20) When acidified $K_2Cr_2O_7$ solution is added to Sn^{2+} salts then Sn^{2+} changes to________.

(A) Sn (B) Sn^{3+} (C) Sn^{4+} (D) Sn^{+}

(21) Why is HCl not used to make the medium acidic in oxidation reactions of $KMnO_4$ in acidic medium ?

(A) Both HCl and $KMnO_4$ act as oxidising agents.

(B) $KMnO_4$ +oxidises HCl into Cl_2 which is also an oxidising agent.

(C) $KMnO_4$ is a weaker oxidising agent than HCl.

(D) $KMnO_4$ acts as a reducing agent in the presence of HCl.

(22) The colour of the coordination compounds depends on the crystal field splitting. What will be the correct order of absorption of wave-length of light in the visible region, for the complexes, $[Co(NH_3)_6]^{3+}$, $[Co(CN)_6]^{3-}$, $[Co(H_2O)_6]^{3+}$

(A) $[Co(CN)_6]^{3-} > [Co(NH_3)_6]^{3+} > [Co(H_2O)_6]^{3+}$

(B) $[Co(NH_3)_6]^{3+} > [Co(H_2O)_6]^{3+} > [Co(CN)_6]^{3-}$

(C) $[Co(H_2O)_6]^{3+} > [Co(NH_3)_6]^{3+} > [Co(CN)_6]^{3-}$

(D) $[Co(CN)_6]^{3-} > [Co(NH_3)_6]^{3+} > [Co(H_2O)_6]^{3+}$

(23) When 1 mole of $CrCl_3 \cdot 6H_2O$ is treated with excess of $AgNO_3$, 3 mol of AgCl are obtained. The

(A) $[CrCl_3(H_2O)_3] \cdot 3H_2O$ (B) $[CrCl_2(H_2O)_4] \cdot 2H_2O$

(C) $[CrCl(H_2O)_5] Cl_2 \cdot H_2O$ (D) $[Cr(H_2O)_6] Cl_3$

(24) The CFSE for octahedral $[CoCl_6]^{4-}$ is 18,000 cm^{-1}. The CFSE for tetrahedral $[CoCl_4]^{2-}$ will be ....

(A) 18,000 cm^{-1} (B) 16,000 cm^{-1} (C) 8,000 cm^{-1} (D) 20,000 cm^{-1}

(25) The compounds $[Co(SO_4)(NH_3)_5]Br$ and $[Co(SO_4)(NH_3)_5]Cl$ represent________.

(A) Linkage isomerism (B) Ionisation isomerism

(C) Coordination isomerism (D) No isomerism

(26) Which of the following species is not expected to be a ligand ?

(A) NO (B) NH_4^+ (C) $NH_2CH_2CH_2NH_2$ (D) CO

(27) IUPAC name of $[Pt(NH_3)_2 Cl(NO_3)]$ is :

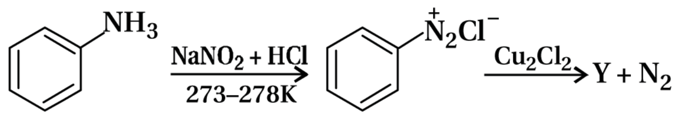
(A) Platinum diaminechloronitrite

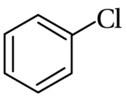
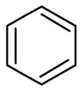
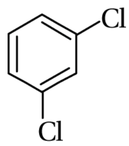
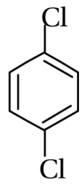
(B) Chloronitrito-N-ammineplatinum (II)

(C) Diamminechloridonitrito-N-platinum (II)

(D) Diamminechloronitrito-N-platinate (II)

(28) Identify the compound Y in the following reaction.



(A)  (B)  (C)  (D) 

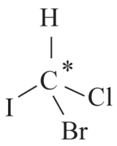
(29) Which reagent will you use for the following reaction ?

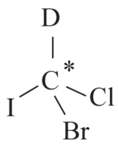
$CH_3CH_2CH_2CH_3 \rightarrow CH_3CH_2CH_2CH_2CH_2Cl \rightarrow CH_3CH_2CHClCH_3$

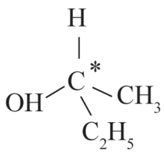
(A) Cl_2 / UV light (B) NaCl + H_2SO_4

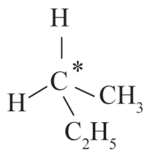
(C) Cl_2 gas in dark (D) Cl_2 gas in the presence of iron in dark

(30) In which of the following molecules carbon atom marked with asteris (*) is


(i)

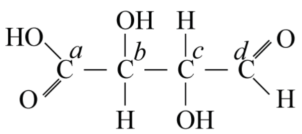

(ii)


(iii)


(iv)

(A) (i), (ii), (iii), (iv) (B) (i), (ii), (iii) (C) (ii), (iii), (iv) (D) (i), (iii), (iv)

(31) Which of the carbon atoms present in the molecule given below are asymmetric ?



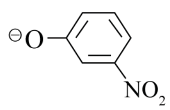
(A) a, b, c, d (B) b, c (C) a, d (D) a, b, c

(32) IUPAC name of m-cresol is.

(A) 3-methylphenol (B) 3-chlorophenol

(C) 3-methoxyphenol (D) benzene-1,3-diol

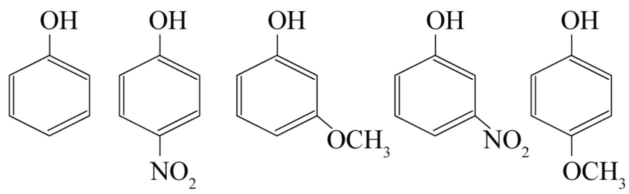
(33) Which of the following species can act as the strongest base?

(A) $^{\ominus}OH$ (B) $^{\ominus}OR$ (C) $^{\ominus}OC_6H_5$ (D) 

(34) Phenol is less acidic than.

(A) ethanol (B) o-nitrophenol (C) o-methylphenol (D) o-methoxyphenol

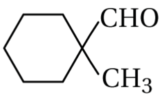
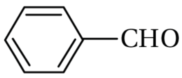
(35) Mark the correct order of decreasing acid strength of the following.

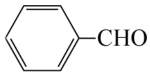


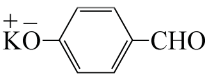
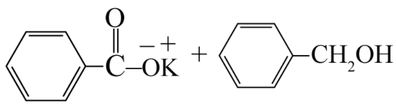
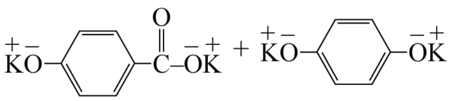
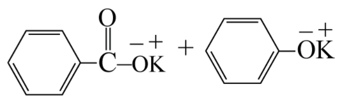
I II III IV V

(A) (V) > (IV) > (II) > (I) > (III)
(B) (II) > (IV) > (I) > (III) > (V)
(C) (IV) > (V) > (III) > (II) > (I)
(D) (V) > (IV) > (III) > (II) > (I)

(36) Cannizzaro's reaction is not given by....

(A) 
(B) 
(C) HCHO
(D) CH_3CHO

(37) Which product is formed when the compound  is treated with concentrated aqueous KOH solution ?

(A) 
(B) 
(C) 
(D) 

(38) Which of the following compounds will give butanone on oxidation with alkaline $KMnO_4$ solution ?

(A) Butan-1-ol
(B) Butan-2-ol
(C) Both (a) and (b)
(D) None of these

(39) In Clemmensen reduction carbonyl compounds is treated with

(A) zinc amalgam + HCl
(B) sodium amalgam + HCl
(C) zinc aalgam + nitric acid
(D) sodium amalgam + HNO_3

(40) Compounds A and C in the following reaction are ....

$$CH_3CHO \xrightarrow[\text{(ii) } H_2O]{\text{(i) } CH_3MgBr} (A) \xrightarrow[\Delta]{H_2SO_4} (B) \xrightarrow[\text{oxidation}]{\text{Hydroboration}} (C)$$

(A) identical
(B) positional isomers
(C) functional isomers
(D) optical isomers

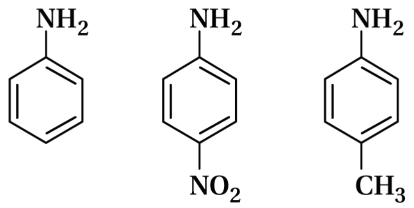
(41) The best reagent for converting-2-phenylpro-panamide into 1-phenylethana-mine is.

(A) excess H_2
(B) NaOH / Br_2
(C) $NaBH_4$/methanol
(D) $LiAlH_4$ / ether

(42) Hoffmann bromamide degradation reaction is shown by.

(A) $ArNH_2$ (B) $ArCONH_2$ (C) $ArNO_2$ (D) $ArCH_2NH_2$

(43) The correct increasing order of basic strength for the following compounds is



(I) (II) (III)

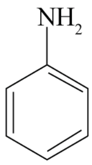
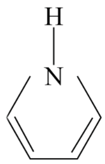
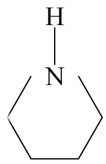
(A) II < III < I (B) III < I < II (C) III < II < I (D) II < I < III

(44) The reaciont $Ar\overset{+}{N}_2Cl^- \xrightarrow{Cu/HCl} ArCl + N_2 + CuCl$ is named as_______.

(A) Sandmeyer reaction (B) Gatterman reaction

(C) Claisen reaction (D) Carbylamine reaction

(45) Among the following amines the strongest Bronsted base is

(A)  (B) NH_3 (C)  (D) 

(46) Surcrose (cane sugar) is a disaccharide. One molecule of sucrose on hydrolysis gives.

(A) 2 molecules of glucose

(B) 2 molecules of glucose + 1 molecule of fructose

(C) 1 molecule of glucose + 1 molecule of fructose

(D) 2 molecules of fructose.

(47) Which of the following acids is a vitamin ?

(A) Aspartic acid (B) Ascorbic acid (C) Adipic acid (D) Saccharic acid

(48) DNA and RNA contain four bases each. Which of the following bases is not present in RNA ?

(A) Adenine (B) Uracil (C) Thymine (D) Cytosine

(49) Which of the following B group vitamins can be stored in our body ?

(A) Vitamin B_1 (B) Vitamin B_2 (C) Vitamin B_6 (D) Vitamin B_{12}

(50) Nucleic acids are the polymers of

(A) nucleosides (B) nucleotides (C) bases sugars

PART – B

SECTION - A

- **Answer questions. Each question 2 marks. (Any 8) (16)**

(51) A 5% solution (by mass) of cane sugar in water has freezing point of 271 K. Calculate the freezing point of 5% glucose in water if freezing point of pure water is 273.15 K.

(52) Explain elctrolysis of sulphuric acid.

(53) The cell in which the following reaction occures :
$2Fe^{3+}(aq) + 2I^{-}(aq) \rightarrow 2Fe^{2+}(aq) + I_2(s)$ had $E^0_{cell} = 0.236$ V at 298 K. Calculate the standard Gibbs energy and the equilibrium constant of the cell reaction.

(54) Identify the reaction order from each of the following rate constants.
(i) $k = 2.3 \times 10^{-5}$ L mol^{-1} s^{-1}
(ii) $k = 3 \times 10^{-4}$ s^{-1}

(55) Write short note on alloys.

(56) What are metal carbonyls ? Explain with examples.

(57) Explain β – Elimination reaction with example

(58) What is organo metallic compound ? Explain grignard reagent.

(59) Give reason: carboxylic acids are more acidic than alcohols and phenols.

(60) Describe the following : Cannizzaro reaction

(61) Write short note on : Gabriel phthalimide synthesis.

(62) Explain the Zwitter ion formation in an amino acid compounds.

SECTION - B

- **Answer questions. Each question 3 marks. (Any 6) (18)**

(63) Explain kohlrausch Law with example.

(64) Write a note on corrosion of iron.

(65) The rate constant for the decomposition of hydrocarbons is 2.418×10^{-5} s^{-1} 546 K. If the energy of activation is 179.9 kJ/mol. What will be the value of pre-exponntial factor.

(66) $K_2Cr_2O_7$ from chromite ore.

(67) $[NiCl_4]^{2-}$ is paramagnetic while $[Ni(CO)]_4$ is diamagnetic though both are tetrahedral. Why ?

(68) Write the mechanism of the reaction of HI with methoxymethane.

(69) Describe the following : Cross aldol condensation

(70) Write structures of different isomers corresponding to the molecular formula, C_3H_9N. Write IUPAC names of the isomers which will liberate nitrogen gas on treatment with nitrous acid.

(71) Note on : Cyclic structure of glucose.

SECTION - C

- **Answer questions. Each question 4 marks. (Any 4)** **(16)**

(72) What is Osmosis ? What is Osmotic Pressure ? Derive it's formula.

(73) What is meant by positive and negative deviations from Raoult's law and how is the sign of $\Delta_{mix}H$ related to positive and negative deviations from Raoult's law ?

(74) The rate constants of a reaction at 500 K and 700 K are 0.02 s^{-1} and 0.07 s^{-1} respectively. Calculate the values of E_A and A.

(75) Indicate the types of isomerism exhibited by the following complexes and draw the structures for these isomers :

(i) $K[Cr(H_2O)_2(C_2O_4)_2]$ (ii) $[Co(en)_3]Cl_3$

(iii) $[Co(NH_3)_5(NO_2)](NO_3)_2$ (iv) $[Pt(NH_3)(H_2O)Cl_2]$

(76) Explain Halogenation, Nitration, Sulphonation and Friedel crafts reaction of chlorobenzene.

(77) Write the equations involved in the following reactions :

(i) Reimer - Tiemann reaction (ii) Kolbe's reaction

Best of Luck

Std : 12th (Science)

Subject : Chemistry

PAPER SET - 3

Marks : 100

PART - A (MCQ)

- **There are 50 objective type (M.C.Q.) questions in Part-A and all questions are compulsory.**

(1) Under identical condition which solution has highest osmetic pressure ?

(A) 1 M $BaCl_2$ (B) 1 M NaCl (C) 1 M $FeCl_3$ (D) 1 M glucose

(2) Which of the following aqueous solution has highest boiling point ?

(A) 0.1M KNO3 (B) 0.1M Urea

(C) 0.1M $K_4[Fe(CN)_4]$ (D) 0.1M NH_4NO_3

(3) What is the osmotic pressure (p) of 0.02M solution of NaCl. ?

(A) 0.01 RT (B) 0.02 RT (C) 0.04 RT (D) 0.002 RT

(4) What is the normality of aqueous solution of H_2SO_4 having pH = 1.

(A) 0.1 N (B) 0.05 N (C) 1 N (D) 0.5 N

(5) Which solution is isotonic with 6% w/v aqueous solution of urea ?

[Molar mass of Urea = 60 gm. mol^{-1}]

(A) 0.1 M NaCl (B) 0.5 M NaCl (C) 0.25 M NaCl (D) 1 M NaCl

(6) How much electricity in terms of Faraday is required to reduced 2 mol of MnO^-_4 into Mn^{2+} ?

(A) 10 (B) 5 (C) 3 (D) 6

(7) How much electricity is required in Faraday to produce 40.0 g of Al from molten Al_2O_3 ? (Atomic mass of Al = 27 g mol^{-1})

(A) 4.44 (B) 2.96 (C) 0.225 (D) 1.48

(8) $\dot{\wedge}_{m(HAc)}$ is equal to_________.

(A) $\dot{\wedge}_{m(KCl)} + \dot{\wedge}_{m(KAc)} - \dot{\wedge}_{m(HCl)}$ (B) $\dot{\wedge}_{m(KCl)} + \dot{\wedge}_{m(NaAc)} - \dot{\wedge}_{m(NaCl)}$

(C) $\dot{\wedge}_{m(ACH)} + \dot{\wedge}_{m(KAC)} - \dot{\wedge}_{m(NaAC)}$ (D) $\dot{\wedge}_{m(KCl)} + \dot{\wedge}_{m(NaAc)} - \dot{\wedge}_{m(NaCl)}$

(9) If l = length, R = Resistance and A = Area of cross section, then

(A) $R \propto \frac{A}{Al}$ (B) $R \propto \frac{A}{l}$ (C) $R \propto \frac{l}{A}$ (D) $R \propto lA$

(10) Which of the following is correct as a Nearest equation for the given electrochemical cell ?

$Mg_{(s)} \mid Mg^{2+}_{(aq)}(0.1\,M) \mid\mid Cl^{-}_{(aq)}(0.01\,M) \mid\mid Cl2_{(g)(1\,bar)} \mid Pt_{(s)}$

(A) $E_{cell} = E^0_{cell} - \frac{0.059}{2} \log \frac{[Cl^-]^2}{[Mg^{2+}]}$ (B) $E_{cell} = E^0_{cell} - \frac{0.059}{2} \log \frac{[Mg^{2+}]}{[Cl^-]^2}$

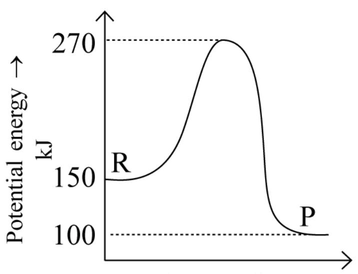
(C) $E_{cell} = E^0_{cell} - \frac{0.059}{2} \log \frac{1}{[Mg^{2+}][Cl^-]^2}$ (D) $E_{cell} = E^0_{cell} - \frac{0.059}{2} \log [Mg^{2+}][Cl^-]^2$

(11) Which substance is reduced in dry cell ?

(A) NH_4^+ (B) graphite (C) MnO_2 (D) Zn

(12) For R $\rightarrow$ P reaction, following graph in given

What will be enthalpy change for the given reaction ?



Reaction coordinate $\rightarrow$

(A) 120 kJ (B) 50 kJ (C) – 50 kJ (D) 170 kJ

(13) What is the slope of graph ln k $\rightarrow \frac{1}{T}$?

(A) $\frac{-2.303\,Ea}{R}$ (B) $-\frac{Ea}{R}$ (D) $\frac{2.303R}{Ea}$ $-$

(C) E_a

(14) For a reaction, K = 4.5 × 10^{-4}L $mol^{-1}s^{-1}$. What is order of reaction ?

(A) Zero (B) Second (C) First (D) Third

(15) Instantaneous rate of reaction for the reaction 3A + 2B $\rightarrow$ 5C is________.

(A) $-\frac{1}{3}\frac{d[A]}{dt} = -\frac{1}{2}\frac{d[B]}{dt} = +\frac{1}{2}\frac{d[C]}{dt}$ (B) $+\frac{1}{3}\frac{d[A]}{dt} = -\frac{1}{2}\frac{d[B]}{dt} = +\frac{1}{5}\frac{d[C]}{dt}$

(C) $\frac{1}{3}\frac{d[A]}{dt} - \frac{1}{2}\frac{d[B]}{dt} + \frac{1}{5}\frac{d[C]}{dt}$ (D) $\frac{1}{3}\frac{d[A]}{dt}$ $\frac{1}{2}\frac{d[B]}{dt}$ $\frac{1}{5}\frac{d[C]}{dt}$

(16) Reaction $3ClO^- \rightarrow ClO_3^- + 2Cl^-$ occurs in following two steps :

(i) $ClO^- + ClO^- \xrightarrow{K_1} ClO_2^- + Cl^-$ (Slow step)

(ii) ClO ClO K_2 ClO_3 Cl (Fast step)

then the rate of given reaction =___________.

(A) $K_1[ClO^-]$ (B) $K_1[ClO^-]^2$ (C) $K_2[ClO_2^-][ClO^-]$ (D) $K_2[ClO^1]^3$

(17) Which compound's aqueous solution is colourless ?

(A) $CuSO_4$ (B) $MnSO_4$ (C) $NiSO_4$ (D) $ZnSO_4$

(18) Colour of $K2MnO_4$ is________.

(A) Green (B) Violet (C) Blue (D) Red

(19) Which of the following is amphoteric oxide ?

Mn_2O_7, CrO_3, Cr_2O_3, CrO, V_2O_5, V_2O_4

(A) Mn_2O_7, CrO (B) CrO_3, V_2O_4 (C) V_2O_5, Cr_2O_3 (D) Cr_2O_3, Mn_2O_7

(20) Which product is obtained during reaction of MnO_4^- with I^- in faintly alkaline condition ?

(A) I_2 (B) IO_3^- (C) IO^- (D) IO_4^-

(21) In which of the following ions, d-d transition is not possible ?

(A) Ti^{4+} (B) Cr^{3+} (C) Mn^{2+} (D) Cu^{2+}

(22) Number of possible isomers for $[Cr(H_2O)_2(C_2O_4)_2]^-$ are__________.

(A) 3 (B) 4 (C) 2 (D) 6

(23) Which is correct relation for high spin complex ?

(A) $\Delta_0 > P$ (B) $\Delta_0 < P$ (C) $\Delta_0 = P$ (D $\Delta_0 \geq P$

(24) Molecular formula of Tetraamineaquachlorido cobalt (III) chloride is________.

(A) $[Co(NH_3)_4(H_2O)]Cl_3$ (B) $[Co(NH_3)_4(H_2O)Cl]Cl_2$

(C) $[Co(NH_3)_4(H_2O)Cl]Cl_3$ (D) $[Co(NH_3)_4(H_2O)Cl]_3Cl_2$

(25) Which transition of electron will be observed in the following, when Ti^{3+} ion having complex absorbs visible light of certain wavelength ?

(A) $t_{2g}^1 e_g^1 \rightarrow t_{2g}^1 e_g^2$ (B) $t_{2g}^2 e_g^0 \rightarrow t_{2g}^1 e_g^1$ (C) $t_{2g}^0 e_g^1 \rightarrow t_{2g}^1 e_g^0$ (D) $t_{2g}^1 e_g^0 \rightarrow t_{2g}^2 e_g^1$

(26) Which type of Isomerism in isomers $[Co(NH_3)_5(SO_4)]$ Brand $[Co(NH_3)_5Br]SO_4$?

(A) Linkage (B) Ionisation (C) Coordination (D) Solvate

(27) In which of the following ions, d-d transition is not possible ?

(A) Ti^{4+} (B) Cr^{3+} (C) Mn^{2+} (D) Cu^{2+}

(28) Which compound will give yellow precipitate on reaction with sodium hypoiodite ?

(A) isobutal alcohol (B) tert butal alcohol

(C) n–' butal alcohol (D) sec- butal alcohol

(29) Which compound has highest value of pka ?

(A) p-cresol (B) phenol (C) o-nitrophenol (D) m-nitrophenol

(30) Which product is obtained between reaction of CH3ONa and (CH3)3 CBr ?

(A) Only Alkene (B) Only Ether

(C) Both alkene and Ether (D) Alcohol

(31) Which of the following processes does not give benzene as a product ?

(A) C_6H_5COONa + Sodalime $\xrightarrow{\Delta}$ (B) $C_6H_5N_2^+Cl^- + H_3PO_2 + H_2O \longrightarrow$

(C) $C_6H_5OH + Zn \xrightarrow{\Delta}$ (D) $C_6H_5OH + H_2CrO_4 \xrightarrow{[O]^2}$

(32) What is the correct order of acidity of compound (I), (II) and (III) ?

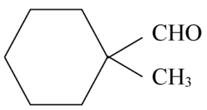
(I) 4 - Nitro benzonic acid

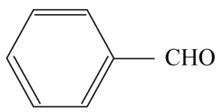
(II) 4 - Methoxy benzoic acid

(III) Benzoic acid

(A) I > II > III (B) I > III > II (C) I < II < III (D) I < III < II

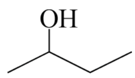
(33) Which of the following compound does not give cannizzaro reaction ?

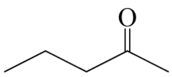
(A) 

(B) 

(C) HCHO (D) CH_3CHO

(34) Which of the followingg compounds will not give 'Iodoform' by reaction with ''Sodium hypo iodide'' ?

(A) $CH_3 - CHO$ (B) 

(C) 

(D) $CH_3–CH_2–CO–CH_2–CH_3$

(35) Which of the following acid has highest pKa value?

(A) $C_6H_5CH_2COOH$ (B) O_2NCH_2COOH (C) FCH_2COOH (D) $NCCH_2COOH$

(36) $C_6H_5CH_2MgBr \xrightarrow[\text{(ii)} H_3O^+]{\text{(i)} CO_2/\text{Ether}}$ 'X' $\xrightarrow[\Delta]{NaOH + CaO}$ 'Y' What is the final product in this reaction ?

(A) $C_6H_5CH_2OH$ (B) $C_6H_5CH_2CH_3$ (C) C_6H_6 (D) $C_6H_5CH_3$

(37) Which of the following is the correct order of acidic strength ?

(A) $CH_3COOH > ClCH_2COOH > Cl_2CHCOOH > Cl_3{\cdot}C{\cdot}COOH$

(B) $Cl_3{\cdot}C{\cdot}COOH > Cl_2CH{\cdot}COOH > Cl{\cdot}CH_2COOH > CH_3COOH$

(C) $CH_3COOH > Cl_3{\cdot}C{\cdot}COOH > Cl_2CH{\cdot}COOH > Cl{\cdot}CH_2COOH$

(D) $CH_3COOH > ClCH_2COOH > Cl_2{\cdot}CH{\cdot}COOH > Cl_3{\cdot}C{\cdot}COOH$

(38) How many π - electrons are present in phthalimide ?

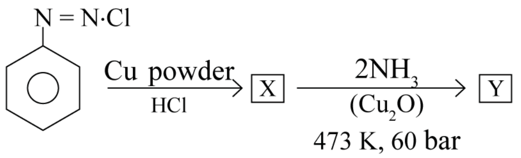
(A) 6 (B) 12 (C) 5 (D) 10

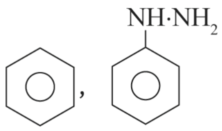
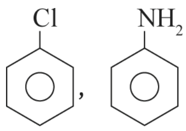
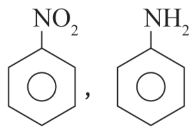
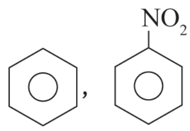
(39) Identify the compound 'C' from following reaction ?

$$CH_3COOH \xrightarrow[\Delta]{NH_3} A \xrightarrow{Br_2+NaOH} B \xrightarrow[HCl]{NaNO_2} C$$

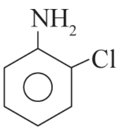
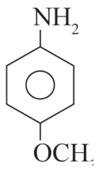
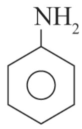
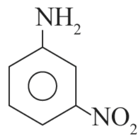
(A) $CH_3-CH_2N_2^+Cl^-$ (B) CH_3-CH_2OH (C) CH_3OH (D) $CH_3-CH_2-NH_2$

(40) Identify X and Y in the following reaction.



(A) 
(B) 
(C) 
(D) 

(41) Which of the following compound is the most basic ?

(A) 
(B) 
(C) 
(D) 

(42) The main product of which of the following reactions gives tertiary sulphoamide with benzene sulphonyl chloride ?

(A) $C_6H_5Cl + 2NH_3 \xrightarrow[\substack{443\ K \\ 60bar}]{Cu_2O}$ (B) $CH_3CH_2NO_2 \xrightarrow{LiAlH_4}$

(C) $CH_3CH_2NC \xrightarrow{LiAlH_4}$ (D) $CH_3CONH_2 \xrightarrow[\Delta]{Br_2/NaOH}$

(43) Which of the following acid axists as zwitter ion ?

(A) picric acid (B) salicylic acid (C) sulphanilic acid (D) Adipic acid

(44) Compare boiling point of isomeric alkyl amines.

(A) $1° > 2° > 3°$ (B) $1° > 2° < 3°$ (C) $1° < 2° < 3°$ (D) $1° < 2° > 3°$

(45) Which one acts as nonreducing sugar ?

(A) glucose (B) lactose (C) sucrose (D) maltose

(46) Which vitamin must be supplied regularly in diet ?

(A) Vitamin - D (B) Vitamin - K (C) Vitamin - E (D) Vitamin - C

(47) Select proper statement from following true (T) and False (F) Statements.

(i) Pentose sugar + base $\rightarrow$ Nucleotide

(ii) Nucleotide + phosphate $\rightarrow$ Nucleoside

(iii) DNA contains four bases A, G, C and T

(iv) RNA contains four bases A, G, C and U

(A) FTFT (B) FTTT (C) FFTT (D) TTTT

(48) Which one is a purine base ?

(A) Uracil (B) Thymine (C) Cytosine (D) Guanine

(49) What is the chemical name of vitamin B_1 ?

(A) Riboflavin (B) Pyridoxine (C) Thiamine (D) α-tocoferol

(50) What is the chemical name of vitamin B_6 ?

(A) Cyano cobal amine (B) pyri doxine

(C) Thiamine (D) Riboflavin

PART – B

SECTION - A

- **Answer questions. Each question 2 marks. (Any 8)** **(16)**

(51) What is Isotonic, hypertonic and hypotonic Solution ?

(52) H_2S, a toxic gas with rotten egg like smell, is used for the qualitative analysis. If the solubility of H_2S in water at STP is 0.195 m, calculate Henry's law constant.

(53) Explain standard hydrogen electrode. (SHE)

(54) Write down cell reaction of Ni–Cd cel

(55) The decomposition of NH_3 on platinum surface is zero order reaction. What are the rates of N_2 and H_2 if $k = 2.5 \times 10^{-4}$ mol L^{-1} s^{-1} ?

(56) Explain the lanthanoid contraction.

(57) Write down limitation of Crystal field.

(58) Write down difference between SN^1 and SN^2 reaction.

(59) Short note: Lucas test

(60) Write on note on Cannizzaro reaction.

(61) Short note : Sulphonation of Aniline.

(62) Short note of : Sucrose.

SECTION - B

- **Answer questions. Each question 3 marks. (Any 6)** **(18)**

(63) 45 g of ethylene glycol ($C_2H_6O_2$) is mixed with 600 g of water. Calculate (a) the freezing point depression and (b) the freezing point of the solution.

(64) Resistance of a conductivity cell filled with 0.1 mol L^{-1} KCl solution is 100Ω. If the resistance of the same cell when filled with 0.02 mol L^{-1} KCl solution is 520Ω. calculate the conductivity and molar conductivity of 0.02 mol L^{-1} KCl solution. The conductivity of 0.1 mol L^{-1} KCl solution is 1.29 S/m.

(65) The half-life for radioactive decay of ^{14}C is 5730 years. An archaeological artifact containing wood had only 80% of the ^{14}C found in a living tree. Estimate the age of the sample.

(66) Explain crystal field splitting in octahedral Complex

(67) Give Fittig, Wurtz – Fittig and Grignard reaction of haloarene.

(68) Write only chemical reactions to obtain 1°, 2° and 3° alcohols from aldehyde and ketone compounds..

(69) Write a short note : Aldol condensation.

(70) Short note on : Sandmeyer reaction and Gattermann reaction.

(71) Explain the structure of nucleic acid compounds.

SECTION - C

- **Answer questions. Each question 4 marks. (Any 4)** **(16)**

(72) 19.5 g of CH_2FCOOH is dissolved in 500 g of water. The depression in the freezing point of water observed is 1.0° C. Calculate the van't Hoff factor and dissociation constant of fluoroacetic acid.

(73) The rate constant for the first order decomposition of H_2O_2 is given by the following equation : $\log k = 14.34 - 1.25 \times 10^4 \, K/T$

Calculate E_a for this reaction and at what temperature will its half-period be 256 minutes ?

(74) Describe the preparation of potassium permanganate. How does the acidified permanganate solution react with (i) iron(II) ions (ii) SO_2 and (iii) oxalic acid ? Write the ionic equations for the reactions.

(75) Explain Werner's Theory for Co-ordination Compound.

(76) What is stereochemistry ? Explain plane polarised light, optical activity and optical isomers.

(77) Explain the effect of substituents on the acidity of carboxylic acids.

Best of Luck

Std : 12^{th} (Science)

Subject : Physics

PAPER SET - 1

Marks : 100

PART - A (MCQ)

- **There are 50 objective type (M.C.Q.) questions in Part-A and all questions are compulsory.**

(1) If 10^9 electrons move out of a body to another body every second. how much time (in years) is required to get a

(A) 100 (B) 200 (C) 250 (D) 300

(2) An electric dipole with dipole moment 4×10^{-9} C m is aligned at 30° with the direction of a uniform electric field of magnitude 5×10^4 N C^{-1} Calculate the magnitude of the torque acting on the dipole.

(A) 10^{-5} Nm (B) 10^{-2} Nm (C) 10^{-4} Nm (D) 10^{-3} Nm

(3) A point charge of 2.0 μC is at the centre of a cubic Gaussian surface 9.0 cm on edge. What is the net electric flux through the surface?

(A) 4.166×10^6 N m^6C^{-1} (B) 7.24×10^4 N m^2C^{-1}

(C) 8.34×10^5 N m^2C^{-1} (D) 2.26×10^5 N m^2C^{-1}

(4) A regular hexagon of side 10 cm has a charge 5μC at each of its vertices. Calculate the potential at the centre of the hexagon.

(A) 9.2×10^6 V (B) 7.4×10^5 V (C) 4.2×10^5 V (D) 2.7×10^6 V

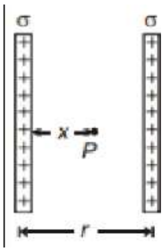
(5) The capacity of a parallel plate condenser is 5μF . When a glass plate is placed between the plates of the conductor, its potential becomes $1/8^{th}$ of the original value. The value of dielectric constant will be

(A) 1.6 (B) 5 (C) 8 (D) 40

(6) A capacitor when filled with a dielectric K = 3 has charge Q_0, voltage V_0 and field E_0. If the dielectric is replaced with another one having K = 9 the new values of charge, voltage and field will be respectively

(A) $3Q_0, 3V_0, 3E_0$ (B) $Q_0, 3V_0, 3E_0$ (C) $Q_0, \frac{V_0}{3}, 3E_0$ (D) $Q_0, \frac{V_0}{3}, \frac{E_0}{3}$

(7) For two infinitely long charged parallel sheets, the electric field at P will be



(A) $\frac{\sigma}{2x} - \frac{\sigma}{2(r-x)}$ (B) $\frac{\sigma}{2\varepsilon_0 x} + \frac{\sigma}{2\pi(r-x)\varepsilon_0}$

(C) $\frac{\sigma}{\varepsilon_0}$ (D) 0

(8) The storage battery of a car has an emf of 12 V. If the internal resistance of the battery is 0.4 Ω, what is the maximum current (in A) that can be drawn from the battery?

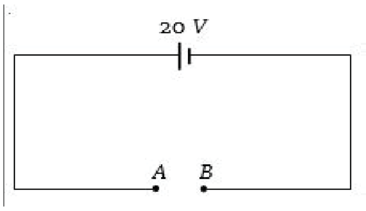
(A) 18 (B) 25 (C) 30 (D) 36

(9) A steady current flows in a metallic conductor of non-uniform cross-section. The quantity/ quantities constant along the length of the conductor is/are

(A) Current, electric field and drift speed (B) Drift speed only

(C) Current and drift speed (D) Current only

(10) In the shown circuit, what is the potential difference across A and B V



(A) 50 (B) 45 (C) 30 (D) 20

(11) The internal resistance of a cell is the resistance of

(A) Electrodes of the cell (B) Vessel of the cell

(C) Electrolyte used in the cell (D) Material used in the cell

(12) A long straight wire carries a current of 35 A. What is the magnitude of the field B at a point 20 cm from the wire?

(A) 4.5×10^{-6}T (B) 2.5×10^{-4}T (C) 3.5×10^{-5}T (D) 7.5×10^{-5}T

(13) What is the magnitude of magnetic force per unit length (in N m^{-1}) on a wire carrying a current of 8 A and making an angle of 30o with the direction of a uniform magnetic field of 0.15 T ?

(A) 0.8 (B) 0.6 (C) 1.2 (D) 1.6

(14) A short bar magnet placed with its axis at 30° with a uniform external magnetic field of 0.25 T experiences a torque of magnitude equal to 4.5×10^{-2} J. What is the magnitude of magnetic moment (in J T^{-1}) of the magnet?

(A) 0.64 (B) 0.36 (C) 1.32 (D) 0.86

(15) Lenz's law is consequence of the law of conservation of

(A) Charge (B) Momentum (C) Mass (D) Energy

(16) The magnetic flux through a circuit of resistance R changes by an amount $\Delta\phi$ in time Δt, Then the total quantity of electric charge Q, which passing during this time through any point of the circuit is given by

(A) $Q = \frac{\Delta\phi}{\Delta t}$ (B) $Q = \frac{\Delta\phi}{\Delta t} \times R$ (C) $Q = -\frac{\Delta\phi}{\Delta t} + R$ (D) $Q = \frac{\Delta\phi}{R}$

(17) A cylindrical bar magnet is kept along the axis of a circular coil. If the magnet is rotated about its axis, then

(A) A current will be induced in a coil

(B) No current will be induced in a coil

(C) Only an e.m.f. will be induced in the coil

(D) An e.m.f. and a current both will be induced in the coil

(18) A magnet is brought towards a coil (i) speedly (ii) slowly then the induced e.m.f. induced charge will be respectively

(A) More in first case / More in first case

(B) More in first case/Equal in both case

(C) Less in first case/More in second case

(D) Less in first case/Equal in both case

(19) The dimensions of magnetic flux are

(A) $MLT^{-2}A^{-2}$ (B) $ML^2T^{-2}A^{-2}$ (C) $ML^2T^{-1}A^{-2}$ (D) $ML^2T^{-2}A^{-1}$

(20) The total charge induced in a conducting loop when it is moved in magnetic field depends on

(A) The rate of change of magnetic flux (B) Initial magnetic flux only

(C) The total change in magnetic flux (D) Final magnetic flux only

(21) A charged 30 μF capacitor is connected to a 27 mH inductor. What is the angular frequency of free oscillations of

(A) 5.63×10^3 rad/s (B) 1.11×10^3 rad/s

(C) 7.25×10^3 rad/s (D) 9.42×10^3 rad/s

(22) An ac voltage is applied to a resistance R and an inductor L in series. If R and the inductive reactance are both equal to 3Ω, the phase difference between the applied voltage and the cureent in the circuit is

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{2}$ (D) Zero

(23) (23) A small signal voltage V (t) = $V_0 \sin^{\omega}t$ t is applied across an ideal capacitor C

(A) Over a full cycle the capacitor C does not consume any energy from the voltage source.

(B) Current I(t) is in phase with voltage V (t).

(C) Current I(t) leads voltage V (t) by180o.

(D) Current I(t) lags voltage V (t) by 90o.

(24) Which of the following units denotes the dimension $\frac{ML^2}{Q^2}$ where Q denotes the electric charge ?

(A) weber (Wb) (B) Wb/m^2 (C) henry (H) (D) H/m^2

(25) Wavelength of light of frequency 100 Hz

(A) 2×10^6m (B) 3×10^6m (C) 4×10^6m (D) 5×10^6m

(26) Pick out the longest wavelength from the following types of radiations

(A) Blue light (B) γ – rays (C) X – rays (D) Red light

(27) If $\vec{E}$ and $\vec{B}$ are the electric and magnetic field vectors of E.M. waves then the direction of propagation of E.M.

(A) $\vec{E}$ (B) $\vec{B}$ (C) $\vec{E} \times \vec{B}$ (D) None of these

(28) The refractive index of a certain glass is 1.5 for light whose wavelength in vacuum is 6000 Å. The wavelength of this light when it passes through glass is .... Å

(A) 4000 (B) 6000 (C) 9000 (D) 15000

(29) A plane glass slab is kept over various coloured letters, the letter which appears least raised is

(A) Blue (B) Violet (C) Green (D) Red

(30) The phenomenon utilised in an optical fibre is

(A) Refraction (B) Interference

(C) Polarization (D) Total internal reflection

(31) If the angle of a prism is 60° and angle of minimum deviation is 40°, then the angle of refraction will be

(A) 30 (B) 20 (C) 3 (D) 4

(32) In the interference pattern, energy is

(A) Created at the position of maxima (B) Destroyed at the position of minima

(C) Conserved but is redistributed (D) None of the above

(33) In Young's double slits experiment, the position of 5^{th} bright fringe from the central maximum is 5 cm. The distance between slits and screen is 1 m and wavelength of used monochromatic light is 600 nm. The separation

(A) 60 (B) 48 (C) 12 (D) 36

(34) The time taken by a photoelectron to come Out after the photon strikes is approximately

(A) 10^{-4}s (B) 10^{-10}s (C) 10^{-16} (D) 10^{-1}s

(35) In the Davisson and Germer experiment, the velocity of electrons emitted from the electron gun can be increased by

(A) increasing the potential difference between the anode and filament

(B) decreasing the potential difference between the anode and filament

(C) increasing the filament current

(D) decreasing the filament current

(36) A radiation of energy 'E' falls normally on a perfectly reflecting surface. The momentum transferred to the surface is (C = Velocity of light)

(A) $\frac{E}{C}$ (B) $\frac{2E}{C}$ (C) $\frac{2E}{C^2}$ (D) $\frac{E}{C^2}$

(37) A 15.0 eV photon collides with and ionizes a hydrogen atom. If the atom was originally in the ground state (ionization potential = 13.6 eV), what is the kinetic energy of the ejected electron ? .... eV

(A) 1.4 (B) 13.6 (C) 15 (D) 28.6

(38) The ratio of the energies of the hydrogen atom in its first to second excited state is

(A) 1/4 (B) 4/9 (C) 9/4 (D) 4

(39) Which of the following transitions in a hydrogen atom emits photon of the highest frequency

(A) n = 2 to n = 1 (B) n = 1 to n = 2 (C) n = 2 to n = 6 (D) n = 6 to n = 2

(40) The energy of electron in first excited state of H–atom is –3.4 eV. its kinetic energy is ...... eV

(A) –3.4 (B) +3.4 (C) –6.8 (D) 6.8

(41) The total energy of an electron in the nth stationary orbit of the hydrogen atom can be obtained by

(A) $E_n = -13.6 \times n^2$ eV (B) $E_n = \frac{13.6}{n^2}$ eV

(C) $E_n = -\frac{13.6}{n^2}$ eV (D) $E_n = -\frac{1.36}{n^2}$ eV

(42) The neutron was discovered by

(A) Marie Curie (B) Pierre Curie (C) James Chadwick (D) Rutherford

(43) The mass number of a nucleus is

(A) Always less than its atomic number

(B) Always more than its atomic number

(C) Always equal to its atomic number

(D) Sometimes more than and sometimes equal to its atomic number

(44) The mass of an α - particle is

(A) Less than the sum of masses of two protons and two neutrons

(B) Equal to mass of four protons

(C) Equal to mass of four neutrons .

(D) Equal to sum of masses of two protons and two neutrons

(45) In nuclear reactions, we have the conservation of

(A) Mass only (B) Energy only

(C) Momentum only (D) Mass, energy and momentum

(46) Suppose a pure Si crystal has 5×10^{28} atoms/m^3. It is doped by 1 ppm concentration of pentavalent As. Calculate the number of electrons and holes. Given that $n_i = 1.5 \times 10^{16} m^{-3}$

(A) $4.5 \times 10^6 m^{-3}$ (B) $4.5 \times 10^9 m^{-3}$ (C) $2.7 \times 10^{-9} m^{-3}$ (D) $3.8 \times 10^6 m^{-3}$

(47) In an n–type silicon, which of the following statement is true

(A) Electrons are majority carriers and trivalent atoms are the dopants.

(B) Electrons are minority carriers and pentavalent atoms are the dopants

(C) Holes are minority carriers and pentavalent atoms are the dopants.

(D) Holes are majority carriers and trivalent atoms are the dopants.

(48) In an unbiased p – n junction, holes diffuse from the p–region to n–region because

(A) they move across the junction by the potential difference.

(B) hole concentration in pâˆ’region is more as compared to n–region.

(C) free electrons in the n-region attract them.

(D) All the above

(49) When a forward bias is applied to a p–n junction, it

(A) Raises the potential barrier (B) Reduce the majority carrier current to zero

(C) Lower the potential barrier (D) None of the above

(50) (50) In an unbiased n – p junction electrons diffuse from n–region to p–region because

(A) holes in p–region attract them

(B) electrons travel across the junction due to potential difference

(C) only electrons move from n to p region and not the vice-versa

● (D) electron concentration in n–region is more compared to that in p–region

PART – B

SECTION - A

μ

Answer questions. Each question 2 marks. (Any 8) (16)

(1) Give characteristics of electric field lines.

(2) The electrostatic force on a small sphere of charge 0.4 C due to another small sphere of charge –0.8 mC in air is 0.2 N.

(a) What is the distance between the two spheres ?

(b) What is the force on the second sphere due to the first ?

(3) Write limitations of Ohm's law.

(4) A short bar magnet placed with its axis at 30° with a uniform external magnetic field of 0.25 T experiences a torque of magnitude equal to 4.5×10^{-2} J. What is the magnitude

of magnetic moment of the magnet ?

(5) Define inductance, give its units and write factors on which its value depends.

(6) A 100 Ω resistor is connected to a 220 V, 50 Hz ac supply.

(a) What is the rms value of current in the circuit ?

(b) What is the net power consumed over a full cycle ?

(7) What is linear magnification ? Obtain equation of linear magnification for concave mirror.

(8) In a Young's double-slit experiment, the slits are separated by 0.28 mm and the screen is placed 1.4 m away. The distance between the central bright fringe and the fourth bright fringe is measured to be 1.2 cm. Determine the wavelength of light used in the experiment.

(9) Explain emission line spectra and absorption spectra.

(10) The work function of caesium is 2.14 eV. Find (a) the threshold frequency for caesium, and (b) the wavelength of the incident light if the photocurrent is brought to zero by a stopping potential of 0.60 V.

(11) Draw conclusion from the main two features of the graph of atomic mass number against the binding energy per nucleon.

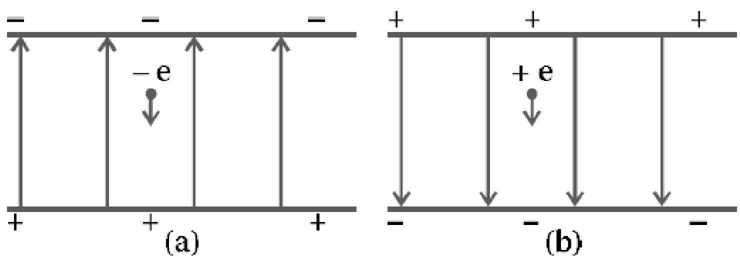
(12) Why is it required to add impurity in pure semiconductor ? Mention it's condition and explain what are impure semiconductors ?

SECTION - B

- **Answer questions. Each question 3 marks. (Any 6)** **(18)**

(13) An electron falls through a distance of 1.5 cm in a uniform electric field of magnitude 2.0×10^4 NC^{-1} figure

(a) The direction of the field is reversed keeping its magnitude unchanged and a proton falls through the same distance figure (b). Compute the time of fall in each case. Contrast the situation with that of 'free fall under gravity'.



(14) What is series connection of cell ? Derive equation of equivalent emf of two cells with emf ε_1 and ε_2 connected in series.

(15) Discuss similarities and differences of Biot-Savart law with Coulomb's law.

(16) A long solenoid with 15 turns per cm has a small loop of area 2.0 cm^2 placed inside the solenoid normal to its axis. If the current carried by the solenoid changes steadily from 2.0 A to 4.0 A in 0.1 s, what is the induced emf in the loop while the current is changing ?

(17) A 15.0 μF capacitor is connected to a 220 V, 50 Hz source. Find the capacitive reactance and the current (rms and peak) in the circuit. If the frequency is doubled, what happens to the capacitive reactance and the current ?

(18) Light from a point source in air falls on a spherical glass surface (n = 1.5 and radius of curvature = 20 cm). The distance of the light source from the glass surface is 100 cm. At what position the image is formed ?

(19) In accordance with the Bohr's model, find the quantum number that characterises the earth's revolution around the sun in an orbit of radius 1.5×10^{11} m with orbital speed 3×10^4 m/s. (Mass of earth = 6.0×10^{24} kg.)

(20) Describe Hallwachâ€™s and Lenard's observation regarding photoelectric effect in short.

(21) Explain the Young's experiment, arrangement and experiment to produce stationary interference pattern.

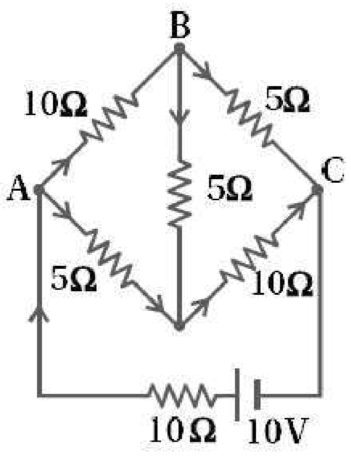
SECTION - C

- **Answer questions. Each question 4 marks. (Any 4)** **(16)**

(22) Discuss the power in AC circuit with only an inductor.

(23) Obtain an equation of current for AC voltage applied to an inductor and draw a graph of V and I.

(24) Determine the current in each branch of the network shown in figure.



(25) Derive the formula for the electric potential due to an electric dipole at a point from it.

(26) A compound microscope consists of an objective lens of focal length 2.0 cm and an eyepiece of focal length 6.25 cm separated by a distance of 15 cm. How far from the objective should an object be placed in order to obtain the final image at

(a) the least distance of distinct vision (25 cm), and

(b) at infinity ? What is the magnifying power of the microscope in each case ?

(27) The Q value of a nuclear reaction A + b → C + d is defined by

$Q = [\ m_A + m_b - m_C - m_d]c^2$ where the masses refer to the respective nuclei. Determine from the given data the Q-value of the following reactions and state whether the reactions are exothermic or endothermic.

(i) ${}^{1}_{1}H + {}^{3}_{1}H \rightarrow {}^{2}_{1}H + {}^{2}_{1}H$

(ii) ${}^{12}_{6}C + {}^{12}_{6}C \rightarrow {}^{20}_{10}Ne + {}^{4}_{2}He$

Atomic masses are given to be :

$m({}^{2}_{1}H) = 2.014102u$

$m({}^{3}_{1}H) = 3.016049u$

$m({}^{12}_{6}H) = 12.000000u$

$m({}^{20}_{10}Ne) = 19.992439\ u$

(28) Explain the forward characteristics of the p-n junction diode by drawing circuit and graph.

Best of Luck

Std : 12^{th} (Science)

Subject : Physics

PAPER SET - 2

Marks : 100

PART - A (MCQ)

- **There are 50 objective type (M.C.Q.) questions in Part-A and all questions are compulsory.**

(1) A charged particle is released from rest in a region of uniform electric and magnetic fields which are parallel to each other. The particle will move on a

(A) straight line (B) circle (C) helix (D) cycloid

(2) A man runs towards mirror at a speed of 15 m/s.What is the speed of his image.... m/s

(A)7.5 (B) 15 (C) 30 (D) 45

(3) The magnetic susceptibility of an ideal diamagnetic substance is

(A) −1 (B) 0 (C) +1 (D) ∞

(4) If $\vec{E}$ and $\vec{B}$ are the electric and magnetic field vectors of E.M. waves then the direction of propagation of E.M. wave is along the direction of

(A) $\vec{E}$ (B) $\vec{B}$ (C) $\vec{E}$ 2 $\vec{B}$ (D) None of these

(5) When photon of energy 3.8 eV falls on metallic surface of work function 2.8 eV , then the kinetic energy of emitted electrons are........... eV

(A) 1 (B) 6.6 (C) 0 to 1 (D) 2.8

(6) Carbon, silicon and Germanium atoms have four valence electrons each. Their valence and conduction band are separated by energy band gaps represented by $(Eg)_C \cdot (Eg)_{Si}$ and $(Eg)_{Ge}$ respectively. Which one of the following relationship is true in their case

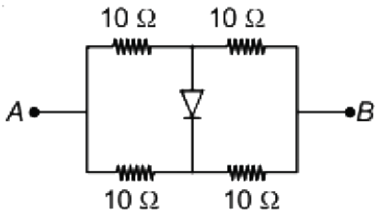
(A) $(E_g)_C > (E_g)_{Si}$ (B) $(E_g)_C = (E_g)_{Si}$ (C) $(E_g)_C < (E_g)_{Ge}$ (D) $(E_g)_C > (E_g)_{Si}$

(7) The electric and the magnetic field, associated with an e.m. wave, propagating along the + z – axis, can be represented by

(A) $\left[\vec{E} = E_0\hat{i}, \vec{B} = B_0\hat{j}\right]$ (B) $\left[\vec{E} = E_0\hat{k}, \vec{B} = B_0\hat{i}\right]$

(C) $\left[\vec{E} = E_0\hat{j}, \vec{B} = B_0\hat{i}\right]$ (D) $\left[\vec{E} = E_0\hat{j}, \vec{B} = B_0\hat{k}\right]$

(8) Four equal resistors, each of resistance 10 Ω, are connected as shown in the circuit diagram. The equivalent resistance between A and B is .... Ω



(A) 5 (B) 10 (C) 20 (D) 40

(9) A concave mirror of focal length f (in air) is immersed in water ($\mu = 4/3$). The focal length of the mirror in water

(A) f (B) $\frac{4}{3}f$ (C) $\frac{3}{4}f$ (D) $\frac{7}{3}f$

(10) As the n (number of orbit) increases, the difference of energy between the consecutive energy levels

(A) Remains the same

(B) Increases

(C) Decreases

(D) Sometimes increases and sometimes decreases

(11) The work function of a metal is 4.2eV, its threshold wavelength will be ....

(A) 4000 (B) 3500 (C) 2955 (D) 2500

(12) The energy of electromagnetic wave in vacuum is given by the relation

(A) $\frac{E^2}{2\varepsilon_0} + \frac{B^2}{2\mu_0}$ (B) $\frac{1}{2}\varepsilon_0 E^2 + \frac{1}{2}\mu_0 B^2$ (C) $\frac{E^2 + B^2}{c}$ (D) $\frac{1}{2}\varepsilon_0 E^2 + \frac{B^2}{2\mu_0}$

(13) A 3.0 cm wire carrying a current of 10 A is placed inside a solenoid perpendicular to its axis. The magnetic field inside the solenoid is given to be 0.27 T . What is the magneticforce on the wire ?

(A) 1.6×10^{-3} N (B) 0.9×10^{-2} N

(C) 8.1×10^{-2} N (D) 0.3×10^{-2} N

(14) If in a photoelectric cell, the wavelength of incident light is changed from 4000Å to 3000Å then change in stopping

(A) 0.66 (B) 1.03 (C) 0.33 (D) 0.49

(15) When light travels from one medium to the other of which the refractive index is different, then which of the

(A) Frequency, wavelength and velocity (B) Frequency and wavelength

(C) Frequency and velocity (D) Wavelength and velocity

(16) The flux linked with a coil at any instant 't' is given by $\phi = 10t^2 - 50t + 250$ The induced emf at t = 3 s is.......... V

(A) 10 (B) 190 (C) −10 (D) 190

(17) The momentum of a photon of energy 1 MeV in kg m/s will be

(A) 5×10^{-22} (B) 0.33×10^{6} (C) 7×10^{-24} (D) 10^{-22}

(18) How many times does the electron go round the first Bohr orbit in a second ?

(A) 6.57×10^{5} (B) 6.57×10^{10} (C) 6.57×10^{13} (D) 6.57×10^{15}

(19) In an intrinsic semiconductor, the density of conduction electron's is 7.07×10^{15} m^{-3}. When it is doped with indium, the density of holes becomes 5×10^{22} m^{-3}. Find the density of conduction electrons in doped semiconductor

(A) Zero (B) $1 \times 10^{9} m^{-3}$

(C) $7 \times 10^{15} m^{-3}$ (D) $5 \times 10^{22} m^{-3}$

(20) The photoelectric work function for a metal surface is 4.125 eV . The cut-off wavelength for this surface is .... Å

(A) 4125 (B) 2062.5 (C) 3000 (D) 6000

(21) If the mass of neutron is 1.7×10^{-27} kg, then the de-Broglie wavelength of neutron of energy 3 eV is ($h = 6.6 \times 10^{-34}$ J · sec)

(A) 1.6×10^{-10}m (B) 1.65×10^{-11}m (C) 1.4×10^{-10}m (D) 1.4×10^{-11}m

(22) In an LCR circuit, the resonating frequency is 500 kHz. If the value of L is doubled and value of C is decreased to $\frac{1}{8}$ times of its initial values, then the new resonating frequency in kHz will be ....

(A) 250 (B) 500 (C) 1000 (D) 2000

(23) A closely wound solenoid 80 cm long has 5 layers of windings of 400 turns each. The diameter of the solenoid is 1.8 cm. If the current carried is 8.0 A, estimate the magnitude of B inside the solenoid near its centre.

(A) 2.5×10^{-2} T (B) 5.5×10^{-2} T (C) 9.3×10^{-2} T (D) 7.4×10^{-3} T

(24) If a photon has velocity c and frequency v, then which of following represents its wavelength

(A) $\frac{hv}{c^2}$ (B) $\frac{hv}{c}$ (C) $\frac{hc}{E}$ (D) hv

(25) Q – value of the decay ${}^{22}_{11}Na \rightarrow {}^{22}_{10}Ne + e_{+} + \upsilon$ is

(A) $\left[m\left({}^{22}_{11}Na\right) - m\left({}^{22}_{10}Ne\right)\right]c^2$ (B) $\left[m\left({}^{22}_{11}Na\right) - m\left({}^{22}_{10}Ne\right) - m_e\right]c^2$

(C) $\left[m\left({}^{22}_{11}Na\right) - m\left({}^{22}_{10}Ne\right) - 2m_e\right]c^2$ (D) $\left[m\left({}^{22}_{11}Na\right) - m\left({}^{22}_{10}Ne\right) - 3m_e\right]c^2$

(26) 4eV is the energy of the incident photon and the work function in 2eV. What is the stopping potential V

(A) 2 (B) 4 (C) 6 (D) $2\sqrt{2}$

(27) In an apparatus, the electric field was found to oscillate with an amplitude of 18V /m. The magnitude of the oscillating magnetic field will be

(A) 4×10^{-6}T (B) 6×10^{-8}T (C) 9×10^{-9}T (D) 11×10^{-11}T

(28) A wire of length 1 m moving with velocity 8 m/s at right angles to a magnetic field of 2 T. The magnitude of induced emf, between the ends of wire will be V

(A) 20 (B) 8 (C) 12 (D) 16

(29) Which of the following is incorrect statement regarding photon

(A) Photon exerts no pressure (B) Photon energy is hv

(C) Photon rest mass is zero (D) None of these

(30) If the magnetic field in a plane electromagnetic wave is given by

$\vec{B} = 3 \times 10^{-8} \sin(1.6 \times 10^{3}x + 48 \times 10^{10}t)\hat{j}$ T

then what will be expression ofr electric field ?

(A) $\vec{E} = \left(9 \sin(1.6 \times 10^{3}x + 48 \times 10^{10}t)\, \hat{k}\ V / m\right)$

(B) $\vec{E} = \left(3.8 \times 10^{-8}(1.6 \times 10^{3}x + 48 \times 10^{10}t)\, \hat{i}\ V / m\right)$

(C) $\vec{E} = \left(60 \sin(1.6 \times 10^{3}x + 48 \times 10^{10}t)\, \hat{k}\ V / m\right)$

(D) $\vec{E} = \left(3 \times 10^{-8} \sin(1.6 \times 10^{3}x + 48 \times 10^{10}t)\, \hat{j}\ V / m\right)$

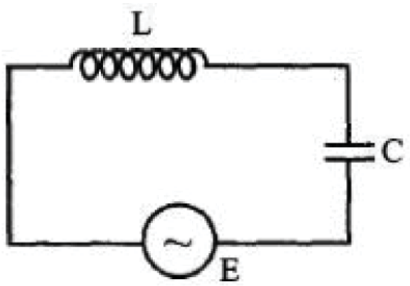
(31) In Young's double slit experiment, the distance between the slits is reduced to half and the distance between the

(A) Will not change (B) Will become half (C) Will be doubled (D) Will become four times

(32) A radio transmitter operates at a frequency of 880 kHz and a power of 10 kW . The number of photons emitted per second are

(A) 1.72×10^{31} (B) 1327×10^{34} (C) 13.27×10^{34} (D) 0.075×10^{-34}

(33) In the circuit shown here, the voltage across L and C are respectively 300 V and 400 V. The voltage E of the ac source is .... Volt.



(A) 400 (B) 500 (C) 100 (D) 700

(34) The energy equivalent of 0.5 g of a substance is .... J

(A) 0.5×10^{13} (B) 4.5×10^{16} (C) 4.5×10^{13} (D) 1.5×10^{13} F

(35) If particles are moving with same velocity, then maximum de-Broglie wavelength will be for

(A) Neutron (B) Proton (C) β-partile (D) α-partile

(36) A negative charge is coming towards the observer. The direction of the magnetic field produced by it will be (as seen by observer)

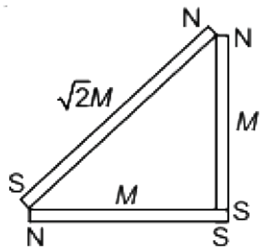
(A) Clockwise

(B) Anti-clockwise

(C) In the direction of motion of charge

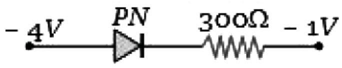
(D) In the direction opposite to the motion of charge

(37) The magnetic moment of the arrangement shown in the figure is............. M



(A) 0 (B) $2\sqrt{2}$ (C) 2 (D) 1

(38) What is the current in the circuit shown below ....amp



(A) 0 (B) 10^{-2} (C) 1 (D) 0.10

(39) When a forward bias is applied to a p – n junction, it

(A) Raises the potential barrier (B) Reduce the majority carrier current to zero

(C) Lower the potential barrier (D) None of the above

(40) Energy in the sun is generated mainly by

(A) Fusion of radioactive material (B) Fission of helium atoms

(C) Chemical reaction (D) Fusion of hydrogen atoms

(41) A coil of resistance 400Ω is placed in a magnetic field. If the magnetic flux ϕ (ωb) linked with the coil varies with time t (sec) as $\phi = 50t^2 + 4$. The current in the coil at t = 2 sec is...........A.

(A) 0.5 (B) 0.1 (C) 2 (D) 1

(42) The number of photo-electrons emitted per second from a metal surface increases when

(A) The energy of incident photons increases

(B) The frequency of incident light increases

(C) The wavelength of the incident light increases

(D) The intensity of the incident light increases

(43) For an AC circuit the potential difference and current are given by $V = 10\sqrt{2}\sin\omega t$ (in V) and $I = 2\sqrt{2}\cos\omega t$ (in V) respectively. The power dissipated in the instrument is............W

(A) 20 (B) 40 (C) $40\sqrt{2}$ (D) 0

(44) Which of the following is not Electro-maganetic waves?

(A) Heat rays (B) γ-rays (C) β-rays (D) X-rays

(45) Which is the correct ascending order of wavelengths?

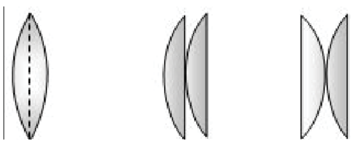
(A) $\lambda_{\text{visible}} < \lambda_{\text{X-ray}} < \lambda_{\text{gamma-ray}} < \lambda_{\text{microwave}}$

(B) $\lambda_{\text{gamma-ray}} < \lambda_{\text{X-ray}} < \lambda_{\text{visible}} < \lambda_{\text{microwave}}$

(C) $\lambda_{\text{X-ray}} < \lambda_{\text{gamma-ray}} < \lambda_{\text{visible}} < \lambda_{\text{microwave}}$

(D) $\lambda_{\text{microwave}} < \lambda_{\text{visible}} < \lambda_{\text{gamma-ray}} < \lambda_{\text{X-ray}}$

(46) Two similar plano-convex lenses are combined together in three different ways as shown in the adjoining figure. The ratio of the focal lengths in three cases will be



(A) 2 : 2 : 1 (B) 1 : 1 : 1 (C) 1 : 2 : 2 (D) 2 : 1 : 1

(47) A beam of electrons is moving with constant velocity in a region having electric and magnetic fields of strength 20 V m^{-1} and 0.5T at right angles to the direction of motion of the electrons. What is the velocity of the electrons .... ms^{-1}

(A) 20 (B) 40 (C) 8 (D) 5.5

(48) Electromagnetic waves are transverse in nature is evident by

(A) Polarization (B) Interference (C) Reflection (D) Diffraction

(49) If the magnetic field of a plane electromagnetic wave is given by

(The speed of light = 3×10^8 m/s)

$$B = 100 \times 10^{-8} \sin\left[2\pi \times 2 \times 10^{15}\left(t - \frac{\pi}{c}\right)\right] T$$

then the maximum electric field associated with it is

(A) 6×10^4 N/C (B) 3×10^4 N/C (C) 4×10^4 N/C (D) 4.5×10^4 N/C

(50) The photoelectric threshold for a certain metal surface is 330 Å. What is the maximum kinetic energy of the photoelectrons emitted, if radiations of wavelength 1100 Å are used ?

(A) 1 eV (B) 2 eV (C) 7.5 eV (D) No electron is emitted

PART – B

SECTION - A

- **Answer questions. Each question 2 marks. (Any 8)** **(16)**

(1) If 10^9 electrons move out of a body to another body every second, how much time is required to get a total charge of 1 C on the other body ?

(2) The electrostatic force on a small sphere of charge 0.4 mC due to another small sphere of charge –0.8 mC in air is 0.2 N.

(a) What is the distance between the two spheres ?

(b) What is the force on the second sphere due to the first ?

(3) Explain the position (orientations) of a dipole in stable equilibrium, unstable equilibrium and in maximum unstable equilibrium.

(4) Give the explanation of Gauss's law for magnetic field.

(5) Two concentric circular coils, one of small radius r_1 and the other of large radius r_2, such that $r_1 << r_2$, are placed co-axially with centres coinciding. Obtain the mutual inductance of the arrangement.

(6) A pure inductor of 25.0 mH is connected to a source of 220 V. Find the inductive reactance and rms current in the circuit if the frequency of the source is 50 Hz.

(7) Write information regarding electromagnetic waves in short.

(8) Compare and contrast the interference and diffraction.

(9) When and why the p-n junction is called a forward bias ? and describe the changes at such bias.

(10) Explain the phenomenon of nuclear fission.

(11) Using the formula for the radius of n^{th} orbit $r_n = \frac{n^2h^2\varepsilon_0}{\pi mZe^2}$ derive an expression for the total energy of electron in n^{th} Bohr's orbit.

(12) Find the

(a) maximum frequency, and

(b) minimum wavelength of X-rays produced by 30 kV electrons.

Use the following values if necessary

(1) $c = 3 \times 10^8\ ms^{-1}$, $m_e = 9.1 \times 10^{-31}kg$

(2) $h = 6.63 \times 10^{-34}$ Js, $m_n = m_p = 1.67 \times 10^{-27}kg$

(3) $e = 1.6 \times 10^{-19}$ c, $k_B = 1.38 \times 10^{-23}\ JK^{-1}$

SECTION - B

- **Answer questions. Each question 3 marks. (Any 6)** **(18)**

(13) Obtain the equation of electric field by dipole at a point on equator of dipole.

(14) What is Wheatstone bridge ? Explain its principle.

(15) State and explain Biot-Savart law for the magnetic field produced by a current element. Give the direction of magnetic field and define the unit of it.

(16) Derive the equation $E = -Bl\upsilon$ of a motional emf with the help of a suitable example.

(17) Obtain an equation of current for AC voltage applied to an inductor and draw a graph of V and I.

(18) Obtain the equation of power for combination of lenses.

(19) In Young's double-slit experiment using monochromatic light of wavelength λ the intensity of light at a point on the screen where path difference is λ, is K units. What is the intensity of light at a point where path difference $\frac{\lambda}{3}$?

(20) Explain effect of frequency of incident radiation in the experiment of photoelectric effect.

(21) It is found experimentally that 13.6 eV energy is required to separate a hydrogen atom into a proton and an electron. Compute the orbital radius and the velocity of the electron in a hydrogen atom.

SECTION - C

- **Answer questions. Each question 4 marks. (Any 4)** **(16)**

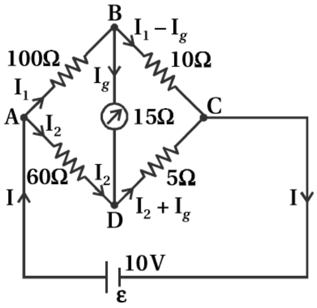
(22) (a) Determine the electrostatic potential energy of a system consisting of two charges 7 μC and – 2 μC (and with no external field) placed at (–9 cm, 0, 0) and (9 cm, 0, 0) respectively.

(b) How much work is required to separate the two charges infinitely away from each oth

(c) Suppose that the same system of charge is now placed in an external electric field $E = A\left(\frac{1}{r^2}\right)$; $A = 9 \times 10^5\ NC^{-1}m^2$. What would the electrostatic energy of the configuration be ?

(23) The four arms of a Wheatstone bridge figure have the following resistances :

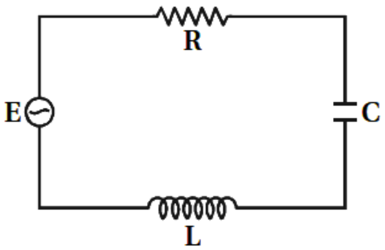
AB = 100 Ω, BC = 10 Ω, CD = 5Ω VG DA = 60 Ω



A galvanometer of 15Ω resistance is connected across BD. Calculate the current through the galvanometer when a potential difference of 10 V is maintained across AC.

(24) Figure shows a series LCR circuit connected to a variable frequency 230 V source.

L = 5.0 H, C = 80 μF, R = 40 Ω



(a) Determine the source frequency which drives the circuit in resonance.

(b) Obtain the impedance of the circuit and the amplitude of current at the resonating frequency.

(c) Determine the rms potential drops across the three elements of the circuit. Show that the potential drop across the LC combination is zero at the resonating frequency.

(25) A person with a normal near point (25 cm) using a compound microscope with objective of focal length 8.0 mm and an eyepiece of focal length 2.5 cm can bring an object placed at 9.0 mm from the objective in sharp focus. What is the separation between the two lenses ? Calculate the magnifying power of the microscope,

(26) How long can an electric lamp of 100 W be kept glowing by fusion of 2.0 kg of deuterium ? Take the fusion reaction as ${}^{2}_{1}H + {}^{2}_{1}H \rightarrow {}^{3}_{2}He + n + 3.27$ MeV

(27) Explain the use of the junction diode as a half wave rectifier by drawing a circuit and draw input and output waves.

Best of Luck

Std : 12^{th} (Science)

Subject : Physics

PAPER SET - 3

Marks : 100

PART - A (MCQ)

- **There are 50 objective type (M.C.Q.) questions in Part-A and all questions are compulsory.**

(1) Two reasons for using soft iron as the material for electromagnets

(A) high permeability and low retentivity (B) low permeability and low retentivity

(C) low permeability and low retentivity (D) low permeability and high retentivity

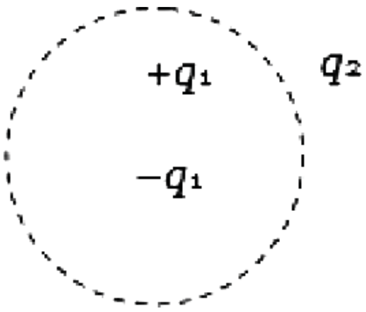
(2) The magnetic moment of a diamagnetic atom is

(A) > 1 (B) < 1 (C) 0 (D) 1

(3) Which of the following shown particle nature of light

(A) Refraction (B) Interference (C) Polarization (D) Photoelectric effect

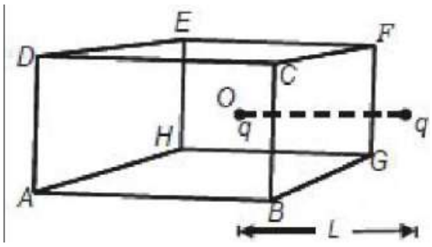
(4) Consider the charge configuration and spherical Gaussian surface as shown in the figure. When calculating the flux of the electric field over the spherical surface the electric field will be due to



(A) q_2 (B) Only the positive charges

(C) All the charges (D) $+q_1$ and $-q_1$

(5) A charged particle q is placed at the centre O of cube of length L (ABCDEFGH). Another same charge q is placed at a distance L from O.Then the electric flux through BGFC is



(A) $q/4\pi\varepsilon_0 L$ (B) zero (C) $q/2\pi\varepsilon_0 L$ (D) $q/3\pi\varepsilon_0 L$

(6) A parallel plate capacitor with air between the plates has a capacitance C. If the distance between the plates is doubled and the space between the plates is filled with a dielectric of dielectric constant 6 , then the capacitance will become

(A) 3C (B) $\frac{C}{3}$ (C) 12C (D) $\frac{C}{6}$

(7) The electric field intensity at a point in vacuum is equal to

(A) Zero

(B) Force a proton would experience there

(C) Force an electron would experience there

(D) Force a unit positive charge would experience there

(8) If particles are moving with same velocity, then maximum de-Broglie wavelength will be for

(A) Neutron (B) Proton (C) α - particle (D) β - particle

(9) If a magnet of pole strength m is divided into four parts such that the length and width of each part is half that of initial one, then the pole strength of each part will be

(A) m/4 (B) m/2 (C) m/8 (D) 4m

(10) Two lenses of power −15 D and +5 D are in contact with each other. The focal length of the combination is cm

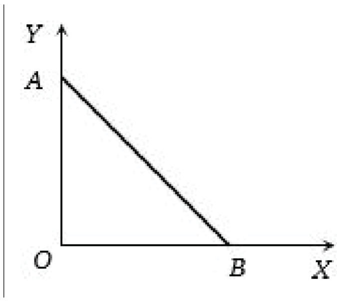
(A) +10 (B) −20 (C) −10 (D) +20

(11) A circular loop of flexible conducting material is kept in a magnetic field directed perpendicularly into its plane. By holding the loop at diametrically opposite points its is suddenly stretched outwards, then

(A) No current is induced in the loop (B) Anti-clockwise current is induced

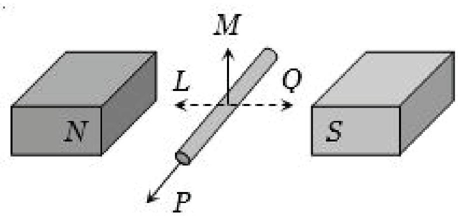
(C) Clockwise current is induced (D) Only e.m.f. is induced

(12) As per this diagram a point charge +q is placed at the origin O. Work done in taking another point charge −Q from the point A [co-ordinates (0, a)] to another point B [co-ordinates (a, 0)] along the straight path AB is



(A) Xg I (B) $\left(\frac{-qQ}{4\pi\varepsilon_0}\frac{1}{a^2}\right)\sqrt{2}a$ (C) $\left(\frac{qQ}{4\pi\varepsilon_0}\frac{1}{a^2}\right)\frac{a}{\sqrt{2}}$ (D) $\left(\frac{qQ}{4\pi\varepsilon_0}\frac{1}{a^2}\right)\sqrt{2}a$

(13) An electric potential difference will be induced between the ends of the conductor shown in the diagram, when the conductor moves in the direction



(A) P (B) Q (C) L (D) M

(14) The energy equivalent of 0.5 g of a substance is .... J

(A) 0.5×10^{13} (B) 4.5×10^{16} (C) 4.5×10^{13} (D) 1.5×10^{13}

(15) Huygen's principle is applicable to

(A) only light waves (B) only sound waves

(C) Only mechanical waves (D) for all the above waves

(16) A point Q lies on the perpendicular bisector of an electrical dipole of dipole moment p. If the distance of Q from the dipole is r (much larger than the size of the dipole), then electric field at Q is proportional to

(A) p^{-1} and r^{-2} (B) p and r^{-2} (C) p^{-1} and r^{-2} (D) p and r^{-3}

(17) An electric dipole of moment p is placed in an electric field of intensity E. The dipole acquires a position such that the axis of the dipole makes an angleθ¸ with the direction of the field. Assuming that the potential energy of the dipole to be zero when $\theta = 90°$, the torque and the potential energy of the dipole will respectively be

(A) $pE\sin\theta$, $-pE\cos\theta$ (B) $pE\sin\theta$, $2-PE\cos\theta$

(C) $pE\sin\theta$, $-2pE\cos\theta$ (D) $pE\cos\theta$, $-pE\cos\theta$

(18) If 50 joule of work must be done to move an electric charge of 2 C from a point, where potential is −10 volt to another point, where potential is V volt, the

(A) 5 (B) −15 (C) +15 (D) +10

(19) Which of the following rays has the maximum

(A) Gamma rays (B) Blue light (C) Infrared rays (D) Ultraviolet rays

(20) Wavelength of light of frequency 100 Hz

(A) 2×10^6 m (B) 3×10^6 m (C) 4×10^6 m (D) 5×10^6 m

(21) The work function of a substance is 4.0 eV . The longest wavelength of light that can cause photoelectron emission from this substance is

(A) 540 (B) 400 (C) 310 (D) 220

(22) The distance between the two plates of a parallel plate capacitor is doubled and the area of each plate is halved. If C is its initial capacitance, its final

(A) 2C (B) C/2 (C) 4C (D) C/4

(23) The work function for metals A, B and C are respectively 1.92eV, 2.0eV and 5eV . According to Einstein's equation, the metals which will emit photo electrons for a radiation of wavelength 4100 is/are

(A) None of these (B) A only (C) A and B only (D) All the three metals

(24) When ultraviolet rays are incident on metal plate, then photoelectric effect does not occurs. It occurs by the incidence of

(A) X-rays (B) Radio wave (C) Infrared rays (D) Green house effect

(25) At absolute zero, Si acts as

(A) non metal (B) metal (C) semiconductor (D) none of these

(26) The reverse biasing in a PN junction diode

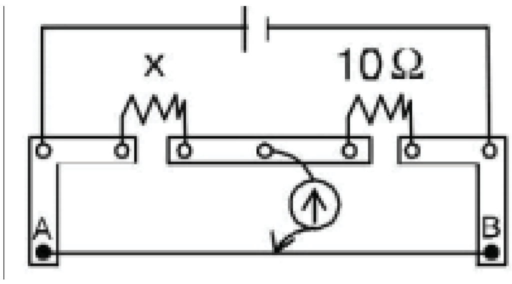
(A) Decreases the potential barrier

(B) Increases the potential barrier

(C) Increases the number of minority charge carriers

(D) Increases the number of majority charge carriers

(27) A meter bridge is set up as shown, to determine an unknown resistance'X' using a standard 10 ohm resistor. The galvanometer shows null point when tapping-key is at 52 cm mark. The endcorrections are 1 cm and 2 cm respectively for the ends A and B. The determined value of 'X' is .... Ω



(A) 10.2 (B) 10.6 (C) 10.8 (D) 11.1

(28) Two infinitely long parallel conducting plates having surface charge densities $+\sigma$and $-\sigma$ respectively, are separated by a small distance. The medium between the plates is vacuum. If Îµ0 is the dielectric permittivity of vacuum, then the electric field in the region between the plates is

(A) 0 volts/meter (B) $\frac{\sigma}{2\varepsilon_0}$ volts / meter

(C) $\frac{\sigma}{\varepsilon_0}$ volts / meter (D) $\frac{2\sigma}{\varepsilon_0}$ volts / meter

(29) A steady current flows in a metallic conductor of

(A) Current, electric field and drift speed (B) Drift speed only

(C) Current and drift speed (D) Current only

(30) If in a plano-convex lens, radius of curvature of convex surface is 10 cm and the focal length of lens is 30 cm, the refractive index of the material of the lens will be ....

(A) 1.5 (B) 1.66 (C) 1.33 (D) 3

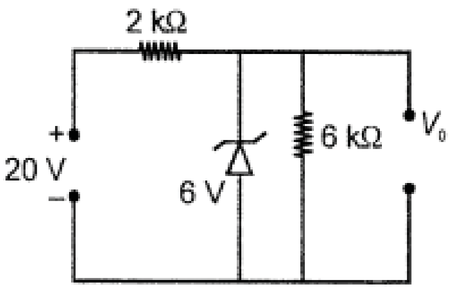
(31) Neglecting variation of mass with velocity, the wavelength associated with an electron having a kinetic energy E is proportional to

(A) $E^{\frac{1}{2}}$ (B) E (C) $E^{-\frac{1}{2}}$ (D) E^{-2}

(32) In Bohr's model of the hydrogen atom, the ratio between the period of revolution of an electron in the orbit of n = 1 to the period of revolution of the electron in the orbit n = 2 is

(A) 1 : 2 (B) 2 : 1 (C) 1 : 4 (D) 1 : 8

(33) What is the value of output voltage V_0 is ....V in the circuit shown in the figure?



(A) 6 (B) 14 (C) 20 (D) 26

(34) A wire of resistance x ohm is drawn out, so that its length is increased to twice its original length, and its new resistance becomes 20 Ω, then x will be .... Ω

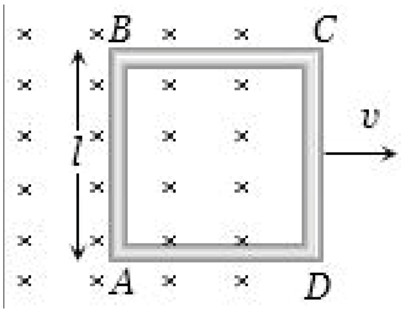
(A) 5 (B) 10 (C) 15 (D) 20

(35) Which of the following phenomena can explain

(A) Photoelectric effect (B) Interference

(C) Diffraction (D) Polarisation

(36) A conducting square loop of side l and resistance R moves in its plane with a uniform velocity υ perpendicular to one of its sides. A magnetic induction B constant in time and space, pointing perpendicular and into the plane at the loop exists everywhere with half the loop outside the field, as shown in figure. The induced e.m.f. is



(A) zero (B) $R\upsilon B$ (C) $\upsilon Bl/R$ (D) υBl

(37) Number of electrons in one coulomb of charge will be

(A) 5.46×10^{29} (B) 6.25×10^{18} (C) $1.6 \times 10^{+19}$ (D) 9×10^{11}

(38) +2 C and +6 C two charges are repelling each other

(A) 4 N (Attractive) (B) 4 N (Repulsive) (C) 8 N (Repulsive) (D) Zero

(39) The number of photo-electrons emitted per second from a metal surface increases when

(A) The energy of incident photons increases

(B) The frequency of incident light increases

(C) The wavelength of the incident light increases

(D) The intensity of the incident light increases

(40) An electron enters between two horizontal plates separated by 2 mm and having a potential difference of 1000 V . The force on electron is

(A) 8×10^{-22} N (B) 8×10^{-14} N (C) 8×10^{9} N F (D) 8×10^{14} N

(41) Consider four conducting materials copper, tungsten, mercury and aluminium with resistivity ρ_C, ρ_T, ρ_M and ρ_A espectively Then:

(A) $\rho_A > \rho_T > \rho_C$ (B) $\rho_C > \rho_A > \rho_T$ (C) $\rho_A > \rho_M > \rho_C$ (D) $\rho_M > \rho_A > \rho_C$

(42) If a dipole of dipole moment $\vec{p}$ is placed in a uniform $\vec{E}$ electric field then torque acting on it is given by

(A) $\vec{\tau} = \vec{p} \cdot \vec{E}$ (B) $\vec{\tau} = \vec{p} \times \vec{E}$ (C) $\vec{\tau} = \vec{p} + \vec{E}$ (D) $\vec{\tau} = \vec{p} - \vec{E}$

(43) A plane electromagnetic wave of frequency 50 MHz travels in free space along the positive x–direction. At a particular point in space and time, $\vec{E} = 6.3\hat{j}\,V/m$. The corresponding magnetic field $\vec{B}$, at that point will be

(A) $18.9 \times 10^{-8}\ \hat{k}T$ (B) $2.1 \times 1010^{-8}\ \hat{k}T$

(C) $6.3 \times 10^{-8}\ \hat{k}T$ (D) $18.9 \times 10^{8}\ \hat{k}T$

(44) In a compound microscope magnification will be large, if the focal length of the eye piece is

(A) Large (B) Smaller

(C) Equal to that of objective (D) Less than that of objective

(45) Threshold wavelength for photoelectric effect on sodium is 5000 Å. Its work function is

(A) 1J (B) 3×10^{-19}J (C) 4×10^{-19}J (D) 2×10^{-19}J

(46) A ferromagnetic material is heated above its curie temperature. Which one is a correct statement

(A) Ferromagnetic domains are perfectly arranged

(B) Ferromagnetic domains becomes random

(C) Ferromagnetic domains are not influenced

(D) Ferromagnetic material changes itself into diamagnetic material

(47) The maximum magnification that can be obtained with a convex lens of focal length 2.5 cm is (the least distance of distinct vision is 25 cm)

(A) 10 (B) 0.1 (C) 62.5 (D) 11

(48) A capacitor having capacitance C is charged to a voltage V . It is then removed and connected in parallel with another identical capacitor which is uncharged. The new charge on each capacitor is now

(A) CV (B) CV/2 (C) 2CV (D) CV/4

(49) The total charge induced in a conducting loop when it is moved in magnetic field depends on

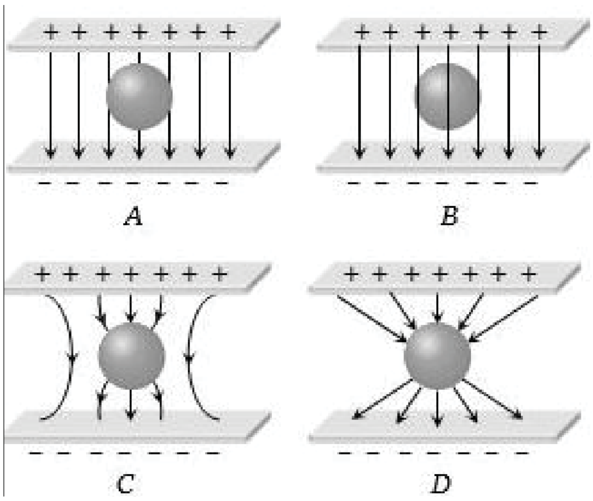
(A) The rate of change of magnetic flux

(B) Initial magnetic flux only

(C) The total change in magnetic flux

(D) Final magnetic flux only

(50) An uncharged sphere of metal is placed in between two charged plates as shown. The lines of force look like



(A) A (B) B (C) C (D) D

PART – B

SECTION - A

- **Answer questions. Each question 2 marks. (Any 8)** **(16)**

(1) Obtain the expression of electric field by a straight wire of infinite length and with linear charge density 'λ'

(2) A system has two charges qA = 2.5questions $q_A = 2.5 \times 10^{-7}$C VG $q_B = -2.5 \times 10^{-7}$C located at points A : (0, 0, –15 cm) and B : (0, 0, +15 cm), respectively. What are the total charge and electric dipole moment of the system ?

(3) Write limitations of Ohm's law.

(4) Write the difference between electric field lines and magnetic field lines.

(5) Define unit of self-inductance. On which factors self-inductance depends.

(6) A charged 30 μF capacitor is connected to a 27 mH inductor. What is the angular frequency of free oscillations of the circuit ?

(7) Obtain $\delta = i + e - A$ for equilateral prism.

(8) In a Young's double-slit experiment, the slits are separated by 0.28 mm and the screen is placed 1.4 m away. The distance between the central bright fringe and the fourth bright fringe is measured to be 1.2 cm. Determine the wavelength of light used in the experiment.

(9) The work function of caesium is 2.14 eV. Find (a) the threshold frequency for caesium, and (b) the wavelength of the incident light if the photocurrent is brought to zero by a stopping potential of 0.60 V.

(10) Explain the limitations of Bohr atomic model.

(11) From the relation $R = R_0 A^{1/3}$, where R_0 is a constant and A is the mass number of a nucleus, show that the nuclear matter density is nearly constant (i.e. independent of A).

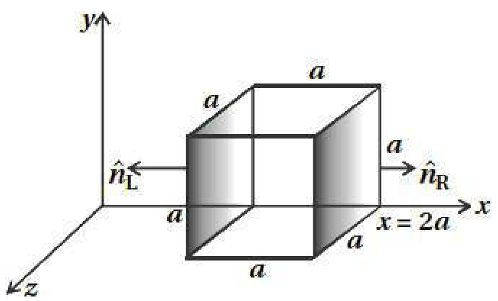
(12) Suppose a pure Si crystal has 5×10^{28} atoms m^{-3}. It is doped by 1 ppm concentration of pentavalent As. Calculate the number of electrons and holes.

Given that $n_i = 1.5 \times 10^{16}\ m^{-3}$

SECTION - B

● **Answer questions. Each question 3 marks. (Any 6)** **(18)**

(13) The electric field components in figure are $E_x = \alpha x^{1/2}, E_y = E_z = 0$ in which $\alpha = 800 N / Cm^{1/2}$ Calculate (a) the flux through the cube and (b) the charge within the cube. Assume thata = 0.1 m. [$\varepsilon_0 = 8.854 \times 10^{-12}\ C^2N^{-1}m^{-2}$]



(14) A heating element using nichrome connected to a 230 V supply draws an initial current of 3.2 A which settles after a few seconds to a steady value of 2.8 A. What is the steady temperature of the heating element if the room temperature is 27.0°C ? Temperature coefficient of resistance of nichrome averaged over the temperature range involved is $1.70 \times 10^{-4}\,°C^{-1}$.

(15) Apply Biot-Savart law to find the magnetic field due to a circular current carrying loop at a point on the axis of the loop.

(16) (a) Obtain the expression for the magnetic energy stored in a solenoid in terms of magnetic field B, area A and length l of the solenoid. (b) How does this magnetic energy compare with the electrostatic energy stored in a capacitor ?

(17) A sinusoidal voltage of peak value 283 V and frequency 50 Hz is applied to a series LCR circuit in which R = 3 W, L = 25.48 mH and C = 796 μF. Find

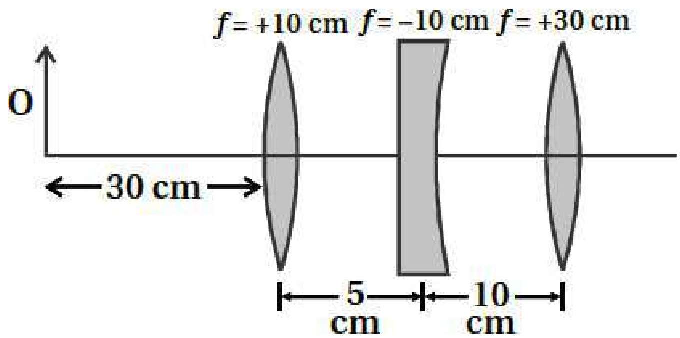
(a) the impedance of the circuit;

(b) the phase difference between the voltage across the source and the current;

(c) the power dissipated in the circuit; and

(d) the power factor

(18) Find the position of the image formed by the lens combination given in the figure.



(19) A beam of light consisting of two wavelengths, 650 nm and 520 nm, is used to obtain interference fringes in a Young's double-slit experiment.

(a) Find the distance of the third bright fring on the screen from the central maximum for wavelength 650 nm.

(b) What is the least distance from the central maximum where the bright fringes due wavelengths coincide ?

(20) Write Einstei's explanation of photoelectric effect and derive Einstein's equation.

(21) In accordance with the Bohr's model, find the quantum number that characterises the earth's revolution around the sun in an orbit of radius 1.5×10^{11} m with orbital speed 3×10^4 m/s. (Mass of earth = 6.0×10^{24} kg.)

SECTION - C

- **Answer questions. Each question 4 marks. (Any 4)** **(16)**

(22) Derive the formula for the electric potential due to an electric dipole at a point from it.

(23) Obtain the relation of phase between instantaneous current and voltage with the help of phase diagram for series LCR circuit.

(24) A beam of light converges at a point P. Now a lens is placed in the path of the convergent beam 12cm from P. At what point does the beam converge if the lens is (a) a convex lens of focal length 20cm, and (b) a concave lens of focal length 16cm ?

(25) What is nuclear fusion ? Explain giving its nuclear equations and write down the definition of thermonuclear fusion.

(26) Explain the use of junction diode as a full wave rectifier by drawing circuit diagram and draw the form of input and output waves.

(27) Explain the parallel combination of the cells in detail.

Best of Luck

Std. : 12^{th} Science **MATHS** Total Marks : 100 Time : 3 Hr.

✻ Choose the correct option. **(50)**

PART-I

(1) Relation R = {(1, 2), (2, 1)} is defined on set {1, 2, 3} then relation R is ....

(A) Only symmetric

(B) Symmetric and transitive but not reflexive

(C) Only reflexive (D) Equivalance

(2) Function f : R $\rightarrow$ R, $f(x) = x + 3$; f is defined then ....

(A) Many-one and onto (B) one-one and not onto

(C) one-one and onto (D) No one-one and not onto

(3) Binary operation $*$ defined on Z, $a * b = a^3 + b^3$ then $(2 * 3)^2$ = ...

(A) 35 (B) 1225 (C) 925 (D) 1625

(4) $\tan^{-1}\sqrt{3} - \sec^{-1}(-2)$ = ...

(A) $\frac{\pi}{3}$ (B) $-\frac{\pi}{3}$ (C) $\frac{2\pi}{3}$ (D) $\frac{\pi}{2}$

(5) $\sin\left(2\tan^{-1}\frac{4}{5}\right)$ = ...

(A) $\frac{24}{25}$ (B) $\frac{20}{29}$ (C) $\frac{8}{41}$ (D) $\frac{40}{41}$

(6) Range of $\left[\tan^{-1}x\right]$ function is .... ([ ] is greatest integer part function)

(A) {−2, −1, 0, 1} (B) {−2, −1, 0, 1, 2}

(C) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (D) {−1, 0, 1, 2}

(7) $\cos\left(\cos^{-1}\left(\cos\frac{7\pi}{3}\right)\right)$ = ...

(A) $\frac{7\pi}{3}$ (B) $\frac{\pi}{3}$ (C) $\frac{1}{2}$ (D) $-\frac{1}{2}$

(8) If $\begin{bmatrix} 3x+7 & 5 \\ y+1 & x+2 \end{bmatrix} = \begin{bmatrix} 4 & y-2 \\ 8 & 1 \end{bmatrix}$ then value of x + y = ...

(A) 6 (B) −6 (C) 4 (D) 8

(9) If A = $\left[a_{ij}\right]_{mxn}$ is square matrix then ....

(A) mn = 2 (B) m > n (C) m = n (D) m < n

(10) For 3 x 4 matrix A, If A'B and BA' is defined then matrix B is ....

(A) 3 x 4 (B) 4 x 4 (C) 3 x 3 (D) 4 x 3

(11) If A is 3 x 3 skew symmetric matrix, then |A| = ...

(A) 1 (B) 3 (C) −1 (D) 0

(12) If $\begin{vmatrix} x & 3 \\ 1 & x \end{vmatrix} = \begin{vmatrix} 4 & 5 \\ -1 & 2 \end{vmatrix}$ then value of x ....

(A) ± 4 (B) 4 (C) −4 (D) $\pm\sqrt{6}$

(13) $\begin{vmatrix} \sin 40^{o} & -\cos 40^{o} \\ \sin 50^{o} & \cos 50^{o} \end{vmatrix} = ...$

(A) 0 (B) 1 (C) −1 (D) $\sin 10^{o}$

(14) $\begin{vmatrix} 1 & yz & x(y+z) \\ 1 & zx & y(z+x) \\ 1 & xy & z(x+y) \end{vmatrix}$ = ...

(A) 0 (B) 1 (C) xyz (D) $xy + yz + zx$

(15) $\frac{d}{dx}\left(4\sin^{3}x - 3\sin x\right) = ...$

(A) $-3\sin 3x$ (B) $3\cos 3x$ (C) $-3\cos 3x$ (D) $-\cos 3x$

(16) $\frac{d}{dx}\left(e^{a\log x}\right) = ...$

(A) a^{x} (B) $a^{x}\log a$ (C) 0 (D) $a \cdot x^{a-1}$

(17) Derivative of $\sin^{-1}x$ with respect to $\cos^{-1}x$ ....

(A) −1 (B) 1 (C) 0 (D) $\frac{1}{\sqrt{1-x^{2}}}$

(18) In the circle of radius 4 cm, rate of change of the area with respect to r is ....cm^2/cm

(A) 6π (B) 8π (C) 12π (D) 16π

(19) From the given below which function is strictly decreasing on $\left(0, \frac{\pi}{2}\right)$?

(A) $\sin x$ (B) $\cos x$ (C) $\tan x$ (D) $\cos 3x$

(20) $x - y + 1 = 0$ is tangent of the curve $y^2 = 4x$ at point..........

(A) (1, −2) (B) (1, 2) (C) (−1, 2) (D) (4, 4)

(21) If $f(x) = 3x^2 + 5x + 3$, then approximate value $f(3.02)$ = ...

(A) 45.46 (B) 55.46 (C) 35.46 (D) 77.66

(22) $\int\left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^{2} dx = ... + c$

(A) $1 - \frac{1}{x^{2}}$ (B) $\frac{x^{2}}{2} + 2x - \frac{1}{x^{2}}$ (C) $\frac{x^{2}}{2} - \frac{1}{x^{2}}$ (D) $\frac{x^{2}}{2} + 2x + \log|x|$

(23) $\int \frac{\sqrt{1+\log x}}{x} dx = \text{..........} + c$

(A) $\frac{3}{2}(1+\log x)^{\frac{3}{2}}$ (B) $\frac{2}{3}(1+\log x)^{\frac{3}{2}}$ (C) $2\sqrt{1+\log x}$ (D) $1 + \log x$

(24) $\int \sin^2(2x) dx = A(4x - \sin 4x) + c$; $A^2 =$ ....

(A) 4 (B) $\frac{1}{4}$ (C) 64 (D) $\frac{1}{64}$

(25) $\int \frac{4x-8}{x^2-4x+5} dx = \text{...........} + c$

(A) $4 \log|x^2 - 4x + 5|$ (B) $\frac{1}{2}\log|x^2 - 4x + 5|$

(C) $\log(x^2 - 4x + 5)^2$ (D) $\log|x^2 - 4x + 5|$

(26) $\int e^x \left(\frac{1+\tan x}{\cos x}\right) dx = \text{...........} + c$

(A) $e^x \sec x$ (B) $e^x \tan x$ (C) $e^x \cos x$ (D) $e^x \sin x$

(27) $\int_0^{\frac{\pi}{2}} \sin\theta \sin 2\theta \, d\theta = \text{..........}$

(A) $\frac{4}{3}$ (B) $\frac{3}{4}$ (C) $\frac{2}{3}$ (D) $\frac{1}{3}$

(28) $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{1}{\sin 2x} dx = \text{..........}$

(A) $\frac{1}{2}\log 3$ (B) $\log 3$ (C) $\frac{1}{2}\log 2$ (D) $\log 2$

(29) If $\int_0^k (2x+1) dx = 6$, So k = ...

(A) 3 (B) 4 (C) 5 (D) -2

(30) Area of the bounded region of the circle $x^2 + y^2 = 8$ into first quadrent is ....

(A) 8π (B) 4π (C) 2π (D) 16π

(31) Area of the region bounded by curve $y = \cos x$ and $x = 0$ and $x = \pi$ is ....

(A) 1 (B) 2 (C) 3 (D) 4

(32) Area of the region bounded by parabola $y^2 = 16x$ and line $x = 4$ is ....

(A) $\frac{128}{3}$ (B) $\frac{64}{3}$ (C) 43 (D) $\frac{256}{3}$

(33) Differential equation $\left(\frac{d^3y}{dx^3}\right)^3 = \sqrt{1+\left(\frac{dy}{dx}\right)^4}$; its order and degree are .... and .... respectively.

(A) 3 and 6 (B) 3 and 4 (C) 3 and 3 (D) 6 and 3

(34) General solution of the differential equation $\frac{dy}{dx} = e^{x+y}$ is ....

(A) $e^{x} + e^{y} = c$ (B) $e^{x} + e^{-y} = c$ (C) $e^{-x} + e^{y} = c$ (D) $e^{-x} + e^{-y} = c$

(35) In second order differential equation, number of arbitrary constants in general solution are ....

(A) 0 (B) 1 (C) 2 (D) 3

(36) .... is vector from the given below.

(A) 1000 sec (B) 1000 cm^3 (C) 1000 kg (D) 1000 N

(37) If vector $\vec{a} = x\hat{i} + 2\hat{j} + z\hat{k}$ and $\vec{b} = 2\hat{i} + y\hat{j} + \hat{k}$ are equal then $\left|\vec{a} + \vec{b}\right| =$ ....

(A) 9 (B) 3 (C) 6 (D) 12

(38) Vector of the magnitude $\sqrt{7}$ in the direction of the vector $\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}$ is ....

(A) $\frac{14}{\sqrt{14}}\hat{i} + \frac{21}{\sqrt{14}}\hat{j} - \frac{7}{\sqrt{14}}\hat{k}$ (B) $\frac{2}{\sqrt{7}}\hat{i} + \frac{3}{\sqrt{7}}\hat{j} - \frac{1}{\sqrt{7}}\hat{k}$

(C) $\frac{2}{\sqrt{2}}\hat{i} + \frac{3}{\sqrt{2}}\hat{j} - \frac{1}{\sqrt{2}}\hat{k}$ (D) $\frac{2}{\sqrt{2}}\hat{i} + \frac{3}{\sqrt{2}}\hat{j} + \frac{1}{\sqrt{2}}\hat{k}$

(39) Direction cosine of the vector $\vec{b} = \hat{i} - \hat{j} + 2\hat{k}$ ....

(A) $\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}$ (B) $\frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}$ (C) $\frac{1}{6}, -\frac{1}{6}, \frac{2}{6}$ (D) $\frac{1}{2}, -\frac{1}{2}, 1$

(40) If $\left|\vec{a}\right| = 1, \left|\vec{b}\right| = 2$ and $\vec{a} \cdot \vec{b} = -1$ then angle between $\vec{a}$ and $\vec{b}$ is ....

(A) $\frac{2\pi}{3}$ (B) $\frac{\pi}{3}$ (C) $\frac{5\pi}{6}$ (D) $\frac{\pi}{2}$

(41) Area of parallelogram whose adjacent sides are $\vec{a} = 3\hat{i} + 5\hat{j} - 2\hat{k}$ and $\vec{b} = 2\hat{i} + \hat{j} - 3\hat{k}$ is ....

(A) $\frac{9}{2}\sqrt{3}$ (B) $9\sqrt{3}$ (C) $\sqrt{507}$ (D) $\sqrt{37}$

(42) Angle between two palnes $3x - 6y + 2z = 7$ and $2x + 2y - 2z = 5$ is ....

(A) $\cos^{-1}\frac{5}{7}$ (B) $\sin^{-1}\frac{\sqrt{366}}{21}$ (C) $\sin^{-1}\frac{5\sqrt{3}}{21}$ (D) $\cos^{-1}\frac{\sqrt{366}}{21}$

(43) Distance between the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+4}{6}$ and $\frac{x-3}{2} = \frac{y-3}{3} = \frac{z+5}{6}$ is ....

(A) $\frac{7}{\sqrt{293}}$ (B) $\frac{\sqrt{293}}{7}$ (C) $\frac{\sqrt{293}}{49}$ (D) $\frac{293}{49}$

(44) Cartesian equation of the line is parallel to line $\frac{x+3}{3} = \frac{4-y}{-5} = \frac{z+8}{6}$ and through $(-2, 4, -5)$ is ....

(A) $\frac{x+2}{3} = \frac{y+4}{5} = \frac{z+5}{6}$ (B) $\frac{x+2}{3} = \frac{y-4}{-5} = \frac{z+5}{6}$

(C) $\frac{x+2}{3} = \frac{y-4}{5} = \frac{z+5}{6}$ (D) $\frac{x-3}{-2} = \frac{y+5}{4} = \frac{z-6}{-5}$

(45) A feasible solution of an LP problem.

(A) must satisfy all of the problem's constraints simultaneously.

(B) need not satisfy all of the constraints, only some of them.

(C) must be a corner point of the feasible region.

(D) must optimize the value of the objective function.

(46) The corner points of the feasible region determined by the system of linear constraints are (0, 10), (5, 5), (15, 15), (0, 20). Let z = px + qy ; where p, q > 0. Condition on p and q so that the maximum of Z occurs at both the points (15, 15) and (0, 20) is...........

(A) p = q (B) q = 3p (C) q = 2p (D) p = 2q

(47) maximize of z = 2x + 3y subject to constraints $x + y \leq 3$ and $x + y \geq 6$ and $x \geq 0$, $y \geq 0$ is ....

(A) 18

(B) 12

(C) 17

(D) Feasible region does not exist.

(48) If $P(A) = \frac{7}{13}, P(B) = \frac{9}{13}$ and $P(A \cap B) = \frac{4}{13}$; so P(B/A) = ...

(A) $\frac{4}{7}$ (B) $\frac{4}{9}$ (C) $\frac{7}{9}$ (D) $\frac{1}{7}$

(49) If A and B are indepent events, and P(B) $\neq$ 0 then P(A/B) = ...

(A) 1 (B) 0 (C) P(B) (D) P(A)

(50) Two child in a family. Given that at least one child is boy, then the probability that other is also boy is ....

(A) $\frac{3}{4}$ (B) $\frac{1}{4}$ (C) $\frac{1}{2}$ (D) $\frac{1}{3}$

PART-II

SECTION - (A) : Solve. (Two marks question) (16)

(1) Convert into simplex form of $\tan^{-1}\left(\frac{\cos x}{1-\sin x}\right)$ (where $-\frac{3\pi}{2} < x < \frac{\pi}{2}$)

(2) Find : $\frac{d}{dx}\log_7(\log_7 x)$

(3) Evaluate : $\int \frac{\cos x}{\sqrt{4 - \sin^2 x}} dx$

OR

(3) Evaluate : $\int \frac{x^2 - 1}{x^4 + 6x^2 + 1} dx$

(4) Find area of the region bounded by curve y = x^2 and line y = 4.

(5) Find area of the region bounded by curve x^2 = 4y and line x = 4y − 2

OR

(5) Find area of the region bounded by parabola $y^2 = 4ax$ and its latus-rectum.

(6) If $\vec{a} = 3\hat{i} + 2\hat{j} + 2\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} - 2\hat{k}$ then find perpendicular unit vector of $\left(\vec{a} + \vec{b}\right)$ and $\left(\vec{a} - \vec{b}\right)$

(7) Find distance between plane which is passing through A(3, −1, 2), B(5, 2, 4) and C(−1, −1, 6) and point P(6, 5, 9)

(8) A fair coin rolled thrice. And events A and B are defined as above.

A : 4 exist when die rolled thired time.

B : 6 and 5 appear when a die rolled first and second time.

Given that event B is already exist then find probability of A.

SECTION - (B) : Solve. (Three marks question) **(18)**

(9) Function $f : R^+ \to [-5, \infty)$, $f(x) = 9x^3 + 6x - 5$ is defined. Prove that f is invertible and $f^{-1}(y) = \left(\dfrac{\sqrt{y+6} - 1}{3}\right)$

(10) Matrix $A = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$ represent as a sum of symmetric and skew-symmetric matrix.

(11) If cosy = x cos(a + y) and cosa ≠ ±1, So Prove that, $\dfrac{dy}{dx} = \dfrac{\cos^2(a+y)}{\sin a}$

OR

(11) $y = x^{\tan x} + \sqrt{\dfrac{x^2 + 1}{2}}$ then find $\dfrac{dy}{dx}$

(12) Find equation of the plane which passing through the common line of planes $x + y + z - 1 = 0$ and $2x + 3y - z + 4 = 0$ and parallel to x-axis.

(13) Solve LP problem by graphically. Find Maximum the $z = 20x + 80y$ subject to constraints $2x + 3y \le 6$, $x \ge 0$, $y \ge 0$

(14) In a factory which manufactures bolts, machines A, B and C manufacture respectively 25%, 35% and 40% of the bolts. Of their outputs, 5%, 4% and 2% are respectively defective bolts. A bolts is drawn at random from the product and is found to be defective. What is the probability that it is manufactured by the machine B ?

SECTION - (C) : Solve. (Four marks question) **(16)**

(15) Prove that , $\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = abc\left(1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)$

(16) Show that the altitude of the right circular cone of maximum volume that can be inscribed in a sphere of radius r is $\frac{4r}{3}$.

<u>OR</u>

(16) Function $f(x) = 2x^3 - 15x^2 + 36x + 1$, $x \in [1, 5]$ find local maximum and minimum values and global maximum and minimum values.

(17) Evaluate : $\int_0^{\pi} \frac{x}{a^2 \sin^2 x + b^2 \cos^2 x} dx$

(18) Solve differential equation : $(x\,dy - y\,dx)\,y \sin\left(\frac{y}{x}\right) = (y\,dx + x\,dy)\,x \cos\left(\frac{y}{x}\right)$

:::::

Std. : 12th Science **MATHS** Total Marks : 100 Time : 3 Hr.

✻ **Choose the correct option.** **(50)**

PART-I

(1) Let A = {1, 2, 3}. Then number of relations containing (1, 2) and (1, 3) which are feflective and symmetric but not transitive is ....

(A) 1 (B) 2 (C) 3 (D) 4

(2) f : N x N $\rightarrow$ N, f(m, n) = m + n is ....

(A) One-one and onto (B) One-one but not onto

(C) Many-One and onto (D) Not One-One and not onto

(3) If $a * b = \frac{ab}{100}$ defined on Q^+ then $(2 * 5)^{-1}$ = ...

(A) 1000 (B) 10000 (C) 100000 (D) 0.1

(4) If $\cos\left(2\sin^{-1} x\right) = \frac{1}{9}$, then x = ...

(A) 1 (B) $\frac{2}{3}$ (C) $\frac{1}{3}$ (D) $\frac{1}{2}$

(5) $\tan^{-1}1 + \tan^{-1}2 + \tan^{-1}3$ = ...

(A) π (B) $\frac{\pi}{2}$ (C) $\frac{3\pi}{4}$ (D) $\frac{3\pi}{2}$

(6) $\operatorname{cosec}\left(\tan^{-1}\left(\frac{y+x}{y-x}\right) - \tan^{-1}\frac{x}{y}\right)$ = ...

(A) $\sqrt{2}$ (B) 1 (C) $\frac{2}{\sqrt{3}}$ (D) 2

(7) $\tan\left(\frac{1}{2}\cos^{-1}\frac{\sqrt{5}}{3}\right)$ = ...

(A) $\frac{\sqrt{5}+1}{4}$ (B) $\frac{3-\sqrt{5}}{2}$ (C) $\frac{3+\sqrt{5}}{2}$ (D) $\frac{\sqrt{3}-1}{2\sqrt{2}}$

(8) If matrix A = diag[5 0 3] then A^2 = ...

(A) diag[1 1 1] (B) A' (C) −A (D) diag[25 0 9]

(9) If $A + B = \begin{bmatrix} 3 & 5 \\ 7 & 0 \end{bmatrix}$ and $A - B = \begin{bmatrix} 1 & 3 \\ 1 & 0 \end{bmatrix}$ then B = ...

(A) $\begin{bmatrix} & \\ 3 & 0 \end{bmatrix}$ (B) $\begin{bmatrix} 2 & 2 \\ 6 & 0 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & 3 \\ 1 & 0 \end{bmatrix}$ (D) $\begin{bmatrix} 4 & 8 \\ 8 & 0 \end{bmatrix}$

(10) If $A = \begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix}$ and $A^2 = \begin{bmatrix} a & b \\ b & a \end{bmatrix}$ then a = ... and b = ...

(A) a = 2 , b = 3 (B) a = 12, b = 13 (C) a = 3, b= 2 (D) a = 13, b= 12

(11) From given below which matrix is not non-singular matrix.

(A) $\begin{bmatrix} \sin\theta & \cos\theta \\ -\cos\theta & \sin\theta \end{bmatrix}$ (B) $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ (C) $\begin{bmatrix} 3 & 6 \\ 0 & 2 \end{bmatrix}$ (D) $\begin{bmatrix} 2 & 2 \\ 3 & 3 \end{bmatrix}$

(12) $\begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+x & 1 \\ 1 & 1 & 1+y \end{vmatrix} = ...$

(A) xy (B) $x + y$ (C) $(1 + x)(1 + y)$ (D) $2xy$

(13) $\begin{vmatrix} 1^2 & 2^2 & 3^2 \\ 2^2 & 3^2 & 4^2 \\ 3^2 & 4^2 & 5^2 \end{vmatrix} = ...$

(A) 1 (B) 8 (C) −8 (D) 0

(14) If A + B + C = π then $\begin{vmatrix} \sin(A+B+C) & \sin A & \cos A \\ -\sin A & \tan(A+B+C) & \tan A \\ \cos(B+C) & \tan(B+C) & 0 \end{vmatrix} = ...$

(A) 1 (B) 0 (C) $2\sin^2 A$ (D) $2\cos^2 A$

(15) $\frac{d}{dx}\log(\log x) = ...\ ;\ (x > 0)$

(A) $\frac{1}{\log x}$ (B) $\frac{1}{\log(\log x)}$ (C) $\frac{1}{x\log x}$ (D) $\frac{1}{x}$

(16) $\frac{d}{dx}\left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)^2 = ...$

(A) $1 - \frac{1}{x^2}$ (B) $1 + \frac{1}{x^2}$ (C) $1 - \frac{1}{x}$ (D) $\frac{1}{2\sqrt{x}} + \sqrt{x}$

(17) $\frac{d}{dx}\left(e^{3a}\right) = ...$

(A) e^{3a} (B) $3\ e^{3a}$ (C) 0 (D) $e^{3a}\log a$

(18) A stone is dropped into a quiet lake and waves move in circles at the speed of 5 cm/s. At the instant when the radius of the circular wave is 8 cm then the enclosed area increasing by .... cm^2/s

(A) 80π (B) 160π (C) 40π (D) 400π

(19) Line $y = x$ touches $y = x^2 + bx + c$ at (1, 1) then ....

(A) b = −1, c = 1 (B) b = 1, c = 1

(C) b = −1, c = −1 (D) b = −1, c = 2

(20) Approximate value of $(0.999)^{\frac{1}{10}}$ is ....

(A) 0.9990 (B) 0.09999 (C) 9.999 (D) 0.9999

(21) $x = 3t^2 + 1$, $y = t^3 - 1$ one parametric equation then slope of normal at point t = 1 is ....

(A) -2 (B) 2 (C) $\frac{1}{2}$ (D) $-\frac{1}{2}$

(22) $\int \frac{1}{4x+25} dx = 10k \tan^{-1}\frac{2x}{5} + c$ then k = ...

(A) $\frac{1}{50}$ (B) 100 (C) $\frac{1}{10}$ (D) $\frac{1}{100}$

(23) $\int \frac{x-2}{x^2 - 4x + 5} dx = ...+ c$

(A) $\log|x^2 - 4x + 5|$ (B) $\frac{1}{2}\log\left|x^2 - 4x + 5\right|$

(C) $2\log|x^2 - 4x + 5|$ (D) $\frac{1}{2}\left(x^2 - 4x + 5\right)^2$

(24) $\int e^{x\log a} \cdot e^{x} dx = ... + c$

(A) $a^x e^x$ (B) $\frac{(ae)^x}{(1+\log a)}$ (C) $\frac{e^x}{\log(ae)}$ (D) $\frac{a^x}{1+\log a}$

(25) $\int 4 \sin 2x \cos 3x \, dx = A \cos 5x + B \cos x + c$ then A = ... , B = ...

(A) $-\frac{2}{5}, 2$ (B) $-\frac{2}{5}, -2$ (C) $\frac{2}{5}, 2$ (D) $-\frac{1}{5}, 1$

(26) $\int \frac{1}{\sqrt{25 - 9x^2}} dx = A\sin^{-1}\left(\frac{3x}{5}\right) + c$ then 3A = ...

(A) $\frac{1}{3}$ (B) 1 (C) 3 (D) $\frac{3}{5}$

(27) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(x^3 + \tan^5 x + 1\right) dx$ = ...

(A) 0 (B) 1 (C) 2 (D) π

(28) $\int_0^{2a} \frac{f(x)}{f(x) + f(2a - x)} dx = ...$

(A) a (B) $\frac{a}{2}$ (C) $\frac{a}{4}$ (D) 2a

(29) $\int_{\sec^{-1}e}^{\sec^{-1}2e} \tan x \, dx = ...$

(A) log3e (B) log2e (C) log2 (D) 0

(30) Area of region bounded by curve $y = x|x|$, X-axis and lines $x = 1$ and $x = -1$ is ....

(A) $\frac{2}{3}$ (B) $\frac{4}{3}$ (C) $\frac{1}{3}$ (D) 0

(31) Area bounded by y = 3 − x and X-axis in interval [0, 3] is ....

(A) $\frac{11}{2}$ (B) $\frac{9}{2}$ (C) 5 (D) 4

(32) Area bounded by y = cos$x$, y = sin$x$, Y-axis and $0 \le x \le \frac{\pi}{4}$ is ....

(A) $\sqrt{2}+1$ (B) $\sqrt{2}$ (C) $\sqrt{2}-1$ (D) $2(\sqrt{2}-1)$

(33) Differential equation .... which shows equation of family y = mx.

(m is orbitrary constant)

(A) $xy_1 - y = 0$ (B) $y_1 - y = 0$ (C) $xy_1 + y = 0$ (D) $xy_1 = 0$

(34) Integrating factor of differential equation $\frac{dy}{dx} - y = \cos x$ is ....

(A) $e^{\sin x}$ (B) e^{-x} (C) e^{x} (D) e

(35) General solution of differential equation $dx - \frac{x}{y}dy = 0$ is ....

(A) $xy = c$ (B) $x = cy^2$ (C) $y = cx$ (D) $y = cx^2$

(36) For non-zero vactors $\vec{a}$ and $\vec{b}$; $\vec{a} \times (-2\vec{b})$ = ...

(A) $2(\vec{b} \times \vec{a})$ (B) $2(\vec{a} \times \vec{b})$ (C) $2|\vec{a}|^2$ (D) $-2|\vec{a}|^2$

(37) If vectors $\vec{a} = \lambda\hat{i} + \hat{j} - 2\hat{k}$, $\vec{b} = \hat{i} + \hat{j} + 3\hat{k}$ and $\vec{c} = 8\hat{i} + 5\hat{j}$ are co-planar then λ = ...

(A) −2 (B) 2 (C) 5 (D) −5

(38) $\vec{a}$ and $\vec{b}$ are unit vectors and angle between $\vec{a}$ and $\vec{b}$ is θ then, $|\vec{a} - \vec{b}\cos\theta|$ = ...

(A) $\sin^2\theta$ (B) $\cos\theta$ (C) $\sin\theta$ (D) $\sin\frac{\theta}{2}$

(39) If $|\vec{a}| = 3$, $|\vec{b}| = 4$ and $\vec{a}$ and $\vec{b}$ are perpendicular to each other, then $|\vec{a} \times \vec{b}|$ = ...

(A) 1 (B) 0 (C) 6 (D) 12

(40) Vector of opposite direction to $\vec{a} = \hat{j} - \hat{k}$ with the magnitude $17\sqrt{2}$ is ....

(A) $17\hat{i} - 17\hat{j}$ (B) $17\hat{j} - 17\hat{k}$ (C) $-17\hat{j} - 17\hat{k}$ (D) $-\hat{j} + \hat{k}$

(41) If $|\vec{x}| = 2$, $|\vec{y}| = 1$ and $\vec{x} \cdot \vec{y} = 1$ then $(3\vec{x} - 2\vec{y}) \cdot (2\vec{x} + 5\vec{y})$ = ...

(A) 3 (B) 45 (C) −25 (D) 25

(42) Plane $2x + 3y + 6z - 15 = 0$ makes an angle .... With the X-axis.

(A) $\cos^{-1}\frac{3\sqrt{5}}{7}$ (B) $\cos^{-1}\frac{2}{7}$ (C) $\sin^{-1}\frac{3}{7}$ (D) $\tan^{-1}\frac{2}{7}$

(43) Direction cosines of line through (2, 3, 4) and (4, 4, 6) is ....

(A) 3, 3, 3 (B) $\frac{1}{3}, \frac{1}{3}, \frac{1}{3}$ (C) $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$ (D) 2, 1, 2

(44) Lines $\frac{x-2}{2} = \frac{y+3}{\lambda} = \frac{z-7}{5}$ and $\frac{x-1}{3} = \frac{y-2}{-\lambda} = \frac{z-3}{\lambda}$ are perpendicular to each other then λ = ...

(A) 2 (B) 6 (C) −1 (D) 1

(45) Constraints are $x + y \le 5$, $3x + 3y \ge 21$; $x \ge 0$, $y \ge 0$ then possible feasible region is ....

(A) bounded (B) unbounded

(C) present in 1st and 2nd quadrent (D) does not exists

(46) Objective function $z = 30x - 30y + 2020$. Vertices of feasible bounded region are (15, 0), (15, 15), (10, 20), (0, 20) and (0, 15). Obtain maximum value of Z at .... point.

(A) (15, 0) (B) (0, 20) (C) (15, 15) (D) (10, 20)

(47) For sum of linear programm, ... point is not a point of fasible region bounded by the conditions $x + 2y \ge 10$, $3x + 4y \le 24$ and $x \ge 0$, $y \ge 0$.

(A) (0, 6) (B) (3, 4) (C) (0, 5) (D) (4, 3)

(48) If A and B are indepent events. If $P(A \cup B) = 0.5$ and $P(A) = 0.2$ then P(B) = ...

(A) $\frac{5}{8}$ (B) $\frac{3}{8}$ (C) $\frac{3}{4}$ (D) $\frac{1}{2}$

(49) A faire coins thrown n-times. Probability of head (H) occure at least one time is...........

(A) $1-\left(\frac{1}{6}\right)^n$ (B) $\frac{1}{2}$ (C) $1-\left(\frac{1}{2}\right)^n$ (D) $\left(\frac{1}{2}\right)^n$

(50) If A and B are events then $P(A) = \frac{1}{3}, P(B) = \frac{1}{4}$ and $P(A \cap B) = \frac{1}{5}$ then $P(A'/B')$...........

(A) $\frac{8}{45}$ (B) $\frac{23}{40}$ (C) $\frac{37}{40}$ (D) $\frac{37}{45}$

PART-II

SECTION - (A) : Solve. (Two marks question) (16)

(1) Solve : $2\tan^{-1}(\cos x) = \tan^{-1}(2\operatorname{cosec} x)$

(2) If $y = \sin(\tan^{-1} e^{-x})$ then find $\frac{dy}{dx}$.

(3) Evaluate : $\int \cos 6x \sqrt{1 + \sin 6x}\, dx$

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(3) Evalute : $\int \frac{1}{e^{x+1}} dx$

(4) Find area of bounded region by parabola $4y = 3x^2$ and line $2y = 3x + 12$

(5) Find area which is bounded by $\{(x, y) / 0 \le y \le x^2 + 1, 0 \le y \le x + 1, 0 \le x \le 2\}$

OR

(5) Find area bounded by feasible region by $x^2 + y^2 = 4$ and $(x - 2)^2 + y^2 = 4$

(6) Point R is divied $\overline{PQ}$ which joining P and Q into ratio 2 : 1 (i) internally (ii) externally and positision vactors of P and Q are $\hat{i} + 2\hat{j} - \hat{k}$ and $-\hat{i} + \hat{j} + \hat{k}$ respectively, then find position vector R.

(7) If co-ordinates of A, B, C and D are (1, 2, 3), (4, 5, 7), (−4, 3, −6) and (2, 9, 2) respectively, then find angle between AB and CD

(8) Probability distribution of random variable X as given below.

$X = x$	−2	−1	0	1	2	3
$P(x)$	0.05	0.14	0.23	0.31	0.16	0.11

then find E(X) and $(\sigma_X)^2$.

SECTION - (B) : Solve. (Three marks question) **(18)**

(9) A = {1, 2, 3, ..., 9} is a set. Relation R defined on A x A as (a, b)R(c, d) = a + d = b + c જ્યાં (a, b), (c, d) $\in$ A x A then prove that R is equivalance relation and find equivalence class of [(2, 5)].

(10) If $A = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$ then prove that, $A^n = \begin{bmatrix} \cos n\theta & \sin n\theta \\ -\sin n\theta & \cos n\theta \end{bmatrix}$, $n \in N$

OR

(10) By the use of matrix find commen point of two intersecting lines whose slopes are m_1 and m_2 and intercepts are c_1 and c_2 respectively. (Where $m_1 \neq m_2$)

(11) If $y = \sin^{-1}\left(\frac{2^{x+1}}{1+4^x}\right)$ then find $\frac{dy}{dx}$.

OR

(11) If $x = a(\cos\theta + \log \tan\frac{\theta}{2})$ and $y = a \sin\theta$ then find $\frac{d^2y}{dx^2}$.

(12) A line makes an angles α, β, γ and δ with diagonals of cube then prove that $\sin^2\alpha + \sin^2\beta + \sin^2\gamma + \sin^2\delta = \frac{8}{3}$

(13) One kind of cake requires 200 g of flour and 25 g of fat and another kind of cake requires 100 g of flour and 50 g of fat. Find the maximum number of cakes which can be made from 5 kg of flour and 1 kg of fat assuming that there is no shortage of the other incredients used in making the cakes.

(14) Bag : I contains 3 red and 4 black balls while another bag : II contains 5 rad and 6 black balls. One ball is drawn at random from one of the bags and it is found to be red. Find the probability that it was drawn from Bag : II.

SECTION - (C) : Solve. (Four marks question) **(16)**

(15) Use the product of $\begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix}\begin{bmatrix} -2 & 0 & 1 \\ 9 & 2 & -3 \\ 6 & 1 & -2 \end{bmatrix}$

Find solution of equation pair of $x - y + 2z = 1$, $2y - 3z = 1$, $3x - 2y + 4z = 2$.

(16) A square piece of tin of side 18 cm is to be made into a box without top, by cutting a square from each corner and folding up the flaps to form the box, What should be the side of the square to be cut off so that the volume of the box is the maximum possible.

OR

(16) Find maximum and minimum value of $f(x) = \sin^4 x + \cos^4 x$; $x \in \left[0, \frac{\pi}{2}\right]$

(17) Evaluate : $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin x + \cos x}{\sqrt{\sin 2x}} \, dx$

(18) Find standard solution of the differential equation $\frac{dy}{dx} + y \cot x = 4x \operatorname{cosec} x \; (x \neq 0)$ when $x = \frac{\pi}{2}$ and y = 0.

:::::

Time : 3 Hours **Total Marks : 100**

Time : 1 Hours **Total Marks : 50**

Instructions : (1) There are 50 objective type (MCQ) questions in Part-A and all questions are compulsory.

(2) The questions are serially numbered from 1 to 50 and each carries 1 mark.

(3) Read each question carefully, select proper alternative and answer in the OMR Sheet.

(4) The OMR Sheet is given for answering the questions. The answer of each circle (●) of the correct answer with ball-pen.

(5) Rough work is to be done in the space provided for this purpose in the Test Booklet only.

(6) Set No of question paper printed on the upper most right side of the question paper is to be written in the column provided in the OMR Sheet.

(7) Use of simple calculator and log table is allowed, if required.

(8) Notations used in this question paper have proper meaning

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PART - A

(2) If A = { 1, 2, 3 } then match following subsets of A x A properly.

(1) If $f : R \to R,\ f(x) = (5 - x^5)^{\frac{1}{5}}$, then *(fof) (x)* = ________

(A) $x^{\frac{1}{5}}$ (B) x^5 (C) x (D) $5 - x^5$

Part-A		Part-B	
(I)	R_1 = {(1, 1) (1,2), (2,1)}	(a)	only Symmetric
(II)	R_2 ={(1, 1) (2,2), (3,3), (1,2), (3,1)}	(b)	equivalence
(III)	R_3 ={(1, 1) (2,2), (3,3)}	(C)	only reflexive

(A) (I)→(b), (II)→(a), (III) →(c) (B) (I)→(a), (II)→(c), (III) → (b)

(C) (I)→(c), (II)→(b), (III) →(a) (D) (I)→(a), (II)→(b), (III) → (c)

(3) If $f : N \to N,\ f(x) = 2x + 3$ then ____________

f is not one-one (B) f is onto (C) $f^{-1}(x) = \frac{x-3}{2}$ (D) f^{-1} not defined

(4) $\tan^{-1}\left(\frac{x}{y}\right) - \tan^{-1}\left(\frac{x-y}{x+y}\right) =$ ________

(A)

(5) $Sin\left(\frac{\pi}{3} - Sin^{-1}\left(\frac{-1}{2}\right)\right) =$ ________

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{4}$ (D) $1 - \frac{3\pi}{4}$

(6)

(A) $\frac{1}{2}$ $\frac{1}{3}$ (C) $\frac{1}{4}$ (D)

If $Cos^{-1}\left(\frac{x}{5}\right) + Cosec^{-1}\left(\frac{5}{4}\right) = \frac{\pi}{2}$ then x = ____________

(A) 1 (B) 3 (C) 5 (D) 4

(7) If $x = \frac{1}{3}$ then $Cos\left(2Cos^{-1}x + Sin^{-1}x\right) =$ ___________

(A) $-\sqrt{\frac{8}{9}}$ (B) $-\sqrt{\frac{1}{3}}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{1}{2}$

(8) If $[\,1\ \ 2\ \ 1\,] \begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & 1 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ x \end{bmatrix} = O$ then value of x is equal to = _______.

(A) 1 (B) 2 (C) -1 (D) -2

(9) If $A = \begin{bmatrix} \text{Cos}\,\alpha & -\text{Sin}\,\alpha \\ \text{Sin}\,\alpha & \text{Cos}\,\alpha \end{bmatrix}$ and $A + A^{1} = I$ then $\propto\ =$ ________________

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$ (C) π (D) $\frac{3\pi}{2}$

(10) If A is a square matrix such that $A^2 = A$ then $(I + A)^2 - 7A =$ ________

(A) I (B) I - A (C) A (D) 3A

(11) If A and B are Symmetric matrices of same order, then AB + BA is a ________

(A) Skew symmetric matrix (B) Symmetric matrix

(C) Zero matrix (D) Identify matrix

(12) If the area of the triangle with vertices $(-2, 0)$ $(0, 4)$ $(0, K)$ having 4 Sq. units then K = _____

(A) ± 2 (B) ± 3 (C) 2, 8 (D) 0, 8

(13) If $A = \begin{bmatrix} 1 & \text{Cos}\theta & 1 \\ -\text{Cos}\theta & 1 & \text{Cos}\theta \\ -1 & -\text{Cos}\theta & 1 \end{bmatrix}$, where $0 \le \theta \le 2\pi$ then ________

(A) Det (A) = 0 (B) Det (A) $\in (2, \infty)$ (C) Det(A) $\in (2, 4)$ (D) Det(A) $\in [2, 4]$

(14) If $D = \begin{bmatrix} 0 & i-100 & i-500 \\ 100-i & 0 & 1000-i \\ 500-i & i-1000 & 0 \end{bmatrix}$ then D = ________

(A) 100 (B) 500 (C) 1000 (D) 0

(15) If $f(x) = \begin{cases} \frac{1-Coskx}{x^2} & : x \neq 0 \\ 8 & : x = 0 \end{cases}$ is continues at $x = 0$, then K = ________

(A) ± 1 (B) ± 2 (C) ± 3 (D) ± 4

(16) If $e^x + e^y = e^{x+y}$ then $\frac{dy}{dx}$ = __________________

(A) e^{x-y} (B) e^{y-x} (C) $-e^{y-x}$ (D) $-e^{x-y}$

(17) $\frac{d}{dx}\left(e^{\tan^{-1}x + \cot^{-1}x}\right) =$ ___________ : ($x \in$ R)

(A) 0 (B) 1 (C) e (D) $e^{\frac{\pi}{2}}$

(18) The interval in which $y = x^2 \cdot e^{-x}$ is increasing is _________

(A) $(-\infty, \infty)$ (B) $(-2, 0)$ (C) $(2, \infty)$ (D) $(0, 2)$

(19) The line $y = mx + 1$ is a tangent to the curce $y^2 = 4x$ if the value of m is ______

(A) 1 (B) 2 (C) 3 (D) $\frac{1}{2}$

(20) The normal at the point (2, -2) on the curve $3x^2 - y^2 = 8$ is ____

(A) $x + y = 0$ (B) $x + 2y = -2$ (C) $x - 3y = 8$ (D) $3x + y = 4$

(21) Approximate value of $(31)^{\frac{1}{5}}$ is ________

(A) 2.01 (B) 2.1 (C) 2.0125 (D) 1.9825

(22) $\int_{-1}^{1} \log\left(\frac{2019-x}{2019+x}\right) dx =$ ___________

(A) 0 (B) log 2019 (C) 1 (D) 2.log (2019)

(23) $\int_0^1 \frac{dx}{x+\sqrt{x}} =$ _________

(A) log 2 (B) log 3 (C) -log 2 (D) log 4

(24) $\int_0^2 x\ (2-x)^{\frac{3}{2}}\ dx =$ _____________

(A) $\frac{32\sqrt{2}}{35}$ (B) $\frac{54\sqrt{2}}{7}$ (C) $\frac{35\sqrt{2}}{32}$ (D) $\frac{1}{35\sqrt{2}}$

(25) $\int \sin\ (\log x)\ dx =$ ________ +c

(A) $\frac{x}{2}$ [Cos (log x) - Sin (log x)] (B) $\frac{x}{2}$ [Sin (log x) + Cos (log x)]

(C) $\frac{x}{2}$ [Sin (log x) – Cos (log x)] (D) x [Sin (log x) – Cos (log x)]

(26) $\int \frac{dx}{\sqrt{e^{2x}-1}} =$ __________ + c

(A) $\text{Sin}^{-1}(e^x)$ (B) $\text{Sec}^{-1}(e^x)$ (C) $\tan^{-1}(e^x)$ (D) $\text{Cot}^{-1}(e^x)$

(27) $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{dx}{1+\sqrt{\tan x}}$

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{12}$ (D) 0

(28) $\int_0^1 \tan^{-1}\left(\frac{2x-1}{1+x-x^2}\right) dx$ નું મૂલ્ય ________

(A) 1 (B) 0 (C) -1 (D) $\frac{\pi}{4}$

(29) $\int_0^{\frac{2\pi}{3}} \sqrt{1+Cos 2x}\ dx =$ ________

(A) $-\sqrt{6}$ (B) $-\sqrt{3}$ (C) $\sqrt{\frac{3}{2}}-2\sqrt{2}$ (D) $\frac{1}{\sqrt{2}}\left(4-\sqrt{3}\right)$

(30) Area of the region bounded by $\frac{x^2}{16}+\frac{y^2}{9}=4$

(A) 12 π (B) 24 π (C) 48 π (D) 64 π

(31) Area of the region bounded by the curve $y = Sin\,x$, and x - axis, $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$

(A) 1 (B) 2 (C) 4 (D) π

(32) Smaller area enclosed by the circle $x^2 + y^2 = 4$ and the line $x + y = 2$ is __________

(A) $2(\pi - 2)$ (B) $\pi - 2$ (C) $2\pi - 1$ (D) $2(\pi + 2)$

(33) The order and degree of differential equation $xy\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 - y\left(\frac{dy}{dx}\right)^3 = 0$ is ________

(A) 1 and 2 (B) 1 and 3 (C) 2 and 2 (D) 2 and 1

(34) Function $y = e^{-3x}$ is solution of differential equation _____________

(A) $\frac{dy}{dx} - 3y = 0$ (B) $\frac{d^2y}{dx^2} + \frac{dy}{dx} - 6y = 0$ (C) $\frac{d^2y}{dx^2} - 9y = 0$ (D) $\frac{dy}{dx} - 9y = 0$

(35) The number of arbitrary constants in the particular solution of a differential equation of third order are _______________

(A) 3 (B) 2 (C) 1 (D) 0

(36) Angle θ between two vectors $\vec{a} = \hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$ is ________

(A) $Cos^{-1}\frac{1}{3}$ (B) $-Cos^{-1}\frac{1}{3}$ (C) $-Sin^{-1}\frac{2\sqrt{2}}{3}$ (D) $Sin^{-1}\frac{1}{3}$

(37) For two vectors $\vec{a}$ and $\vec{b}$ $|\vec{a}| = 2$, $|\vec{b}| = 3$ and $\vec{a} \cdot \vec{b} = 4$ then $|\vec{a} - \vec{b}| =$ _____

(A) $\sqrt{3}$ (B) $\sqrt{15}$ (C) 1 (D) $\sqrt{5}$

(38) The value of $\hat{i} \cdot (\hat{j} \times \hat{k}) + \hat{j} \cdot (\hat{i} \times \hat{k}) + \hat{k}(\hat{i} \times \hat{j})$ is __________

(A) 0 (B) -1 (C) (D) 3

(39) If $\vec{a}$ and $\vec{b}$ are two non-zero collinear vectors then ____________ is correct.

(A) $\vec{b} \ne \lambda \vec{a};\ \forall \lambda \in R$ (B) $\bar{a} = \bar{b} = \bar{0}$

(C) The respective components of $\vec{a}$ and $\vec{b}$ are in proportion.

(D) both direction and magnitude of $\vec{a}$ and $\vec{b}$ are different.

(40) If scalar product of vector $\vec{a}$ with vectors $3\hat{i} - 5\hat{k}$, $2\hat{i} + 7\hat{j}$ and $\hat{i} + \hat{j} + \hat{k}$ are respectively $-1, 6, 5$ then $\vec{a}$ = _________

(A) $3\hat{i} + 2\hat{k}$ (B) $3\hat{i} + \hat{j} + 2\hat{k}$ (C) $\hat{i} + 3\hat{j} + 2\hat{k}$ (D) $\hat{i} + \hat{j} + \hat{k}$

(41) For two non-zero vectors $\vec{a}$ and $\vec{b}$ $|\vec{a} + \vec{b}| = |\vec{a}|$ then vectors $2\vec{a} + \vec{b}$ and $\vec{b}$ are ______

(A) Parallel (B) Perpendicular (C) Co-llinear (D) Equal

(42) The coordinates of the foot of the perpendicular drawn from the origin to the plane $2x - 3y + 4z - 6 = 0$ are _______

(A) $\left(\frac{12}{29}, \frac{-18}{29}, \frac{24}{29}\right)$ (B) $\left(\frac{12}{\sqrt{29}}, \frac{-18}{\sqrt{29}}, \frac{24}{\sqrt{29}}\right)$ (C) $\left(\frac{6}{29}, \frac{-9}{29}, \frac{12}{29}\right)$ (D) $\left(\frac{6}{\sqrt{29}}, \frac{-9}{\sqrt{29}}, \frac{12}{\sqrt{29}}\right)$

(43) The angle bet^{n} two line $\frac{x+3}{3} = \frac{y-1}{5} = \frac{z+3}{4}$ and $\frac{x+1}{1} = \frac{4-y}{-1} = \frac{z-5}{2}$ is ________________

(A) $Cos^{-1}\left(\frac{8\sqrt{3}}{13}\right)$ (B) $Cos^{-1}\left(\frac{8}{5\sqrt{3}}\right)$ (C) $Sin^{-1}\left(\frac{8\sqrt{3}}{15}\right)$ (D) $\frac{\pi}{2}$

(44) Distance between the two planes $2x + 3y + 4z - 4 = 0$ અને $4x + 6y + 8z = 12$ is ______

(A) 2 Units (B) 4 Units (C) 8 Units (D) $\frac{2}{\sqrt{29}}$ Units

(45) Objective function of an L.P. problems is ___________.

(A) a constant (B) a function to be optimized (C) an inequality (D) a quadratic equation

(46) In the question of maximum value of $z = 8000x + 12000y$ subject to constraints $9x + 12y \le 180$, $3x + 4y \le 60$, $x + 3y \le 30$, $x \ge 0$, $y \ge 0$ ____ is not a point of feasible region.

(A) (20, 0) (B) (12, 6) (C) (12, 0) (D) (0, 15)

(47) In solving the L.P. problem "Minimize $z = 6x + 10y$ subject to $x \ge 6, y \ge 2,$ $2x + y \ge 10, x \ge 0, y \ge 0$" redundant constraints are_______

(A) $x \ge 6, y \ge 2$ (B) $2x + y \ge 10, x \ge 0, y \ge 0$ (C) $x \ge 6$ (D) $x \ge 6, y \ge 0$

(48) The mean of the numbers obtained on throwing a die having written, 1 on three faces, 2 on two faces and 5 on one face is ______

(A) 1 (B) 2 (C) 5 (D) $\frac{8}{3}$

(49) E, F are independent events and P (E) ≠ O, P(F) ≠ O, then _______ is false.

(A) P(E/F) = P(E) (B) $P(F^{1} / E) = 1 - P(F/E)$

(C) $P(E^{1} / F^{1}) = 1 - P(E)$ (D) $P(E^{1} / F^{1}) = 1 - P(E/F)$

(50) When four letters are inserted in to four covers (one in each)

A = event that only one letters goes to the proper cover.

B = event that exactly three letters go to the proper covers.

C = event that all letters go to proper covers and

	Part-X		Part-Y
(p)	P (A)	(a)	O
(q)	P (B)	(b)	$\frac{1}{24}$
(r)	P (C)	(c)	$\frac{1}{3}$

then ____ is true

(A) p → a, q → c, r → b (B) p → c, q → a, r → b

(C) p → c, q → b, r → a (D) p → b, q → a, r → c

Time : 2 Hours **PART-B** Maximum Marks : 50

Instructions : (1) Write in a clear legible hand writing.

(2) There are three sections in Part-B of the questions paper and total 1 to 18 questions are there.

(3) All the questions are compulsory. Internal options are given.

(4) The numbers at the right side represent the marks of the questions.

(5) Start new section on new page.

(6) Maintain Sequence,.

(7) Use of simple calculator and log table is allowed, if required.

Section-A

- **Answer question No. 1 to 8 as directed. (Each question carry 2marks.)** **[16]**

(Each carries 2 marks)

(1) Prove that $\tan^{-1}\sqrt{x} = \frac{1}{2}Cos^{-1}\left(\frac{1-x}{1+x}\right)$, where $x \in [\,0, 1]$

(2) Differentiate $\sqrt{\frac{(x-3)(x^2+4)}{3x^2+4x+5}}$ w.r.t. x

(3) Find $\int \frac{(x+1)(x+\log x)^2}{x}\,dx$

(4) Find the area of the region bounded by the two parabolas $y = x^2$ and $y^2 = x$.

(5) Find the area bounded by the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and the ordinates

$x = 0$ and $x = ae$, where $b^2 = a^2(1-e^2)$ and $e < 1$.

OR

(5) Find the area of the region bounded by curve $y = 4x^2$ and lines $y = 1$, $y = 4$.

(6) If a unit vector $\vec{a}$, makes angle $\frac{\pi}{3}$ with $\hat{i}$, $\frac{\pi}{4}$ with $\hat{j}$ and an acute angle θ with $\hat{k}$, then find θ and hence the components of $\vec{a}$.

(7) Find the co-ordinates of the point where the line through the points $A(3, 4, 1)$ and $B(5, 1, 6)$ crosses the xy - plane.

(8) Three cards are drawn successively without replacement from a pack of 52 well shuffled cards. What is the probability that first two cards are kings and the third card drawn is an ace ?

OR

(8) Events A and B are such that $P(A)=\frac{1}{2}, P(B)=\frac{7}{12}$ and $P(A^1 \cap B^1)=\frac{1}{4}$. State whether A and B are independent ?

Section-B

- **Answer questions number 9 to 14 as directed. (Each question carry 3 marks)** **[18]**

(9) Consider $f: R^{+} \rightarrow [4, \infty)$, given by $f(x) = x^2 + 4$ show that f is invertible with the inverse f^{-1} of f given by $f^{-1}(y) = \sqrt{y-4}$ where R^{+} is set of all non-negative real numbers.

(10) Solve following system using matrix

$x - y + 2z = 1,\ 2y - 3z = 1,\ 3x - 2y + 4z = 2$

OR

(10) If $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$ then prove that $A^n = \begin{bmatrix} 1+2n & -4n \\ n & 1-2n \end{bmatrix}$ where n is any positive integer.

(11) If $x\sqrt{1+y} + y\sqrt{1+x} = 0$ for, $-1 < x < 1$ then prove that $\frac{dy}{dx} = \frac{-1}{(1+x)^2}$

(12) Fine the vector equation of the plane passing through the intersection of the planes

$\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 6$ and $\vec{r} \cdot (2\hat{i} + 3\hat{j} + \hat{k}) = -5$ and the point (1, 1, 1)

OR

(12) Find the vector equation of the line passing through the point (1, 2, -4)

and perpendicular to the two lines $\frac{x-8}{3} = \frac{y+19}{-16} = \frac{z-10}{7}$ and $\frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5}$

(13) The corner points of the bounded feasible region for L.P. problem are A (0, 4), B (0 , 5) C(3 5), D (5, 3), E (5, 0), F (4, 0) . Obtain the maximum and minimum value of the objective function $z = 10x - 7y + 1900$

(14) If a fair coin is tossed 10 times, find the probability of

(a) exactly six heads (b) atleast six heads (c) atmost six heasds

Section-C

- **Answer question No. 15 to 18 in detail. (Each questio carry 4 marks)** **[16]**

(15) Show that- $\begin{vmatrix} (y+z)^2 & xy & zx \\ xy & (x+z)^2 & yz \\ xz & yz & (x+y)^2 \end{vmatrix} = 2xyz(x+y+z)^3$

(16) An open topped box is to be constructed by removing equal squares from each corner of a 3 metre by 8 metre rectangular sheet of aluminum and folding up the sides. Find the volume of the largest such box.

OR

(16) Show that the height of the cylinder of maximum volume that can be inscribed in a sphere of radius R is $\frac{2R}{\sqrt{3}}$ also find the maximum value.

(17) Prove that - $\int_0^{\frac{\pi}{2}} \log_e(\sin x)\, dx = \frac{-\pi}{2} \log_e 2$

(18) The temperature of a body in a room is 80^0 F. After five minutes the temperature of the body becomes 60^0 F. After another 5 minutes the temperature becomes 50^0 F. What is the temperature of surrounding ? (Newton's law of cooling)

● ● ●

(SCIENCE STREAM)
(CLASS - XII)

Part - A : Time : 1 Hour / Marks : 50
Part - B : Time : 2 Hours / Marks : 50

(Part - A)

Time : 1 Hour] *[Maximum Marks : 50*

Instructions :

1) There are 50 objective type (M.C.Q.) questions in Part - A and all questions are compulsory.
2) The questions are serially numbered from 1 to 50 and each carries 1 mark.
3) Read each question carefully, select proper alternative and answer in the O.M.R. sheet.
4) The OMR sheet is given for answering the questions. The answer of each question is represented by (A) O, (B) O, (C) O, (D) O. Darken the circle ● of the correct answer with ball-pen.
5) Set No. of Question Paper printed on the upper-most right side of the Question Paper is to be written in the column provided in the OMR sheet.

■ Select the most appropriate answers: [4]

1) Where the knowledge is free means ________.
   (A) Knowledge is limited to few people
   (B) Nobody has to pay fee for study
   (C) Everyone has unrestricted access to knowledge
   (D) Free books and open schools for all

2) The real purpose of education is _______.
   (A) To give us position
   (B) To make us something
   (C) To make us somebody
   (D) To enable us think freely

3) The poet has ceased to be ....

(A) the song

(B) the clay

(C) the sword

(D) the dreamer

4) One of the following is not a part of stress control exercises.

(A) Mental quick fixes

(B) Short meditation

(C) Over thinking and shallow breathing

(D) The crown pull

■ **Read the stanza and answer the questions selecting the most appropriate options:[3]**

In
A dark cave
Flows
A dark river
And on
Its stony bank
There sits
a man,
Old
as the sun
or
older
Weaving
a yarn
of
rainbow
and
humming
a song.

5) Here the source of the river is ________.

(A) stony (B) unknown

(C) secret (D) mysterious

6) Here 'river' stands for ________

and 'a man' stands for ________.

(A) water - a person (B) life - God

(C) nature - a human being (D) God - life

7) The man is ________.

(A) watching the river flowing (B) weaving a yarn

(C) humming a song (D) both (B) and (C)

■ **Read the stanza and answer the questions selecting the most appropriate options:[2]**

In other days I used to be
A poet through whose pen
Innumerable songs would come
To win the hearts of men;
But now through new got knowledge
Which I had not had so long
I have ceased to be the poet
And have learned to be the song

8) The poet wants to win the hearts of men ________.

(A) by being a song

(B) by getting new knowledge

(C) by being a poet

(D) by creating many songs

9) The poet was ________ in the past.

(A) a song

(B) a poet

(C) a learned person

(D) a poetry

■ **Read the passage and answer the questions selecting the most appropriate options:[3]**

Derf was feeding questions into his computer. It printed out answers, but his mind was far away. He was thinking about how to get the 'centre'.

Then Derf's computer made a new noise. He looked at it. It hadn't been able to answer the last question. Derf had made a mistake. He should have asked, "How many rooms are needed in the city by the year 2080?" But instead he asked "How?"

It was a mistake. The computer couldn't answer a question like that. It seemed quite upset. Lights went on and off. It was trying hard to find an answer in its memory bank, but it could not. In the end, it printed out a new instruction to Derf. "I don't understand this question. Please ask in a different way."

Derf felt very excited. He had never seen this before. A computer breakdown. Yes! But computer ignorance (a state of being ignorant) no!

10) Derf had made a mistake ________.

(A) by operating wrongly

(B) by printing out a new instruction

(C) by asking 'how' instead of 'how many'

(D) by asking 'how many' instead of 'how'

11) Derf was excited to see __________.

(A) the lights on and off
(B) the breakdown of the computer
(C) the correct answer
(D) the computer working very well

12) The computer ignorance means ________.

(A) the computer knew the answer
(B) the computer could not comprehend the question
(C) the computer had an answer in its memory bank
(D) the computer had vast information in its memory bank

Read the passage and answer the questions selecting the most appropriate options: [2]

That evening the illnatured man had a strange feeling-something which had never felt before. And his wife gave him a strange look as he said, "Peg, Farmer Green has killed me! He said and he would and he has done it."

Yes, the 'enemy' was 'killed' without the loss of a single life or shedding even a single drop of blood. He went in the morning to confess his ingratitude to his kind neighbour, and ask his forgiveness, and the very man had been noted for nothing but his wickedness became friend of all.

13) Farmer Green has shown the best way _______.

(A) to shed the blood
(B) to kill anybody
(C) to change the attitude of a person in a positive way
(D) to confess ingratitude

14) Which of the following 'word' shows good quality?

(A) forgiveness
(B) illnatured
(C) ingratitude
(D) wickedness

■ **Select the most appropriate word to fill in the blanks:** [4]

15) The word ______ is mismatch for Walt.

(A) practical dreamer

(B) industrious

(C) imaginative

(D) ambitious

16) Water ______ should be well calibrated.

(A) production (B) consumption

(C) distribution (D) purification

17) The expression '__________' is superior to headache.

(A) indisposition (B) splitting headache

(C) disposition (D) rheumatism

18) The poet was not _______ in the past.

(A) a poet (B) a dreamer

(C) the sword maker (D) the clay

■ **Select the most appropriate word having the nearest meaning:** [3

19) Tedious : ___________.

(A) exciting (B) interesting

(C) boring (D) enjoyable

20) Equivalent : _______.

(A) parallel (B) different

(C) distinguished (D) same

21) Conceal : _______.

(A) strange (B) hide

(C) show (D) cancell

■ **Identify the function used in the sentence:** **[3]**

22) Jenish was so tired that he could not walk further.

(A) showing reason (B) showing result

(C) showing purpose (D) showing choice

23) If I were you, I would study more.

(A) showing supposition (B) showing condition in past

(C) showing condition in present (D) showing advice

24) My son was used to taking part in sports competition.

(A) Describing a person

(B) Showing past habit

(C) Showing it is not important

(D) Showing present habit

- **Choose the correct option to complete the sentence:** [5]

25) Bhavik is so active __________. (Showing result)

(A) that he could achieve his goal

(B) that he can achieve his goal

(C) because he does regular exercise

(D) as everybody likes him

26) The peon behaves __________. (Making supposition)

(A) as though he were a manager

(B) as a manager does

(C) like a manager

(D) as if he were manager

27) I don't like cricket __________. (Alternative choice)

(A) I like to watch either ODI or IPL

(B) I will play football

(C) Neither ODI nor IPL appeals me

(D) So I will avoid to watch it

28) __________, Komal was not selected for the post. (Indicating contrast)

(A) Instead of preparing well

(B) Inspite of preparing well

(C) Due to preparing well

(D) However she prepared well

29) My friend teaches in the class ________. (Expressing manner of action)

(A) as though he were a teacher

(B) with stylish manner

(C) as a teacher does

(D) as if he were teacher

■ **Select the most appropriate response:** **[4]**

30) Student : Why should I do regular exercise?

Teacher : ____________ (showing purpose)

(A) So that you can improve your health

(B) In order to improve your health

(C) With a view to improving your health

(D) All the above

31) Bharati : What is your opinion about the Taj Mahal?

Parul : ________________ (Comparing thing)

(A) It is so beautiful that I like it

(B) It is the best monuments

(C) It is one of the most wonderful monuments

(D) It is really very wonderful

32) Honey : I can not focus my mind

Teacher : _________. (Giving suggestion)

(A) You must consult a good doctor

(B) You had better do meditation

(C) You can focus your mind by practice

(D) You should do heavy physical exercise

33) Bhavik : What do you do when your father comes home?

Chetak : _______________. (Quick action - showing time)

(A) No sooner does my father comes than I hug him

(B) As soon as my father comes, I hug him

(C) When my father come home, I hug him

(D) No sooner do my father come then I hug him

■ **Choose the correct connector:** [3]

__(34)__ her illness, she went to school __(35)__ the teacher asked her __(36)__ she had consulted a doctor.

34) (A) Because of (B) Due to

(C) Owing to (D) Inspite of

35) (A) because (B) in which

(C) in where (D) where

36) (A) that (B) if

(C) so (D) why

Select the correct arrangement to make a meaning full sentence: [3]

37) costly (1) the smart phone is (2) to buy (3) too (4) for you (5)

(A) 2, 3, 5, 4, 1 (B) 2, 4, 1, 5, 3

(C) 3, 2, 4, 1, 5 (D) 2, 4, 5, 3, 1

38) his car (1) he wants to (2) anyway (3) let him (4) drive (5)

(A) 2, 5 ,1, 4,.3 (B) 4, 2, 5, 1, 3

(C) 4, 1, 5, 3, 2 (D) 4, 5, 1, 3, 2

39) Mrs. Doctor (1) in her work (2) I know (3) is very accurate (4) that (5)

3 5 1 4 2

(A) 3, 5, 1, 4, 2 (B) 1, 4, 2, 5, 3

(C) 3, 1, 5, 4, 2 (D) 1, 3, 5, 4, 2

Select the correct question to get the underlined word/phrase as answer: [3]

40) She takes tea twice in a day.

(A) How many times she takes tea?

(B) How often does she take tea in a day?

(C) How much tea does she take in a day?

(D) When does she take tea?

41) He put the keys in a drawer.

(A) Where he put the keys?

(B) Where does he put the keys?

(C) Where did he put the keys?

(D) In where did he put the keys?

42) Since he is late, the teacher will punish him.

(A) Since when will the teacher punish him?

(B) When will the teacher punish him?

(C) Why will the teacher punish him?

(D) Why the teacher will punish him?

Select the correct sentences: [4]

43) (A) The question paper was so easy that I can solve it.

(B) The question paper was easy enough for me to solve.

(C) The question paper was so easy that I could solve.

(D) The question paper was too easy to solve it.

44) (A) Good as India played, they lost the match.

(B) However well India played, they won the match.

(C) Inspite playing well, India lost the match.

(D) Though India played well, they lost the match.

45) (A) She is used to go to temple daily.

(B) She used to go to temple daily.

(C) She was used to go to temple daily.

(D) She used to going to temple daily.

46) (A) He failed in the exam due to working hard.

(B) He failed in the exam because of he was lazy.

(C) He failed in the exam since he was lazy.

(D) He failed in the exam though he was lazy.

■ Select the title of the read related with the sentence: [4]

47) Your 'Self' is a very complex thing.

(A) For Youth

(B) Can you install LOVE?

(C) Manage Your Stress

(D) Stress Control Exercises

48) We can open a source of optimism with the technique of anticipating success.

(A) Stress Control Exercises (B) Strike Against War

(C) For Youth (D) Manage Your Stress

49) We can achieve strength by means of Love and understanding.

(A) The Heaven of Freedom (B) Can You install LOVE?

(C) Strike Against War (D) No Men Are Foreign

50) Workers have no enemies except their masters.

(A) For Youth (B) Strike Against War

(C) No Men Are Foreign (D) Sojourner Truth

(SCIENCE STREAM)
(CLASS - XII)

(Part - B)

Time : 2 Hours] *[Maximum Marks : 50*

Instructions :

1) **Write in a clear legible handwriting.**
2) **There are four sections in Part - B of the question paper and total 1 to 21 questions are there.**
3) **All questions are compulsory. Internal options are given.**
4) **The numbers at the right side represent the marks of the question.**
5) **Start new section on new page.**
6) **Maintain the sequence.**

SECTION - A

■ **Read the paragraph and answer the questions:** **[8]**

As I was about to leave, he asked me, "Tell me one thing - how does it make any difference to you whether he is Nana or thinks himself to be Nani? How does it change anything whether he eats raita by staying your Nana or refuses to eat by morphing into your Nani? Let him live his life anyway he wants to.

1) What was the doctor's advice? **[1]**

2) Find out one word from the paragraph which means 'transforming from one image to another'. **[1]**

Then, there are ants that domesticate 'cows' and 'milk' them. What we have called 'cows' are a kind of green fly. These are found on rose leaves and on the leaves of beans. They give out a sweet, honey like liquid which the ants relish a lot. So the ants take these green flies to their nests and keep them there. They feed them, protect them from their enemies and they 'milk' them, pressing their sides gently and making them give out their honey.

3) Here' ________ is considered as 'cows' and ________ is considered as 'milk'. **[1]**

4) How do ants milk the cows? **[1]**

Customer Service Rep : Yes, it is. You should receive a message that says it will reinstall for the life of your HEART. Do you see that message?

Customer : Yes, I do. Is it completely installed?

Customer Service Rep : Yes, but remember that you have only the base programme. You need to begin connecting to other HEART in order to get the upgrades.

5) How should the customer get the upgrades of the programme? [1]

6) Which is the base programme? [1]

Walt was a complex man. To the writers, producers and animators who worked with him he was a genius who had an extra ordinary ability to add an extra stroke of imagination to any story or idea. To the millions of people who watched his TV show, he was a warm, kindly personality, bringing fun and pleasure into their homes. To bankers who financed us, I am sure he seemed like a wild man, hell bent for bankruptcy.

7) How was Walt for producers and animators? [1]

8) The bankers thought that Walt was ________________. [1]

9) **Write any one short note focussing on the questions:** [3]

Magic of Monkey's Paw

- How did the couple get the paw?
- What warning did Morris give?
- What was the horrible effect of his first wish?
- What was the second wish and its effect?
- After the third wish of Mr. White what happened?

J. Krishnamurthy's view on freedom

- Is the problem of freedom being simple or complex?
- What is to be free?
- What kind of persons are free?
- For what does our education system encourage us?
- What should be the real function of education?
- Where does the real freedom exist?

SECTION - B

10) Read the following passage and write its summary. **[5]**

Body language is an integral (fundamental) part of the selection criteria at a job interview.

A candidate may not be aware that there is someone among the interview panel who is studying his/her body language. Body language is read by observing the body posture, the eye movements and the rhythm of breathing. One of the skills a human resource manager should possess is the ability to read the body language. This would help him while conducting interviews or imparting training to employees. Reading body language is a skill that could be acquired through training and practice.

Read the news clipping and answer the questions: **[4]**

Ravi Shastri back as Team India head coach

TOI, Mumbai

Ravi Shastri, present head coach of the men's senior national team, has been reappointed by the Indian cricket board for the extended two year term that will conclude in November 2021.

Shastri, who took over as the head coach in July 2017 on a two year contract, was a direct entry into the interview process that took place in Mumbai on Friday.

The Kapil Dev led Cricket Advisory Committee (CAC) which also included former cricketers Anshuman Gaekwad and Shanta Rangaswamy, short - listed three names from the six candidates who had appointed for the role. Shastri, yet again, turned out to be the clear favourite.

Australia's Tom Moody, New Zealand's Mike Hesson, West Indies' Phil Simmons and Mumbai based coaches Lalchand Rajput and Robin Singh alongside Shastri, were the six candidates in a fray after shortlisting process by the BCCI earlier.

Questions :

11) Who was the former team India coach? [1]

12) Who is selected as Team India's new coach? [1]

13) The tenure of the new coach will be for _____ years. [1]

14) How many candidates remained after shortlisting process? [1]

Do as directed: [4]

15) A scientist noticed the following behaviour of ants. He was making a study of ants. The ants had organized the challenging task.

had been

<u>Start like this</u> : The following behaviour of ants was noticed ______________. [2]

16) You can open a source of optimism. You may succeed or not. If you do not work sincerely, you will have to repent a lot. You have to build your career by your own.

would hdd.

<u>Start like this</u> : You could open ______________. [2]

<u>SECTION - C</u>

17) **Render the following dialogue into indirect speech.** [3]

"What a fine dress! When did you buy it, Bharti?" Said Komal

"Last Sunday"

"How much did you pay for it?"

"Only Rs. 2,000".

18) **Study the data and write a paragraph on it in about eighty words:** **[5]**

Independence Day	15th August	got independence from British Empire
Republic Day	26th January	became a Republic Nation in 1950
Gandhi Jayanti	2nd October	birthday the father of Nation Mahatma Gandhi
National Youth Day	12th January	birthday of Swami Vivekananda a great philosopher and Hindu Monk
International Yoga Day	21st June	performed Yoga internationally

19) You are going to deliver a speech on problem of working women. Draft a speech using the given points in about 100 words. [5]

Points : Various fields - various duties - balance between work and family - transportation facilities - obstacles and difficulties - family restrictions, family culture - equal or superior to men.

OR

The Holistic General Hospital, Mehsana needs a full time experienced doctor (physician). You are Ronak B. Patel from 27, Adarsh Society, Station Road, Visnagar. Draft an application on behalf of him addressing to the Managing Trusty, The New Progressive Education Trust, Pooja Building, Near New District Court, Mehsana : 384001.

SECTION - D

20) **Write a paragraph in about 150 words using the key words.** [8]

- My Grand Parents :

Points : Introduction - role of grand parents - imparting values with love - caring and loving nature of grand parents towards their children - condition of nuclear family now - a - days.

OR

- Your favourite cartoon series :

Points : Names of some cartoon series - your favourite series - its characters - your favourite character - the part you like most - what you learn from it - message to children.

OR

- Each one tree one :

Points : Trees, boon for human - relationship - usefulness - duty towards trees - duty towards future generation - support of NGOs and also government.

21) Mr. Chetan Shah writes an email to flipkart.com requesting them to replace a smart phone he has purchased as the piece is not working properly.

Draft an email on his behalf in about 100 words. [5]

OR

You have recently visited a home for the aged with your classmates. Write a report on it in about 100 words.

✪ ✪ ✪